University of Minnesota Macro Notes

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Contents

1 Introduction

This document serves as a collection of concepts covered in graduate level macroeconomics. Specifically, the PhD Macroeconomics sequence ECON 8105 (Kehoe), ECON 8106 (Chari), ECON 8107 (Amador), and ECON 8108 (Luttmer) at the University of Minnesota. If you ever encouter these notes and have questions or spot an error, be sure to reach out to me here. Additionally, I speak a lot in the second person because these notes are written in a manner as to be read by myself in the past.

2 ECON 8105 (Kehoe)

2.1 General Equilibrium

There are two main types of economic modeling strategies that we need to talk about in this section. Arrow-Debreu (AD) representations and Sequential Markets (SM) representations. The crux of the idea is to develop a theory for the dynamic interactions of various households (and other sorts of agents) who face various resource constraints and optimization choices. We pose this question as simultaneously solving multiple maximization problems. The AD and SM representations are techniques of translating the structure of economies into a specific mathematical formulation with prices (formed based off of the relative utility preferences of agents).

In the AD representation, households are able to trade in futures contacts for some single consumption good at time zero (before the economy gets up and running) and there is no other specific mechanism to save the consumption good. The also are often able to trade futures contracts for bonds and capital as well depending on the model. We are often asked to describte the structure of such a market and the official wording is below:

Definition 1. *[Arrow Debreu Market Structure] With an Arrow-Debreu markets structure futures, markets for goods are open in period* 0 *. Consumers trade futures contracts among themselves.*

In the SM representation, households are able to trade one-period-ahead futures contracts for a consumption good in each time period.

Definition 2. *[Sequential Markets Structure] With sequential markets market structure, there are markets for goods and bonds open every period. Consumers trade goods and bonds among themselves.*

For the context of this class, one key detail that always tripped me up is that whenever there is capital, there is also a firm (which often goes unspecified) that utilizes this capital and transforms it into the consumption good. In this case, the market structure shifts to households also interacting with the firm who buys claims on capital and labor in exchange for claims on the consumption good. With sequential markets, there are also bonds that households trade amongst themselves.

2.1.1 The Paradox of General Equilibrium

Finally, I wanted to speak a little about the simultaneity of determining all the components of an equilibrium. It is a feature of the economics that is difficult to really intuit and appreciate in the beginning but comes as you work with more models and do more problems. The main confusion is that a lot of times we discuss economics as a series of cause and effects, whereas in these equilibrium models, everything is intertwined. It seems somewhat paradoxical that both households and firms consider prices to be a truth of the world that is independent of their behavior but somehow is also pinned down exactly through the preferences/characteristics of households and firms.

In the math, the simultaneity is clear, we solve all of the maximization problems and try to find such a solution that fit's all the agents in the economy while also satisfying the market clearing conditions. Qualitatively it took me some time to come to terms with how to interpret it until (I am a big fan of sci-fi) I thought about it in the context of the short story "Stories of your life" (it is also the basis for the movie "Arrival" and also a quick read for anyone who is interested) and how it describes the worldview of the aliens through various mathematical problems where there is an analogy between the physical causal interpretation and the simultaneity interpretation through Fermat's Principle of Least Time.

We do a lot of causal analysis in economics, even using general equilibrium models, and I think overcoming this paradox of how the model pins down each of the important objects versus the exercise of causal analysis that we are trying to do is important for getting a foundation of intuition. It will come as you are more involved in it, but I had a easier time with it when I tried to draw analogies to other concepts. The idea is that yes, everything interacts with everything else through some channel or the other, but we can try to wean out to what extent is each channel important.

2.2 Basic Endowment Model

Consider an economy that consists of two households who live forever and want to maximize their lifetime utility. Let $u_i(\cdot)$ be the in-period utility function for agent $i \in \{1, 2\}$. Given a consumption stream $\{x_t^i\}_{t=0}^{\infty}$ (where each x_t^i represents the amount of consumption that person i does at time t), we can write the total lifetime utility of agent i as

$$\sum_{t=0}^{\infty} \beta^t u_i(x_t^i).$$

The reason as to why we include this β^t discounting is one of these types of details that does not seem immediately obvious but soon becomes so after working a few problems. In short, it is meant to make sure that infinite sequences (since these agents live forever) converge so that lifetime utility is a well defined object.

Now let $\{w_t^i\}_{t=0}^{\infty}$ be the endowment stream of person i, where each w_t^i represents the amount of the consumption good that person i recieves at time t. Each household i is constrained such that they can only consume as much as their lifetime utility allows. This description of the model may seem a little vague right now, but hopefully a description of the equilibria will pin it down.

Definition 3. *[Arrow Debreu Equilibrium for basic endowment] An Arrow Debreu market equilibrium for such an economy as above is a collection of household allocations $\{(\hat{c}_t^1, \hat{c}_t^2)\}_{t=0}^{\infty}$ and prices $\{\hat{p}_t\}$ such that*

- *Household Maximization: Given prices and endowment stream $\{w_t^i\}_{t=0}^{\infty}$, the household allocations solve the household's maximization problem for each i*

$$\begin{aligned} &\max_{\{c_t^i\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u_i(c_t^i) \\ s.t. \quad &(1) \quad \sum_{t=0}^{\infty} \hat{p}_t c_t^i \leq \sum_{t=0}^{\infty} \hat{p}_t w_t^i, \\ &(2) \quad c_t^i \geq 0, \forall t \geq 0. \end{aligned}$$

- *Market Clearing: Given endowment streams $\{(w_t^1, w_t^2)\}_{t=0}^{\infty}$, the household allocations also satisfy*

$$(1) \quad \hat{c}_t^1 + \hat{c}_t^2 = w_t^1 + w_t^2.$$

One key difference between the AD and Sequential Markets formulations is that we need to rule out schemes where an individual can keep on borrowing forever and paying but their loans in the current period by borrowing more from future periods (aka a Ponzi scheme). This means we need to also say that there is some borrowing limit that is sufficiently large so that it wouldn't affect any behavior other than disrupting Ponzi schemes.

Definition 4. *[Sequential Markets Equilibrium for basic endowment] A Sequential Markets equilibrium* *for such an economy as above is a collection of household allocations* $\{(\hat{c}_t^1, \hat{b}_{t+1}^1, \hat{c}_t^2, \hat{b}_{t+1}^2)\}_{t=0}^{\infty}$ *and prices* $\{\hat{r}_t\}$ *such that*

- *Household Maximization: Given prices, endowment stream* $\{w_t^i\}_{t=0}^{\infty}$*, and initial bond holdings* b_0^i*, the household allocations solve the household's maximization problem for each i*

$$\begin{aligned}
&\max_{\{c_t^i, b_{t+1}^i\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u_i(c_t^i) \\
&s.t. \quad (1) \quad c_t^i + b_{t+1}^i \leq \hat{r}_t b_t^i + w_t^i, \forall t \geq 0, \\
&\qquad\;\; (2) \quad c_t^i \geq 0, \forall t \geq 0, \\
&\qquad\;\; (3) \quad b_t^i \geq -B, \text{ for some sufficiently large } B.
\end{aligned}$$

- *Market Clearing: Given endowment streams* $\{(w_t^1, w_t^2)\}_{t=0}^{\infty}$*, the household allocations also satisfy*

$$\begin{aligned}
&(1) \quad \hat{c}_t^1 + \hat{c}_t^2 = w_t^1 + w_t^2, \\
&(2) \quad \hat{b}_t^1 + \hat{b}_t^2 = 0.
\end{aligned}$$

2.2.1 Equivalence of Arrow-Debreu and Sequential Markets

Both of these markets are indeed equivalent after doing some transformations between the sets of equilibria objects. The main idea is that the allocations will always be the same but the prices, given of course that the prices relay different mechanisms of trading, require some transformation. The exact equivalence will be dependent on the economy structure, but the general idea in deriving the equivalence is to take advantage of the euler equations (I'll point it out clearly below but in general this is the equation that we can back out of the households' first order conditions that intertemporaly relates consumption, aka relates c_t to c_{t+1}) and the fact that the allocations are the same across both sets of equilibria. You can find out a specific worked out example of how this equivalence comes to be in the exercises.

Proposition 1. *[Equivalence of AD and Sequential Markets Equilibrium] Consider the AD equilibrium for the economy above, a collection of household allocations household allocations* $\{(\hat{c}_t^1, \hat{c}_t^2)\}_{t=0}^{\infty}$ *and prices* $\{\hat{p}_t\}$*. With this, we can write that the corresponding sequential markets equilibrium of household allocations* $\{(\bar{c}_t^1, \bar{b}_{t+1}^1, \bar{c}_t^2, \bar{b}_{t+1}^2)\}_{t=0}^{\infty}$ *and prices* $\{\bar{r}_t\}$ *is*

- $\bar{c}_t^i = \hat{c}_t^i, \forall t, i$
- $\bar{r}_t = \frac{\hat{p}_{t-1}}{\hat{p}_t} - 1$
- $\bar{b}_0 = 0$*, and* $\bar{b}_{t+1} = w_t^i + (1 + \bar{r}_t) - \bar{c}_t^i$

We can back out the reverse relationship through the relationships decribed above.

2.3 Basic Optimal Growth Model

Now, consider an economy with a representative household and a representative firm. The household now has the ability to sell their labor and capital to the firm which it uses as inputs to create the consumption good via technology $F(\cdot)$. In return, the firm pays the household wages (for labor) and rent (for capital). It will be a bit more clear once we talk about the equilibrium concepts. For this basic version, the firm acts as if it is operating in a perfectly competitive market which means that it makes zero profit and minimizes costs. Assuming that w_t is the wage rate for labor and r_t^k is the rent for capital, the profits for the firm are $p_t F(k_t, \ell_t) - w_t \ell_t - r_t^k k_t$. We can back out the optimality conditions of the firm from this formulation and will use this in our definition of the equilibrium.

Each household has a constant endowment of labor each period which we will normalize to one. This means that they can spend their time on either working for a wage or something else (the effect of which is captured in the utility function).

One final important detail to hash out is how the capital market works. Households invest in new capital at time t which they can rent out to firms at time $t+1$. This capital sticks around over time but depreciates with rate δ. It is NOT the case that capital acts like a bond where households lend a certain amount to the firm in time period t and gets paid with interest in time period $t+1$.

Definition 5. *[Arrow Debreu Equilibrium for basic growth model] An Arrow Debreu market equilibrium for such an economy as above is a collection of household allocations $\{(\hat{c}_t, \hat{k}_{t+1}, \hat{\ell}_{t+1})\}_{t=0}^{\infty}$, firm allocations $\{\hat{k}_t^f, \hat{\ell}_t^f\}_{t=0}^{\infty}$, and prices $\{\hat{p}_t, \hat{r}_t^k, \hat{w}_t\}$ such that*

- *Household Maximization: Given prices and initial level of capital (bestowed by some higher power) k_0, the household allocations solve the household's maximization problem for each i*

$$\begin{aligned} \max_{\{c_t, \ell_t, k_{t+1}\}_{t=0}^{\infty}} & \sum_{t=0}^{\infty} \beta^t u(c_t, \ell_t) \\ s.t. \quad (1) \quad & \sum_{t=0}^{\infty} \hat{p}_t c_t + \hat{p}_t k_{t+1} - \hat{p}_t (1-\delta) k_t \leq \sum_{t=0}^{\infty} \hat{w}_t \ell_t + \hat{r}_t^k k_t, \\ (2) \quad & c_t \geq 0, k_t \geq 0, 0 \leq \ell_t \leq 1, \forall t \geq 0. \end{aligned}$$

- *Firm Maximization: Given prices, firms minimize costs and make zero profits, specifically the firm allocations satisfy*

$$\frac{\hat{r}_t^k}{\hat{p}_t} = \frac{\partial F}{\partial k}(\hat{k}_t^f, \hat{\ell}_t^f)$$

$$\frac{\hat{w}_t}{\hat{p}_t} = \frac{\partial F}{\partial w}(\hat{k}_t^f, \hat{\ell}_t^f)$$

- *Market Clearing: The household and firm allocations also satisfy*

$$\textit{Goods Market:} \quad \hat{c}_t + \hat{k}_{t+1} = F(\hat{k}_t^f, \hat{\ell}_t^f) + (1-\delta)\hat{k}_t, \forall t \geq 0,$$

$$\begin{aligned} \textit{Capital Market:} \quad & \hat{k}_t = \hat{k}_t^f, \forall t \geq 0, \\ \textit{Labor Market:} \quad & \hat{\ell}_t = \hat{\ell}_t^f, \forall t \geq 0. \end{aligned}$$

Of course, in the sequential markets definition we have bonds. So far, we have described a model with a representative consumer which means that the household has no one to trade bonds with and as a result bond holdings are trivially 0 at any point in time. This will change as we add multiple types of households or some other agent, such as a government, that is willing to buy/sell bonds.

Definition 6. *[Sequential Markets Equilibrium for basic growth model] A sequential markets equilibrium for such an economy as above is a collection of household allocations* $\{(\hat{c}_t, \hat{k}_{t+1}, \hat{b}_{t+1}, \hat{\ell}_{t+1})\}_{t=0}^{\infty}$*, firm allocations* $\{\hat{k}_t^f, \hat{\ell}_t^f\}_{t=0}^{\infty}$*, and prices* $\{\hat{p}_t, \hat{r}_t^k, \hat{r}_t^b, \hat{w}_t\}$ *such that*

- *Household Maximization: Given prices and initial level of capital (bestowed by some higher power)* k_0 *and bond holdings* b_0*, the household allocations solve the household's maximization problem for each* i

$$\begin{aligned} \max_{\{c_t, \ell_t, k_{t+1}, b_{t+1}\}_{t=0}^{\infty}} & \sum_{t=0}^{\infty} \beta^t u(c_t, \ell_t) \\ s.t. \quad (1) \quad & c_t + k_{t+1} + \hat{b}_{t+1} \leq (1 - \delta + \hat{r}_t^k) k_t + \hat{w}_t \ell_t + (1 + \hat{r}_t^b) b_t, \\ (2) \quad & c_t \geq 0, k_t \geq 0, 0 \leq \ell_t \leq 1, \forall t \geq 0, \\ (3) \quad & b_t \geq -B, \textit{ for some sufficiently large } B. \end{aligned}$$

- *Firm Maximization: Given prices, firms minimize costs and make zero profits, specifically the firm allocations satisfy*

$$\frac{\hat{r}_t^k}{\hat{p}_t} = \frac{\partial F}{\partial k}(\hat{k}_t^f, \hat{\ell}_t^f)$$

$$\frac{\hat{w}_t}{\hat{p}_t} = \frac{\partial F}{\partial w}(\hat{k}_t^f, \hat{\ell}_t^f)$$

- *Market Clearing: The household and firm allocations also satisfy*

$$\begin{aligned} \textit{Goods Market:} \quad & \hat{c}_t + \hat{k}_{t+1} = F(\hat{k}_t^f, \hat{\ell}_t^f) + (1 - \delta)\hat{k}_t, \forall t \geq 0, \\ \textit{Capital Market:} \quad & \hat{k}_t = \hat{k}_t^f, \forall t \geq 0, \\ \textit{Labor Market:} \quad & \hat{\ell}_t = \hat{\ell}_t^f, \forall t \geq 0, \\ \textit{Bonds Market:} \quad & \hat{b}_{t+1} = 0, \forall t \geq 0. \end{aligned}$$

2.4 Basic Overlapping Generations Model

An overlapping generations model is not necessarily a separate entity from the concept of the topics we have discussed so far. The idea is still to solve for an equilibrium but the main difference from the models we will discuss in this section versus the section above is that households only live for two periods (something that is often extended in a lot of other models in use right now but the basic version we cover will only have two periods of life) and recieve different endowments in their young and old ages.

Specifically, consider a world with time periods $t \in (1, 2, \ldots)$. In each period, there are two households: one old and one young. The old household will die in the next period and the young household will live for one more period and become old. In this sense, we imbue some heterogeneity between households which often time leads to trading taking place. One important piece of notation is that we will designate an allocation or endowment that belongs to an individual born at time t at time t and $t+1$ respectively as c_t^t, c_{t+1}^t. Or in other words, the subscript denotes the time period in which a certain object belongs to and the superscript denotes the time period in which the person was born. We will also say that in period 1, there exists some person who was born in period 0 that starts out as some old guy in period 1. We don't recognize anything that happened in period 0 or before. This is just to make sure that there are always one old and young household in the economy at any time period $t \geq 1$.

Whether or not trading takes place is an important question in this model. If the old generation has a higher income than the younger generation, then they want to consume the entirety of their income given that they will die in the next period and not be able to recoup any returns from bonds; this often times leads to no trading occuring in the model and there are some exercises later in this section that will demonstrate that.

We can also do growth in an overlapping generations model as well, but I won't cover that in this guide since there aren't many differences between the OLG and infinitely lived versions. As with the previous model, let's look at what an AD and Sequential Markets Equilibrium looks like

Definition 7. *[Arrow Debreu Equilibrium for basic OLG model] An Arrow Debreu Equilibrium for such an economy as above is a collection of household allocations $\{(\hat{c}_t^t, \hat{c}_{t+1}^t)\}_{t=0}^{\infty}$ and prices $\{\hat{p}_t\}_{t=0}^{\infty}$ such that*

- *Initial Old Household Maximization:* *Given price $\hat{p}_1$ and endowments (m, w_1^0), the household allocation $\hat{c}_1^0$ solves the following maximization problem*

$$\begin{aligned} \max_{c_1^0} \quad & u(c_1^0) \\ s.t. \quad (1) \quad & \hat{p}_1 c_1^0 \leq \hat{p}_1 w_1^0 + m, \\ (2) \quad & c_1^0 \geq 0. \end{aligned}$$

- *Household Maximization:* *Given prices and endowment structure $\{(w_t^t, w_{t+1}^t\}_{t=0}^{\infty}$, the household allocations solve the series of maximization problems for every $t \geq 1$*

$$\begin{aligned} \max_{c_t^t, c_{t+1}^t} \quad & u(c_t^t, c_{t+1}^t) \\ s.t. \quad (1) \quad & \hat{p}_t c_t^t + \hat{p}_{t+1} c_{t+1}^t \leq \hat{p}_t w_t^t + \hat{p}_{t+1} w_{t+1}^t, \end{aligned}$$

$$(2) \quad c_t^t, c_{t+1}^t \geq 0.$$

- *Market Clearing: The allocations also satisfy*

$$\text{Goods Market:} \quad \hat{c}_t + \hat{c}_{t-1}^t = w_t^t + w_{t-1}^t$$

Then for the sequential markets representation, we add one-period bonds.

Definition 8. *[Sequential Markets Equilibrium for basic OLG model] A Sequential Markets Equilibrium for such an economy as above is a collection of household allocations $\{(\hat{c}_t^t, \hat{c}_{t+1}^t, \hat{b}_{t+1}^t)\}_{t=0}^{\infty}$ and prices $\{\hat{r}_{t+1}\}_{t=0}^{\infty}$ such that*

- *Initial Old Household Maximization: Given price and endowments (m, w_1^0), the household allocation $\hat{c}_1^0$ solves the following maximization problem*

$$\begin{aligned}
&\max_{c_1^0} && u(c_1^0) \\
&s.t. && (1) \quad c_1^0 \leq w_1^0 + m, \\
& && (2) \quad c_1^0 \geq 0.
\end{aligned}$$

- *Household Maximization: Given prices and endowment structure $\{(w_t^t, w_{t+1}^t\}_{t=0}^{\infty}$, the household allocations solve the series of maximization problems for every $t \geq 1$*

$$\begin{aligned}
&\max_{c_t^t, c_{t+1}^t, b_{t+1}^t} && u(c_t^t, c_{t+1}^t) \\
&s.t. && (1) \quad c_t^t + b_{t+1}^t \leq w_t^t, \\
& && (2) \quad c_t^{t+1} \leq w_{t+1}^t + \hat{r}_{t+1} b_{t+1}^t, \\
& && (3) \quad c_t^t, c_{t+1}^t \geq 0.
\end{aligned}$$

- *Market Clearing: The allocations also satisfy*

$$\begin{aligned}
&\text{Goods Market:} && \hat{c}_t + \hat{c}_{t-1}^t = w_t^t + w_{t-1}^t \\
&\text{Bonds Market:} && \hat{b}_2^1 = m, \quad \hat{b}_{t+1}^1 = \Pi_{\tau=2}^t (1 + \hat{r}_\tau) m
\end{aligned}$$

2.5 Other Notes

2.5.1 Algebra and Useful Places to Look

It's tough to talk vaguely about algebra, so the goal is to cover at least one of each type of problem you may face in the prelim. I've found that in most cases, for those models that we are able to solve analytically, we can usually do so by backing out the euler condition from the first order conditions and then plugging in this expression into the market clearing conditions. This usually pins down consumption and from this we can look back to the euler condition to determine prices.

The final note of precaution is to be careful about the inada conditions. The inada conditions are a set of conditions what we look at for the utility function of households (but more generally for any sort of maximization problem) that allow us to disregard any lagrangian maximization edge cases. I will cover at least one problem where the inada conditions don't hold and how we work through it.

Definition 9. *[Inada Conditions] Consider any utility function $U(C_t)$. If $\lim_{C_t \to \infty} \frac{\partial U}{\partial C_t} = 0$ and $\lim_{C_t \to 0} \frac{\partial U}{\partial C_t} = \infty$ then the equilibrium value $C_t^* \in (0, +\infty)$.*

2.5.2 Transfers and the Negishi Method

I will come back to this after prelims given that I don't think he typically asks questions about this.

2.5.3 Pareto Optimality Analysis

Pareto optimality is a descriptor used for a specific feasible allocation that is no worse than any other feasible allocation with respect to overall welfare. We can define welfare in any manner of ways but generally in this class we take it to be some weighted sum of the lifetime utility of all households. There are two types of questions we could be faced with

- Solve for optimal allocations as a social planner based on some pareto weights.
- Identify if a particular competitive equilibrium is optimal or not (usually with respect to OLG models given that these are not complete markets). Prove or disprove.

2.5.4 The Time Varying Constraint

We will see why more clearly in the Chari section and conditions that ensure that dynamic programming works, but we also require that asset accumulation, be it capital or bonds (or any other type of asset), doesn't blow up over time relative to prices. The most generalized version of this is the expression

$$\lim_{T \to \infty} \beta^T \lambda_T a_{T+1} = 0.$$

Here λ_T is the lagrange multiplier on the time T budget constraint. The condition itself says that the present discounted value of wealth must go to zero as $T \to \infty$. There are two ways it can fail which have their own implications:

- $a_{T+1} \to +\infty$: the consumer is dying with leftover wealth — they over-saved, which can't be optimal since they could have consumed more.
- $a_{T+1} \to -\infty$: the consumer is running a Ponzi scheme — borrowing without bound, which lenders won't allow (this is ruled out by the no-Ponzi condition separately).

2.6 Exercises

Note for myself: Go back to each section and make clear which questions they should refer to corresponding to each section!

2.6.1 Prelim: Spring 2025 (Arrow-Debreu Markets, Sequential Markets, Pareto Efficiency, Dynamic Programming)

Consider an economy with a representative consumer with the utility function

$$\sum_{t=0}^{\infty} \beta^t \log c_t$$

where $0 < \beta < 1$. This consumer has an endowment of $\bar{\ell}_t = 1$ units of labor in each period and $\bar{k}_0$ units of capital in period 0. Feasible allocation/production plans satisfy

$$c_t + k_{t+1} \leq \theta k_t^{\alpha} \ell_t^{1-\alpha}.$$

(a) Describe an Arrow-Debreu market structure for this economy, explaining when markets are open, who trades with whom, and so on. Define a sequential markets equilibrium.

SOLUTION: An **Arrow Debreu market structure** for this economy is one where futures markets to trade the consumption good, capital, and labor opens in period 0. The representative consumer buys consumption good futures contracts from the representative firm and sell futures contracts for labor and capital to the firm.

An **Arrow Debreu equilibrium** for this economy is a collection of household allocations $\{\hat{c}_t, \hat{k}_t, \hat{\ell}_t\}_{t=0}^{\infty}$, firm allocations $\{\hat{k}_t^f . \hat{\ell}_t^f\}_{t=0}^{\infty}$, and prices $\{\hat{p}_t, \hat{r}_t, \hat{w}_t\}_{t=0}^{\infty}$ such that:

- Household Maximization: Given prices and initial household capital holdings $\bar{k}_0$, the household allocations satisfy the following maximization problem

$$\begin{aligned} \max_{c_t, k_t, \ell_t} \quad & \sum_{t=0}^{\infty} \beta^t \log c_t \\ s.t. \quad (1) \quad & \sum_{t=0}^{\infty} \hat{p}_t c_t + \hat{p}_t k_{t+1} - \hat{p}_t (1-\delta) k_t \leq \sum_{t=0}^{\infty} \hat{r}_t k_t + \hat{w}_t \ell_t, \\ (2) \quad & c_t, k_t \geq 0, \quad 0 \leq \ell_t \leq 1, \\ (3) \quad & k_0 = \bar{k}_0. \end{aligned}$$

- Firm Problem: Given prices, the firm allocations solve the firm's cost minimization problem satisfy the following two conditions

$$\frac{\hat{r}_t}{\hat{p}_t} = F_k(\hat{k}_t^f, \hat{\ell}_t^f),$$

$$\frac{\hat{w}_t}{\hat{p}_t} = F_\ell(\hat{k}_t^f, \hat{\ell}_t^f).$$

- Market Clearing: The allocations also satisfy the following market clearing conditions

$$\begin{aligned} [Goods] &: \hat{c}_t + \hat{k}_{t+1} = F(\hat{k}_t^f, \hat{\ell}_t^f) + (1-\delta)\hat{k}_t, \\ [Labor] &: \hat{\ell}_t = \hat{\ell}_t^f, \\ [Capital] &: \hat{k}_t = \hat{k}_t^f. \end{aligned}$$

■

(b) Describe a sequential markets structure for this economy, explaining when markets are open, who trades with whom, and so on. Define a sequential markets equilibrium.

Solution: A **sequential markets structure** for this economy is one where markets for goods, labor, one-period-ahead bonds, and capital opens every period. Here the representative consumer trivially trades bonds with themselves, buys the consumption good from the firm and sells their labor and capital to the firm.

A **sequential markets equilibrium** for such an economy as above is a collection of household allocations $\{(\hat{c}_t, \hat{k}_{t+1}, \hat{b}_{t+1}, \hat{\ell}_{t+1})\}_{t=0}^\infty$, firm allocations $\{\hat{k}_t^f, \hat{\ell}_t^f\}_{t=0}^\infty$, and prices $\{\hat{r}_t^k, \hat{r}_t^b, \hat{w}_t\}$ such that

- Household Maximization: Given prices and initial level of capital $\bar{k}_0$ and bond holdings $\bar{b}_0$, the household allocations solve the household's maximization problem

$$\begin{aligned} \max_{\{c_t, \ell_t, k_{t+1}, b_{t+1}\}_{t=0}^\infty} & \sum_{t=0}^\infty \beta^t \log c_t \\ s.t. \quad & (1) \quad c_t + k_{t+1} + \hat{b}_{t+1} \le (1-\delta+\hat{r}_t^k)k_t + \hat{w}_t\ell_t + (1+\hat{r}_t^b)b_t, \\ & (2) \quad c_t \ge 0, k_t \ge 0, 0 \le \ell_t \le 1, \forall t \ge 0, \\ & (3) \quad b_t \ge -B, \text{ for some sufficiently large } B, \\ & (4) \quad k_0 = \bar{k}_0, b_0 = \bar{b}_0. \end{aligned}$$

- Firm Maximization: Given prices, firms minimize costs and make zero profits, specifically the firm allocations satisfy

$$\hat{r}_t^k = F_k(\hat{k}_t^f, \hat{\ell}_t^f)$$

$$\hat{w}_t = F_w(\hat{k}_t^f, \hat{\ell}_t^f)$$

- Market Clearing: The household and firm allocations also satisfy

$$\begin{aligned}
\text{Goods Market:} \quad & \hat{c}_t + \hat{k}_{t+1} = F(\hat{k}_t^f, \hat{\ell}_t^f) + (1-\delta)\hat{k}_t, \forall t \geq 0, \\
\text{Capital Market:} \quad & \hat{k}_t = \hat{k}_t^f, \forall t \geq 0, \\
\text{Labor Market:} \quad & \hat{\ell}_t = \hat{\ell}_t^f, \forall t \geq 0, \\
\text{Bonds Market:} \quad & \hat{b}_{t+1} = 0, \forall t \geq 0.
\end{aligned}$$

■

(c) Carefully state a proposition or propositions that establish the essential equivalence of the equilibrium concept in part a with that in part b. Be sure to specify the relationships between the objects in the Arrow-Debreu equilibrium and those in the sequential markets equilibrium.

SOLUTION: Let's do AD $\rightarrow$ Sequential first. Consider the set of household allocations $\{\hat{c}_t, \hat{k}_t, \hat{\ell}_t\}_{t=0}^{\infty}$, firm allocations $\{\hat{k}_t^f, \hat{\ell}_t^f\}_{t=0}^{\infty}$, and prices $\{\hat{p}_t, \hat{r}_t, \hat{w}_t\}_{t=0}^{\infty}$ to be an Arrow Debreu equilibrium. We can then write the corresponding Sequential Markets equilibrium of household allocations $\{(\bar{c}_t, \bar{k}_{t+1}, \bar{b}_{t+1}, \bar{\ell}_{t+1})\}_{t=0}^{\infty}$, firm allocations $\{\bar{k}_t^f, \bar{\ell}_t^f\}_{t=0}^{\infty}$, and prices $\{\bar{r}_t^k, \bar{r}_t^b, \bar{w}_t\}$ to be:

- $\hat{c}_t = \bar{c}_t$, $\hat{k}_t = \bar{k}_t$, $\hat{\ell}_t = \bar{\ell}_t$, $\hat{\ell}_t^f = \bar{\ell}_t^f$, $\hat{k}_t^f = \bar{k}_t^f$ for all $t \geq 0$.
- $\frac{\hat{w}_t}{\hat{p}_t} = \bar{w}_t$, $\frac{\hat{r}_t}{\hat{p}_t} = \bar{r}_t^k$, and $\bar{r}_t^b = \bar{r}_t^k - 1$ for all $t \geq 0$.
- For the economy above, we have that $b_t = 0$ for all $t \geq 0$.

Now, for the opposite direction, consider the Sequential Markets equilibrium of household allocations $\{(\bar{c}_t, \bar{k}_{t+1}, \bar{b}_{t+1}, \bar{\ell}_{t+1})\}_{t=0}^{\infty}$, firm allocations $\{\bar{k}_t^f, \bar{\ell}_t^f\}_{t=0}^{\infty}$, and prices $\{\bar{r}_t^k, \bar{r}_t^b, \bar{w}_t\}$. We can then write the corresponding Arrow Debreu equilibrium of household allocations $\{\hat{c}_t, \hat{k}_t, \hat{\ell}_t\}_{t=0}^{\infty}$, firm allocations $\{\hat{k}_t^f, \hat{\ell}_t^f\}_{t=0}^{\infty}$, and prices $\{\hat{p}_t, \hat{r}_t, \hat{w}_t\}_{t=0}^{\infty}$ to be:

- $\hat{c}_t = \bar{c}_t$, $\hat{k}_t = \bar{k}_t$, $\hat{\ell}_t = \bar{\ell}_t$, $\hat{\ell}_t^f = \bar{\ell}_t^f$, $\hat{k}_t^f = \bar{k}_t^f$ for all $t \geq 0$.
- Set numeraire $\hat{p}_0 = 1$, then we also have $\hat{p}_t = \Pi_{i=1}^t \frac{1}{\bar{r}_t^k}$

- $\hat{r}_t = \hat{p}_t \bar{r}^k_t$ and $\hat{w}_t = \hat{p}_t \bar{w}_t$

■

(d) Define a Pareto efficient allocation/production plan. Prove either that an Arrow-Debreu allocation/production plan is Pareto efficient or that a sequential markets allocation/production plan is Pareto efficient.

Solution: A pareto efficient allocation/production plan is a set of household alloctions $\{\hat{c}_t, \hat{k}_t, \hat{\ell}_t\}_{t=0}^{\infty}$such that it is feasible, meaning that it satisfies the following condition

$$\hat{c}_t + \hat{k}_{t+1} \leq \theta \hat{k}_t^{\alpha} \hat{\ell}_t^{1-\alpha},$$

and also that there exists no other set of feasible household allocations $\{\bar{c}_t, \bar{k}_t, \bar{\ell}_t\}_{t=0}^{\infty}$ where

$$\sum_{t=0}^{\infty} \beta^t \log \hat{c}_t < \sum_{t=0}^{\infty} \beta^t \log \bar{c}_t.$$

Let's now prove that an AD allocation is Pareto Efficient. To do so, let household allocations $\{\hat{c}_t, \hat{k}_t, \hat{\ell}_t\}_{t=0}^{\infty}$, firm allocations $\{\hat{k}_t^f, \hat{\ell}_t^f\}_{t=0}^{\infty}$, and prices $\{\hat{p}_t, \hat{r}_t, \hat{w}_t\}_{t=0}^{\infty}$ be an AD equilibrium and assume for the sake of contradiction that there exists some other allocation $\{\bar{c}_t, \bar{k}_t, \bar{\ell}_t\}_{t=0}^{\infty}$ that is both feasible and achieves a higher lifetime utility than the allocation from the AD equilibrium.

From the inherent definition of optimization problems, if

$$\sum_{t=0}^{\infty} \beta^t \hat{c}_t < \sum_{t=0}^{\infty} \bar{c}_t$$

then we must have that

$$\sum_{t=0}^{\infty} \hat{p}_t (\bar{c}_t + \hat{k}_{t+1} - (1-\delta)\hat{k}_t) > \sum_{t=0}^{\infty} \hat{r}_t \bar{k}_t + \hat{w}_t \bar{\ell}_t,$$

because if otherwise, then the bar allocations would have solved the optimization problem instead of the hat allocations.

Now, assuming that the firm has the same CRS production as in the feasibility constraint and using the fact that we are assuming competitive markets in which the firm makes 0 profit at equilibrium, we know that

$$\pi_t(\bar{k}_t^f, \bar{\ell}_t^f) = \hat{p}_t \theta (\bar{k}_t^f)^{\alpha} (\bar{\ell}_t^f)^{1-\alpha} - \hat{r}_t \bar{k}_t - \hat{w}_t \bar{\ell}_t \leq 0$$

from the optimality of $\hat{k}_t^f, \hat{\ell}_t^f$. This then let's us write (also assuming that $\delta = 1$)

$$\sum_{t=0}^{\infty} \hat{p}_t(\bar{c}_t + \bar{k}_{t+1}) > \sum_{t=0}^{\infty} \hat{r}_t \bar{k}_t + \hat{w}_t \bar{\ell}_t \geq \sum_{t=0}^{\infty} \hat{p}_t \theta (\bar{k}_t^f)^{\alpha} (\bar{\ell}_t^f)^{1-\alpha} = \sum_{t=0}^{\infty} \hat{p}_t \theta \bar{k}_t^{\alpha} \bar{\ell}_t^{1-\alpha}.$$

This, however, implies that $\bar{c}_t + \bar{k}_{t+1} > \theta \bar{k}_t^{\alpha} \bar{\ell}_t^{1-\alpha}$ which violates the feasibility constraint. We have thus reached a contradiction and indeed it must be that the AD equilibrium is pareto efficient. ■

(e) Write down Bellman's equation that defines the value function for the dynamic programming problem that a Pareto efficient allocation/production plan solves. Guess that the value function has the form $V(k) = a_0 + a_1 \log k$ for some yet-to-be-determined constants a_0 and a_1. Solve for the policy function $k' = g(k)$.

Solution: Note first that because labor ℓ does not enter the household's utility function in any manner, we can immediately claim that they will always use up their entire endowment to produce the consumption good; $\ell_t = 1$ for all $t \geq 0$. Second, since we know that the household's utilty satisfies the inada conditions, we can claim that the feasibility constraint will be binding, i.e. that

$$c_t + k_{t+1} = \theta k_t^{\alpha} \implies c_t = \theta k_t^{\alpha} - k_{t+1}.$$

With these facts in hand, we can write down the Bellman equation

$$\boxed{V(k) = \max_{k'} \left\{ \log(\theta k^{\alpha} - k') + \beta V(k') \right\}.}$$

Next, we guess that $V(k) = a_0 + a_1 \log(k) \implies V'(k) = \frac{a_1}{k}$. With this, we can take the first order condition of the RHS of the expression above in order to find the optimal value of k'.

$$\begin{aligned}
\frac{\partial}{\partial k'} \left[\log(\theta k^{\alpha} - k') + \beta V(k') \right] &= 0 \\
-\frac{1}{\theta k^{\alpha} - k'} + \beta \frac{a_1}{k'} &= 0 \\
k' &= \beta a_1 (\theta k^{\alpha} - k') \\
k' &= \frac{a_1 \beta \theta k^{\alpha}}{1 + a_1 \beta}
\end{aligned}$$

Now we just need to determine what the value of a_1 is to be. To do so, we can plug the optimal k' back into the bellman equation.

$$
\begin{aligned}
V(k) &= \max_{k'} \left\{ \log(\theta k^\alpha - k') + \beta V(k') \right\}, \\
a_0 + a_1 \log(k) &= \log\left(\theta k^\alpha - \frac{a_1 \beta \theta k^\alpha}{1 + a_1 \beta}\right) + \beta a_0 + \beta a_1 \log\left(\frac{a_1 \beta \theta k^\alpha}{1 + a_1 \beta}\right), \\
a_0 + a_1 \log(k) &= \log\left(\frac{\theta k^\alpha}{1 + a_1 \beta}\right) + \beta a_0 + \beta a_1 \log\left(\frac{a_1 \beta \theta k^\alpha}{1 + a_1 \beta}\right), \\
a_0 + a_1 \log(k) &= \beta a_0 + \log\left(\frac{\theta}{1 + a_1 \beta}\right) \beta a_1 \log\left(\frac{a_1 \beta \theta}{1 + a_1 \beta}\right) + \alpha \log(k) + \beta a_1 \alpha \log(k), \\
a_1 &= \alpha + \beta \alpha a_1, \\
a_1 &= \frac{\alpha}{1 - \beta \alpha}.
\end{aligned}
$$

Now, we can finally classify the policy function as

$$
\boxed{k' = g(k) = \alpha \beta \theta k^\alpha.}
$$

■

(f) Use the policy function from part e to calculate the sequential markets equilibrium.

Solution: We know from the get go using the bonds market clearing condition that $\bar{b}_t = 0$ for all $t \geq 0$. We also already know from the explanation above regarding labor that $\hat{\ell}_t^f = \hat{\ell}_t = 1$ for all $t \geq 0$.

Next, since we are given $\bar{k}_0$, we can use this in conjunction with the policy function above to first determine the equilibrium capital is

$$
\hat{k}_t^f = \hat{k}_t = (\alpha \beta \theta)^{\sum_{s=0}^{t-1} \alpha^s} \bar{k}_0^{\alpha^t}.
$$

Now we can back out consumption from the goods market clear condition

$$
\hat{c}_t = F(\hat{k}_t, 1) + (1 - \delta)\hat{k}_t - \hat{k}_{t+1}.
$$

Finally, we have to now compute prices, but these come quite directly from the firm's problem

$$
\begin{aligned}
1 + \hat{r}_t^b = \hat{r}_t^k &= F_k(\hat{k}_t, 1), \\
\hat{w}_t &= F_w(\hat{k}_t, 1).
\end{aligned}
$$

■

2.6.2 Prelim: Spring 2023 (Infinitely lived consumers with leisure and dynamic programming)

Consider an economy in which the representative consumer lives forever. There is a good in each period that can be consumed or saved as capital as well as labor. The consumer's utility function is

$$\sum_{t=0}^{\infty} \beta^t u(c_t, x_t).$$

The consumer is endowed with 1 unit of labor in each period, some of which can be consumed as leisure, x_t, and some of which is supplied as labor, ℓ_t. The consumer is also endowed with $\bar{k}_0$ units of capital in period 0. Feasible allocations satisfy

$$c_t + k_{t+1} - (1-\delta)k_t \leq f(k_t, \ell_t).$$

(a) Formulate the problem of maximizing the representative consumer's utility subject to feasibility conditions as a dynamic programming problem. Write down the appropriate Bellman's equation. Make appropriate assumptions on the parameters β and δ and on the functions u and f that guarantee a solution to Bellman's equation. Cite any results that you need, but do not prove anything.

SOLUTION: The corresponding bellman equation to a social planner's problem is

$$\begin{aligned} V(k) = \max_{k',\ell} &\left\{ u(f(k, 1-\ell) + (1-\delta)k - k', 1-\ell) + \beta V(k') \right\} \\ s.t. \quad & (1) \quad k' \geq 0, \\ & (2) \quad 0 \leq \ell \leq 1, \\ & (3) \quad f(k, 1-\ell) + (1-\delta)k - k' \geq 0. \end{aligned}$$

To have a solution to the bellman's equation, we must have that $\beta \in (0,1)$, $\delta \in (0,1]$, u is continuous and strictly increasing, and f is also continuous and strictly increasing in both k and l. In addition, we also need that $f(0,\ell) = 0$ for any ℓ with an existence of some maximum sustainable capital $\bar{k} > 0$. These conditions satisfy Assumptions 4.3 and 4.4 in SLP with let's us use Theorem 4.6 in SLP to claim that there is a fixed point that is a solution to the Bellman's equation. ■

(b) Define a sequential markets equilibrium for this economy. Suppose that you have solved the dynamic programming problem in part (a). Explain carefully how to calculate the sequential markets equilibrium.

SOLUTION: A sequential markets equilibrium for this economy is a collection of allocations $\{\hat{c}_t, \hat{b}_{t+1}, \hat{k}_{t+1}, \hat{\ell}_t\}_{t=0}^{\infty}$, firm allocations $\{\hat{k}_t^f, \hat{\ell}_t^f\}_{t=0}^{\infty}$, and prices $\{\hat{r}_t^b, \hat{r}_t^k, \hat{w}_t\}_{t=0}^{\infty}$ such that

- Household Optimization: Given prices and initial capital, k_0, and bonds, b_0, the household allocations solve

$$\begin{aligned}
\max_{c_t, b_{t+1}, k_{t+1}, \ell_t} \quad & \sum_{t=0}^{\infty} \beta^t u(c_t, 1-\ell_t) \\
s.t. \quad (1) \quad & c_t + b_{t+1} + k_{t+1} \leq (1+\hat{r}_t^b) b_{t+1} + (1+\hat{r}_t^k - \delta) k_{t+1} + \hat{w}_t \ell_t, \\
(2) \quad & c_t \geq 0,\ k_{t+1} \geq 0,\ 0 \leq \ell_t \leq 1.
\end{aligned}$$

- Firm Optimization: Given prices, firms allocations minimize costs and make zero profits. Specifically the following factor price equations are satisfied

$$\begin{aligned}
\hat{r}_t^k &= f_k\left(\hat{k}_t^f, \hat{\ell}_t^f\right), \\
\hat{w}_t &= f_\ell\left(\hat{k}_t^f, \hat{\ell}_t^f\right).
\end{aligned}$$

- Market Clearing: Given the initial level of capital, the firm and household allocations also satisfy the following conditions for all $t \geq 0$.

$$\begin{aligned}
[Goods] \quad & \hat{c}_t + \hat{k}_{t+1} \leq f\left(\hat{k}_t^f, \hat{\ell}_t^f\right) + (1-\delta)\hat{k}_t, \\
[Capital] \quad & \hat{k}_t = \hat{k}_t^f, \\
[Labor] \quad & \hat{\ell}_t = \hat{\ell}_t^f, \\
[Bonds] \quad & \hat{b}_{t+1} = 0.
\end{aligned}$$

First, note that from the bonds market clearing condition, we know that $\hat{b}_t = 0$ for all $t \geq 1$. Next, we also know from the market clearing conditions that the firm allocations is the same as the household allocations. Now, if we suppose that we have already solved the social planner's dynamic programming problem, then we already know the set of equilibrium household allocations. All we need to do is back out prices which is very simple, we know from the firm's optimality conditions that

$$\begin{aligned}
\hat{r}_t^k &= f_k\left(\hat{k}_t^f, \hat{\ell}_t^f\right) = f_k\left(\hat{k}_t, \hat{\ell}_t\right), \\
\hat{w}_t &= f_\ell\left(\hat{k}_t^f, \hat{\ell}_t^f\right) = f_\ell\left(\hat{k}_t, \hat{\ell}_t\right).
\end{aligned}$$

Finally, from no arbitrage condition (and also the FOCs) we know that

$$1 + \hat{r}_t^b = 1 + \hat{r}_t^k - \delta \implies \hat{r}_t^b = \hat{r}_t^k - \delta.$$

■

(c) Define an Arrow–Debreu equilibrium for this economy. Suppose that you have solved the dynamic programming problem in part (a). Explain carefully how to calculate the Arrow–Debreu equilibrium.

Solution: An AD equilibrium for this economy is a collection of household allocations $\{\hat{c}_t, \hat{k}_{t+1}, \hat{\ell}_t\}_{t=0}^{\infty}$, firm allocations $\{\hat{k}_t^f, \hat{\ell}_t^f\}_{t=0}^{\infty}$, and prices $\{\hat{p}_t, \hat{r}_t^k, \hat{w}_t\}_{t=0}^{\infty}$ such that

- Household Optimization: Given prices and initial level of capital k_0, the household allocations solve

$$\begin{aligned} \max_{c_t, k_{t+1}, \ell_t} \quad & \sum_{t=0}^{\infty} \beta^t u(c_t, 1-\ell_t) \\ s.t. \quad (1) \quad & \sum_{t=0}^{\infty} \hat{p}_t (c_t + \hat{k}_{t+1}) \le \sum_{t=0}^{\infty} \hat{p}_t (1-\delta) k_t + \hat{r}_t^k k_t + \hat{w}_t \ell_t, \\ (2) \quad & c_t \ge 0,\, 0 \le \ell_t \le 1,\, k_t \ge 0. \end{aligned}$$

- Firm Optimization: Given prices, firm allocations minimize costs and make zero profits. Specifically the following factor price equations are satisfied

$$\frac{\hat{r}_t^k}{\hat{p}_t} = f_k\left(\hat{k}_t^f, \hat{\ell}_t^f\right),$$
$$\frac{\hat{\ell}_t}{\hat{p}_t} = f_\ell\left(\hat{k}_t^f, \hat{\ell}_t^f\right).$$

- Market Clearing: Given the initial capital allocations, the firm and household allocations also satisfy the following conditions for all $t \ge 0$.

$$\begin{aligned} [Goods] \quad & \hat{c}_t + \hat{k}_{t+1} \le f\left(\hat{k}_t^f, \hat{\ell}_t^f\right) + (1-\delta)\hat{k}_t, \\ [Capital] \quad & \hat{k}_t = \hat{k}_t^f, \\ [Labor] \quad & \hat{\ell}_t = \hat{\ell}_t^f. \end{aligned}$$

Once again, from the market clearing conditions that the firm allocations is the same as the household allocations. Now, if we suppose that we have already solved the social planner's dynamic programming problem, then we already know the set of equilibrium household allocations. All we need to do is back out prices which is very simple, we know from the firm's optimality conditions that

$$\frac{\hat{r}_t^k}{\hat{p}_t} = f_k\left(\hat{k}_t^f, \hat{\ell}_t^f\right),$$
$$\frac{\hat{\ell}_t}{\hat{p}_t} = f_\ell\left(\hat{k}_t^f, \hat{\ell}_t^f\right).$$

We know, however, from the first order condition for capital that

$$-\hat{p}_t + \hat{p}_{t+1})(1-\delta) + \hat{r}^k_{t+1} = 0 \implies \hat{r}^k_{t+1} = \hat{p}_t - \hat{p}_{t+1})(1-\delta).$$

Combining this expression with the factor prices, we get that

$$\frac{\hat{p}_{t-1}}{\hat{p}_t} - (1-\delta) = f_k\left(\hat{k}^f_t, \hat{\ell}^f_t\right).$$

This is a law of motion for prices, we can set $\hat{p}_0 = 1$ as a numeraire and then determine an expression for $\hat{p}_t$ which we can plug back into the factor price to determine $\hat{r}^k_t$ and $\hat{w}_t$. ■

(d) Suppose now that there are equal amounts of two types of consumers in the economy. The two types of consumers have the same discount factor β. They have different utility functions in each period, $u_1(c,x)$ and $u_2(c,x)$, different endowments of labor in each period, $\bar{\ell}^1$ and $\bar{\ell}^2$, and different initial endowments of capital, $\bar{k}^1_0$ and $\bar{k}^2_0$. Define a sequential markets equilibrium.

Solution: Here, since there is heterogeneity, a sequential markets equilibrium is a collection of household allocations $\{(\hat{c}^1_t, \hat{b}^1_{t+1}, \hat{k}^1_{t+1}, \hat{\ell}^1_t), (\hat{c}^2_t, \hat{b}^2_{t+1}, \hat{k}^2_{t+1}, \hat{\ell}^2_t)\}_{t=0}^{\infty}$, firm allocations $\{\hat{k}^f_t, \hat{\ell}^f_t\}_{t=0}^{\infty}$, and prices $\{\hat{r}^b_t, \hat{r}^k_t, \hat{w}_t\}_{t=0}^{\infty}$ such that

- Household Optimization: Given prices and initial capital, (k^1_0, k^2_0), bonds, (b^1_0, b^2_0), and labor endowments $(\bar{\ell}_1, \bar{\ell}_2)$ the household allocations solve both

$$\begin{aligned}
\max_{c^1_t, b^1_{t+1}, k^1_{t+1}, \ell^1_t} \quad & \sum_{t=0}^{\infty} \beta^t u^1(c^1_t, 1-\ell^1_t) \\
s.t. \quad (1) \quad & c^1_t + b^1_{t+1} + k^1_{t+1} \leq (1+\hat{r}^b_t) b^1_{t+1} + (1+\hat{r}^k_t - \delta) k^1_{t+1} + \hat{w}^1_t \ell_t, \\
(2) \quad & c^1_t \geq 0,\ k^1_{t+1} \geq 0,\ 0 \leq \ell^1_t \leq \bar{\ell}_1.
\end{aligned}$$

  and also

$$\begin{aligned}
\max_{c^2_t, b^2_{t+1}, k^2_{t+1}, \ell^2_t} \quad & \sum_{t=0}^{\infty} \beta^t u_2(c^2_t, 1-\ell^2_t) \\
s.t. \quad (1) \quad & c^2_t + b^2_{t+1} + k^2_{t+1} \leq (1+\hat{r}^b_t) b^2_{t+1} + (1+\hat{r}^k_t - \delta) k^2_{t+1} + \hat{w}^2_t \ell_t, \\
(2) \quad & c^2_t \geq 0,\ k^2_{t+1} \geq 0,\ 0 \leq \ell^2_t \leq \bar{\ell}_2.
\end{aligned}$$

- Firm Optimization: Given prices, firms allocations minimize costs and make zero profits. Specifically the following factor price equations are satisfied

$$\hat{r}_t^k = f_k\left(\hat{k}_t^f, \hat{\ell}_t^f\right),$$
$$\hat{w}_t = f_\ell\left(\hat{k}_t^f, \hat{\ell}_t^f\right).$$

- Market Clearing: Given the initial level of capital, the firm and household allocations also satisfy the following conditions for all $t \geq 0$.

$$\begin{aligned}
[Goods] \quad & \hat{c}_t^1 + \hat{c}_t^2 + \hat{k}_{t+1}^1 + \hat{k}_{t+1}^2 \leq f\left(\hat{k}_t^f, \hat{\ell}_t^f\right) + (1-\delta)\hat{k}_t^1 + (1-\delta)\hat{k}_t^2, \\
[Capital] \quad & \hat{k}_t^1 + \hat{k}_t^2 = \hat{k}_t^f, \\
[Labor] \quad & \hat{\ell}_t^1 + \hat{\ell}_t^2 = \hat{\ell}_t^f, \\
[Bonds] \quad & \hat{b}_{t+1}^1 + \hat{b}_{t+1}^2 = 0.
\end{aligned}$$

■

(e) Does the equilibrium allocation for the economy in part (d) solve a dynamic programming problem like that in part (a)? Carefully explain why or why not. If it does solve such a problem, write down the appropriate Bellman's equation.

Solution: It definitely does, we just need to reframe the social planner's problem. Assuming the social planner has pareto weights α_1 and α_2 the new bellman equation is

$$V(k_1, k_2) = \max_{c_1, c_2, k_1', k_2', \ell_1, \ell_2} \max \left\{\alpha_1 u_1(c_1, 1-\ell_1) + \alpha_2 u_2(c_2, 1-\ell_2) + \beta V(k_1', k_2')\right\}$$
$$\begin{aligned}
s.t. \quad (1) \quad & c_1, c_2, k_1', k_2', \ell_1, \ell_2 \geq 0, \\
(2) \quad & \ell_1 \leq \bar{\ell}_1, \ \ell_2 \leq \bar{\ell}_2, \\
(3) \quad & c_1 + c_2 + k_1' + k_2' \leq f(k_1 + k_2, \ell_1 + \ell_2) + (1-\delta)k_1 + (1-\delta)k_2.
\end{aligned}$$

■

2.6.3 Prelim: Fall 2024 (Pareto efficiency in an overlapping generations economy)

Consider an overlapping generations economy in which there is one good in each period and each generation, except the initial one, lives for two periods. The representative consumer in generation t, $t = 1, 2, \ldots$, has the utility function

$$\log c_t^t + c_{t+1}^t$$

and the endowment $(w_t^t, w_{t+1}^t) = (2, 2)$. [Notice that the utility function is not $\log c_t^t + \log c_{t+1}^t$.] The representative consumer in generation 0 lives only in period 1, has utility function $\log c_1^0$, and has the endowment $w_1^0 = 2$. There is no fiat money.

(a) Define an Arrow–Debreu equilibrium for this economy. Calculate the unique Arrow–Debreu equilibrium.

SOLUTION: An AD equilibrium for this economy is a collection of allocations $(\{\hat{c}_t^t, \hat{c}_{t+1}^t\}_{t=1}^{\infty}, \hat{c}_1^0)$ and prices $\{\hat{p}_t\}_{t=1}^{\infty}$ such that

- Initial Old Optimization: Given price $\hat{p}_1$ and endowment w_1^0, the consumption allocations solve

$$\begin{aligned} &\max_{c_1^0} \quad \log(c_1^0) \\ s.t \quad &(1) \quad c_1^0 \geq 0, \\ &(2) \quad \hat{p}_1 c_1^0 \leq \hat{p}_1 w_1^0. \end{aligned}$$

- General Consumer Optimization: Given prices and endowment streams, the consumption allocations solve

$$\begin{aligned} &\max_{c_t^t c_{t+1}^t} \quad \log(c_t^t) + c_{t+1}^t \\ s.t. \quad &(1) \quad \hat{p}_t c_t^t + \hat{p}_{t+1} c_{t+1}^t \leq \hat{p}_t w_t^t + \hat{p}_{t+1} w_{t+1}^t, \\ &(2) \quad c_t^t, c_{t+1}^t \geq 0. \end{aligned}$$

- Market Clearing: Given endowments, the allocations also satisfy the following condition

$$[Goods] \quad \hat{c}_t^t + \hat{c}_t^{t-1} = w_t^t + w_t^{t-1}.$$

Now to calculate this, let's first setup the lagrangian that some agent born at time period t would solve.

$$\mathcal{L}(c_t^t, c_{t+1}^t, \lambda^t, \mu^t) = \log(c_t^t) + c_{t+1}^t + \lambda^t(\hat{p}_t w_t^t + \hat{p}_{t+1} w_{t+1}^t - \hat{p}_t c_t^t - \hat{p}_{t+1} c_{t+1}^t) + \mu_t(c_{t+1}^t).$$

There are two important features to note straight off the bat. The first is that the utility function of the second period of the consumer's life does not satisfy the inada conditions which means what we are not guaranteed that this will be non-zero and so (since this was a constraint we imposed on the optimization problem) we need to include a lagrange multiplier for this constraint.

Second, since there is no fiat money introduced into this economy via the first initial old consumer, we know that they have no incentive to trade and will be hand to mouth which ensures that the next generation is also hand-to-mouth which cascades into every generation being hand-to-mouth. This then ensures that c_{t+1}^t is non-zero and we don't need to worry about that lagrange multiplier. We already know that

$$\boxed{\hat{c}_1^0 = 2,\ \hat{c}_t^t = 2,\ \hat{c}_{t+1}^t = 2,}$$

let's work on pinning down prices from the optimality conditions.

$$\begin{aligned}
[c_t^t]: &\quad \frac{1}{c_t^t} - \lambda^t \hat{p}_t = 0 \implies \lambda^t = \frac{1}{c_t^t \hat{p}_t} \\
[c_{t+1}^t]: &\quad 1 - \lambda^t \hat{p}_{t+1} = 0 \implies \lambda = \frac{1}{\hat{p}_{t+1}}
\end{aligned}$$

Combining these two expressions and plugging in the fact that we already know the value of consumption we get

$$\hat{p}_{t+1} = 2\hat{p}_t.$$

Setting a numeraire of $\hat{p}_1 = 1$ we get that prices are

$$\boxed{\hat{p}_t = 2^{t-1}.}$$

■

(b) Define a sequential markets equilibrium for this economy. Calculate the unique sequential markets equilibrium.

SOLUTION: An sequential markets equilibrium for this economy is a collection of allocations $\left(\{\hat{c}_t^t, \hat{c}_{t+1}^t, \hat{b}_{t+1}^t\}_{t=1}^{\infty}, \hat{c}_1^0\right)$ and prices $\{\hat{r}_t\}_{t=1}^{\infty}$ such that

- Initial Old Optimization: Given price $\hat{r}_1$, endowment w_1^0, and initial bond holdings b_1^0, the consumption allocations solve

$$\begin{aligned}
\max_{c_1^0} &\quad \log(c_1^0) \\
s.t \quad (1) &\quad c_1^0 \geq 0, \\
(2) &\quad c_1^0 \leq (1 + \hat{r}_1) b_1^0 + w_1^0.
\end{aligned}$$

- General Consumer Optimization: Given prices and endowment streams, the consumption allocations solve

$$\begin{aligned}
\max_{c_t^t, c_{t+1}^t, b_{t+1}^t} \quad & \log(c_t^t) + c_{t+1}^t \\
s.t. \quad (1) \quad & c_t^t + b_{t+1}^t \leq w_t^t, \\
(2) \quad & c_{t+1}^t \leq (1 + \hat{r}_{t+1}) b_{t+1}^t + w_{t+1}^t, \\
(3) \quad & c_t^t, c_{t+1}^t \geq 0.
\end{aligned}$$

- Market Clearing: Given endowments, the allocations also satisfy the following condition

$$\begin{aligned}
[Goods] \quad & \hat{c}_t^t + \hat{c}_t^{t-1} = w_t^t + w_t^{t-1}, \\
[Bonds] \quad & \hat{b}_{t+1}^t = \hat{b}_t^{t-1}.
\end{aligned}$$

Now to calculate the equilibrium note the same set of logic that we underwent before. The problem setup implies that $b_1^0 = 0$ which means that everyone is a hand-to-mouth consumer. With this we know from the get go that

$$\boxed{\hat{c}_1^0 = 2,\ \hat{c}_t^t = 2,\ \hat{c}_{t+1}^t,\ \hat{b}_{t+1}^t = 0.}$$

Now let's back out prices, here are the first order conditions

$$\begin{aligned}
[c_t^t]: \quad & \frac{1}{c_t^t} - \lambda_t^t = 0, \\
[c_{t+1}^t]: \quad & 1 - \lambda_{t+1}^t = 0, \\
[b_{t+1}^t]: \quad & -\lambda_t^t + (1 + \hat{r}_{t+1}) \lambda_{t+1} = 0
\end{aligned}$$

Putting these three expressions together we get that

$$\boxed{\hat{r}_t = -\frac{1}{2}.}$$

■

(c) Define a Pareto efficient allocation. Either prove that the equilibrium allocation in part (a) is Pareto efficient or prove that it is not.

Solution: First, note that a *feasible allocation* is any allocation $\left(\hat{c}_1^0, \{\hat{c}_t^t, \hat{c}_{t+1}^t\}_{t=1}^{\infty}\right)$ such that

$$\hat{c}_t^t + \hat{c}_t^{t-1} = w_t^t + w_t^{t-1}.$$

A *Pareto efficient* allocation is then any feasible allocation $\left(\hat{c}_1^0, \{\hat{c}_t^t, \hat{c}_{t+1}^t\}_{t=1}^\infty\right)$ such that there does not exist any other feasible allocation $\left(\bar{c}_1^0, \{\bar{c}_t^t, \bar{c}_{t+1}^t\}_{t=1}^\infty\right)$ where both of the statements below are true

$$\forall t, \quad \log(\hat{c}_t^t) + \hat{c}_t^t \leq \log(\bar{c}_t^t) + \bar{c}_t^t \text{ and } \log(\hat{c}_1^0) \leq \log(\bar{c}_1^0),$$

$$\exists t, \qquad \log(\hat{c}_t^t) + \hat{c}_t^t < \log(\bar{c}_t^t) + \bar{c}_t^t \text{ or } \log(\hat{c}_1^0) < \log(\hat{c}_1^0).$$

The allocation in part (a) is not pareto efficient and to prove so we can find some other allocation that is pareto improving. Specifically, consider the allocation

$$\bar{c}_1^0 = 3,\ \bar{c}_t^t = 1,\ \bar{c}_{t+1}^t = 3.$$

This allocation is feasible because we know that for any $t \geq 1$ we have that

$$\bar{c}_t^t + \bar{c}_t^{t-1} = 1 + 3 = 4 = w_t^t + w_t^{t-1}.$$

Aditionally, consumers prefer this allocation because

$$\begin{aligned} \log(\hat{c}_1^0) &< \log(\bar{c}_1^0) \\ \iff \log(2) &< \log(3) \\ \iff 2 &< 3. \end{aligned}$$

$$\begin{aligned} \log(\hat{c}_t^t) + \hat{c}_{t+1}^t &< \log(\bar{c}_t^t) + \bar{c}_{t+1}^t \\ \iff \log(2) + 2 &< \log(1) + 3 \\ \iff \log(2) &< \log(1) + 1 \\ \iff \log(2) - \log(1) &< 1 \\ \iff \log(2) &< 1 \\ \iff 2 &< e. \end{aligned}$$

Since both of the concluding statements are true, we know that the competitive equilibrium above is not pareto efficient. ■

(d) Suppose now that there is a storage technology available in every period t, $t = 1, 2, \ldots$, that transforms one unit of the good in period t into $\gamma > 0$ units of the good in period $t+1$. Define a sequential markets equilibrium for this economy. Under what conditions on γ will the storage technology be used in equilibrium?

SOLUTION: In this world, a sequential markets equilibrium is is a collection of allocations $\left(\{\hat{c}_t^t, \hat{c}_{t+1}^t, \hat{b}_{t+1}^t, \hat{s}_{t+1}^t\}_{t=1}^{\infty}, \hat{c}_1^0\right)$ and prices $\{\hat{r}_t\}_{t=1}^{\infty}$ such that

- Initial Old Optimization: Given price $\hat{r}_1$, endowment w_1^0, initial bond holdings b_1^0, and initial storage holdings s_1^0 the consumption allocations solve

$$\begin{aligned} &\max_{c_1^0} && \log(c_1^0) \\ s.t &\quad (1) && c_1^0 \geq 0, \\ &\quad (2) && c_1^0 \leq (1+\hat{r}_1)b_1^0 + w_1^0 + \gamma s_1^0. \end{aligned}$$

- General Consumer Optimization: Given prices and endowment streams, the consumption allocations solve

$$\begin{aligned} &\max_{c_t^t, c_{t+1}^t, b_{t+1}^t, s_{t+1}^t} && \log(c_t^t) + c_{t+1}^t \\ s.t. &\quad (1) && c_t^t + b_{t+1}^t + s_{t+1}^t \leq w_t^t, \\ &\quad (2) && c_{t+1}^t \leq (1+\hat{r}_{t+1})b_{t+1}^t + w_{t+1}^t + \gamma s_{t+1}^t, \\ &\quad (3) && c_t^t, c_{t+1}^t, s_{t+1}^t \geq 0. \end{aligned}$$

- Market Clearing: Given endowments, the allocations also satisfy the following condition

$$\begin{aligned} [Goods] &\quad \hat{c}_t^t + \hat{c}_t^{t-1} + \hat{s}_{t+1}^t + \hat{s}_t^{t-1} = w_t^t + w_t^{t-1}, \\ [Bonds] &\quad \hat{b}_{t+1}^t = \hat{b}_t^{t-1}. \end{aligned}$$

The storage technology will be used at some point t if $\gamma \geq 1 + \hat{r}_t$. ■

2.6.4 Prelim: Spring 2021 (Pareto Efficiency in Overlapping Generations Models)

Consider an overlapping generations economy in which there is one good in each period and each generation, except the initial one, lives for two periods. The representative consumer in generation t, $t = 1, 2, \ldots$, has the utility function

$$\log c_t^t + \log c_{t+1}^t$$

and the endowment $(w_t^t, w_{t+1}^t) = (w_1, w_2)$. The representative consumer in generation 0 lives only in period 1, prefers more consumption to less, and has the endowment $w_1^0 = w_2$. There is no fiat money.

a. Define an Arrow–Debreu equilibrium for this economy. Calculate the unique Arrow–Debreu equilibrium. [You do not have to prove that equilibrium is unique.]

Solution: An AD equilibrium for this economy is a collection of allocations $\left(\{\hat{c}_t^t, \hat{c}_{t+1}^t\}_{t=1}^{\infty}, \hat{c}_1^0\right)$ and prices $\{\hat{p}_t\}_{t=1}^{\infty}$ such that

- Initial Old Optimization: Given price $\hat{p}_1$ and endowment w_1^0, the consumption allocations solve

$$\begin{aligned} \max_{c_1^0} \quad & \log(c_1^0) \\ s.t \quad (1) \quad & c_1^0 \geq 0, \\ (2) \quad & \hat{p}_1 c_1^0 \leq \hat{p}_1 w_1^0. \end{aligned}$$

- General Consumer Optimization: Given prices and endowment streams, the consumption allocations solve

$$\begin{aligned} \max_{c_t^t c_{t+1}^t} \quad & \log(c_t^t) + \log(c_{t+1}^t) \\ s.t. \quad (1) \quad & \hat{p}_t c_t^t + \hat{p}_{t+1} c_{t+1}^t \leq \hat{p}_t w_t^t + \hat{p}_{t+1} w_{t+1}^t, \\ (2) \quad & c_t^t, c_{t+1}^t \geq 0. \end{aligned}$$

- Market Clearing: Given endowments, the allocations also satisfy the following condition

$$[Goods] \quad \hat{c}_t^t + \hat{c}_t^{t-1} = w_t^t + w_t^{t-1}.$$

Because there is no fiat money and the initial old has preferences that is strictly increasing in their consumption, we know that everyone will be hand-to-mouth in this economy. In other words, we know that the allocations will be

$$\boxed{\hat{c}_1^0 = w_2,\ \hat{c}_t^t = w_1,\ \hat{c}_{t+1}^t = w_2.}$$

Now let's recover prices using the first order conditions,

$$\begin{aligned} [c_t^t]: \quad & \frac{1}{c_t^t} - \lambda^t \hat{p}_t = 0, \\ [c_{t+1}^t]: \quad & \frac{1}{c_{t+1}^t} - \lambda^t \hat{p}_{t+1} = 0. \end{aligned}$$

Putting together both of these conditions, we get that $c_t^t \hat{p}_t = c_{t+1}^t \hat{p}_{t+1}$. Since we already know the consumption allocations, we can set $\hat{p}_1 = 1$ as the numeraire and write

$$\frac{\hat{p}_{t+1}}{\hat{p}_t} = \frac{w_1}{w_2} \implies \boxed{\hat{p}_t = \left(\frac{w_1}{w_2}\right)^{t-1}.}$$

■

b. Define a sequential markets equilibrium for this economy. State a proposition or propositions that relate the unique sequential markets equilibrium to the unique Arrow–Debreu equilibrium. Calculate the unique sequential markets equilibrium. [You do not have to prove that equilibrium is unique.]

SOLUTION: An sequential markets equilibrium for this economy is a collection of allocations $\left(\{\hat{c}_t^t, \hat{c}_{t+1}^t, \hat{b}_{t+1}^t\}_{t=1}^{\infty}, \hat{c}_1^0\right)$ and prices $\{\hat{r}_t\}_{t=1}^{\infty}$ such that

- Initial Old Optimization: Given price $\hat{r}_1$, endowment w_1^0, and initial bond holdings b_1^0, the consumption allocations solve

$$\begin{aligned} &\max_{c_1^0} \quad \log(c_1^0) \\ s.t \quad &(1) \quad c_1^0 \geq 0, \\ &(2) \quad c_1^0 \leq (1+\hat{r}_1)b_1^0 + w_1^0. \end{aligned}$$

- General Consumer Optimization: Given prices and endowment streams, the consumption allocations solve

$$\begin{aligned} &\max_{c_t^t, c_{t+1}^t, b_{t+1}^t} \quad \log(c_t^t) + \log(c_{t+1}^t) \\ s.t. \quad &(1) \quad c_t^t + b_{t+1}^t \leq w_t^t, \\ &(2) \quad c_{t+1}^t \leq (1+\hat{r}_{t+1})b_{t+1}^t + w_{t+1}^t, \\ &(3) \quad c_t^t, c_{t+1}^t \geq 0. \end{aligned}$$

- Market Clearing: Given endowments, the allocations also satisfy the following condition

$$\begin{aligned} [Goods] \quad & \hat{c}_t^t + \hat{c}_t^{t-1} = w_t^t + w_t^{t-1}, \\ [Bonds] \quad & \hat{b}_{t+1}^t = \hat{b}_t^{t-1}. \end{aligned}$$

The equivalence between an AD equilibrium and a sequential markets equilibrium is as follows: consider we have the AD equilibrium of allocations $\left(\{\hat{c}_t^t, \hat{c}_{t+1}^t\}_{t=1}^{\infty}, \hat{c}_1^0\right)$ and prices $\{\hat{p}_t\}_{t=1}^{\infty}$. The corresponding sequential markets equilibrium of allocations

$\left(\{\bar{c}_t^t, \bar{c}_{t+1}^t, \bar{b}_{t+1}^t\}_{t=1}^\infty, \bar{c}_1^0\right)$ and prices $\{\bar{r}_t\}_{t=1}^\infty$ is

- $\left(\{\bar{c}_t^t, \bar{c}_{t+1}^t\}_{t=1}^\infty, \bar{c}_1^0\right) = \left(\{\hat{c}_t^t, \hat{c}_{t+1}^t\}_{t=1}^\infty, \hat{c}_1^0\right)$.
- $\bar{r}_t = \frac{\hat{p}_t}{\hat{p}+t+1} - 1$.
- $\bar{b}_{t+1} = w_t^t - \bar{c}_t^t$.

∎

c. Define a Pareto efficient allocation for this economy.

SOLUTION: First, note that a *feasible allocation* is any allocation $\left(\hat{c}_1^0, \{\hat{c}_t^t, \hat{c}_{t+1}^t\}_{t=1}^\infty\right)$ such that

$$\hat{c}_t^t + \hat{c}_t^{t-1} = w_t^t + w_t^{t-1}.$$

A *Pareto efficient* allocation is then any feasible allocation $\left(\hat{c}_1^0, \{\hat{c}_t^t, \hat{c}_{t+1}^t\}_{t=1}^\infty\right)$ such that there does not exist any other feasible allocation $\left(\bar{c}_1^0, \{\bar{c}_t^t, \bar{c}_{t+1}^t\}_{t=1}^\infty\right)$ where both of the statements below are true

$$\forall t, \quad \log(\hat{c}_t^t) + \hat{c}_t^t \leq \log(\bar{c}_t^t) + \bar{c}_t^t \text{ and } \log(\hat{c}_1^0) \leq \log(\hat{c}_1^0),$$

$$\exists t, \quad \log(\hat{c}_t^t) + \hat{c}_t^t < \log(\bar{c}_t^t) + \bar{c}_t^t \text{ or } \log(\hat{c}_1^0) < \log(\hat{c}_1^0).$$

∎

d. Prove that the equilibrium allocation is Pareto efficient if $w_2 > w_1$.

SOLUTION: Assume for the sake of contradiction that there exists another feasible allocation $\bar{e} = \left(\bar{c}_1^0, \{\bar{c}_t^t, \bar{c}_{t+1}^t\}_{t=1}^\infty\right)$ that Pareto dominates $\hat{e} = \left(\hat{c}_1^0, \{\hat{c}_t^t, \hat{c}_{t+1}^t\}_{t=1}^\infty\right)$. There are two cases to consider.

(a) <u>Claim 1:</u> If $\log \bar{c}_t^t + \log \bar{c}_{t+1}^t \geq \log \hat{c}_t^t + \log \hat{c}_{t+1}^t$, then

$$\hat{p}_t \bar{c}_t^t + \hat{p}_{t+1} \bar{c}_{t+1}^t \;\geq\; \hat{p}_t \hat{c}_t^t + \hat{p}_{t+1} \hat{c}_{t+1}^t \;=\; \hat{p}_t w_1 + \hat{p}_{t+1} w_2.$$

Similarly, $\log \bar{c}_1^0 \geq \log \hat{c}_1^0$ implies $\hat{p}_1 \bar{c}_1^0 \geq \hat{p}_1 \hat{c}_1^0 = \hat{p}_1 w_2$.

Proof. Suppose $\log \bar{c}_t^t + \log \bar{c}_{t+1}^t \geq \log \hat{c}_t^t + \log \hat{c}_{t+1}^t$ but, for contradiction,

$$\hat{p}_t \bar{c}_t^t + \hat{p}_{t+1} \bar{c}_{t+1}^t < \hat{p}_t w_1 + \hat{p}_{t+1} w_2.$$

Then we can construct a new allocation. Define

$$\bar{\bar{c}}_t^t \;=\; \bar{c}_t^t \;+\; \frac{\hat{p}_t w_1 + \hat{p}_{t+1} w_2 - \hat{p}_t \bar{c}_t^t - \hat{p}_{t+1} \bar{c}_{t+1}^t}{\hat{p}_t} \;>\; \bar{c}_t^t,$$

and set $\bar{\bar{c}}_{t+1}^t = \bar{c}_{t+1}^t$. Then

$$\begin{aligned} \log \bar{\bar{c}}_t^t + \log \bar{\bar{c}}_{t+1}^t &> \log \bar{c}_t^t + \log \bar{c}_{t+1}^t \\ &\geq \log \hat{c}_t^t + \log \hat{c}_{t+1}^t, \end{aligned}$$

with budget constraint satisfied:

$$\begin{aligned} \hat{p}_t \bar{\bar{c}}_t^t + \hat{p}_{t+1} \bar{\bar{c}}_{t+1}^t &= \hat{p}_t \bar{c}_t^t + \hat{p}_t w_1 + \hat{p}_{t+1} w_2 - \hat{p}_t \bar{c}_t^t - \hat{p}_{t+1} \bar{c}_{t+1}^t + \hat{p}_{t+1} \bar{c}_{t+1}^t \\ &= \hat{p}_t w_1 + \hat{p}_{t+1} w_2. \end{aligned}$$

But this contradicts $(\hat{c}_t^t, \hat{c}_{t+1}^t)$ being the solution to generation t's maximization problem. □

(b)

(c) <u>Claim 2:</u> If $\log \bar{c}_t^t + \log \bar{c}_{t+1}^t > \log \hat{c}_t^t + \log \hat{c}_{t+1}^t$, then

$$\hat{p}_t \bar{c}_t^t + \hat{p}_{t+1} \bar{c}_{t+1}^t \quad \hat{p}_t \hat{c}_t^t + \hat{p}_{t+1} \hat{c}_{t+1}^t \;=\; \hat{p}_t w_1 + \hat{p}_{t+1} w_2.$$

Similarly, $\log \bar{c}_1^0 > \log \hat{c}_1^0$ implies $\hat{p}_1 \bar{c}_1^0 > \hat{p}_1 \hat{c}_1^0 = \hat{p}_1 w_2$.

This is a similar proof as above.

Next, since $\bar{e}$ Pareto dominates $\hat{e}$, we know that

- $\log \bar{c}_1^0 \geq \log \hat{c}_1^0$ and
- $\log \bar{c}_t^t + \log \bar{c}_{t+1}^t \geq \log \hat{c}_t^t + \log \hat{c}_{t+1}^t$ for all $t \geq 1$,

with at least one strict inequality. By the claims above, it must be that

$$\hat{p}_t \bar{c}_t^t + \hat{p}_{t+1} \bar{c}_{t+1}^t \;\geq\; \hat{p}_t w_1 + \hat{p}_{t+1} w_2 \quad \forall\, t \geq 1,$$

and $\hat{p}_1 \bar{c}_1^0 \geq \hat{p}_1 w_2$, with at least one strict inequality. Adding these budget constraints up over all $t \geq 1$:

$$\sum_{t=1}^{\infty} \hat{p}_t \left(\bar{c}_t^{t-1} + \bar{c}_t^t \right) \;>\; \sum_{t=1}^{\infty} \hat{p}_t (w_1 + w_2).$$

Since $\hat{p}_t = (w_1/w_2)^{t-1}$ in the Arrow–Debreu equilibrium, we compute

$$\sum_{t=1}^{\infty} \hat{p}_t (w_1 + w_2) = (w_1 + w_2) \sum_{t=1}^{\infty} \left(\frac{w_1}{w_2}\right)^{t-1} = (w_1 + w_2) \cdot \frac{1}{1 - (w_1/w_2)^{-1}},$$

which converges since $w_2 > w_1 \implies w_1/w_2 < 1$ Thus,

$$\sum_{t=1}^{\infty} \hat{p}_t \left(\bar{c}_{t+1}^{t-1} + \bar{c}_t^t\right) \; > \; (w_1 + w_2) \cdot \frac{1}{1 - (w_1/w_2)^{-1}}. \tag{i}$$

On the other hand, multiplying the feasibility condition in each period t,

$$\bar{c}_t^{t-1} + \bar{c}_t^t \; \leq \; w_2 + w_1,$$

by $\hat{p}_t$ and summing:

$$\sum_{t=1}^{\infty} \hat{p}_t \left(\bar{c}_t^{t-1} + \bar{c}_t^t\right) \; \leq \; \sum_{t=1}^{\infty} \hat{p}_t (w_2 + w_1) = (w_1 + w_2) \cdot \frac{1}{1 - (w_1/w_2)^{-1}}. \tag{ii}$$

Inequality (ii) contradicts (i). Therefore no such Pareto-dominating allocation $\bar{e}$ exists, and the equilibrium allocation $\hat{e}$ is Pareto efficient. ■

e. Prove that the equilibrium allocation is not Pareto efficient if $w_2 < w_1$.

Solution: The hand-to-mouth equilibrium above is not Pareto efficient because we can find some other allocation that is pareto dominates it. Specifically, consider the allocation

$$\left(\bar{c}_1^0, \{\bar{c}_t^t, \bar{c}_{t+1}^t\}_{t=1}^{\infty}\right) = \left(\frac{w_1 + w_2}{2}, \left\{\frac{w_1 + w_2}{2}, \frac{w_1 + w_2}{2}\right\}\right).$$

This was we have that, for the initial old household

$$\begin{aligned} &\log(\hat{c}_1^0) < \log(\bar{c}_1^0) \\ \iff &w_2 < \frac{w_1 + w_2}{2} \\ \iff &w_2 < w_1, \end{aligned}$$

and for the general household,

$$\begin{aligned} &\log(\hat{c}_t^t) + \log(\hat{c}_{t+1}^t) < \log(\bar{c}_t^t) + \log(\bar{c}_{t+1}^t) \\ \iff &\log(w_1) + \log(w_2) < \log((w_1 + w_2)/2) + \log((w_1 + w_2)/2) \end{aligned}$$

$$
\begin{aligned}
\iff w_1 w_2 &< \frac{w_1^2 + 2w_1w_2 + w_2^2}{4} \\
\iff 0 &< w_1^2 - 2w_1w_2 + w_2^2 \\
\iff 0 &< (w_1 - w_2)^2 \\
\iff w_1 &\neq w_2.
\end{aligned}
$$

As such, the equilibrium allocation is not pareto efficient. ■

f. Is the equilibrium allocation Pareto efficient if $w_2 = w_1$? Explain your answer, but you do not need to prove anything.

SOLUTION: Yes it is pareto efficient, we already know that the consumption allocations are hand-to-mouth which means that consumption in the first period of life is w_1 and consumption in the second period of life is w_2. Since the household has log utility in both periods and consumption is perfectly smooth, we know that the equilibrium allocation when $w_1 = w_2$ is indeed Pareto efficient. ■

2.6.5 Prelim: Spring 2019 (Overlapping generations with production)

Consider an overlapping generations economy in which the representative consumer in generation t, $t = 1, 2, \ldots$, has preferences over the consumption of the single good in each of the two periods of her life given by the utility function

$$u(c_t^t, c_{t+1}^t) = \log c_t^t + \beta \log c_{t+1}^t.$$

This consumer is endowed with quantities of labor $(\ell_t^t, \ell_{t+1}^t) = (\ell_1, \ell_2)$. In addition, there is a generation 0 whose representative consumer lives only in period 1 and has the utility function

$$u^0(c_1^0) = \log c_1^0,$$

and the endowment of $\ell_1^0 = \ell_2$ units of labor and $k_1^0 = \bar{k}_1$ units of capital. This consumer also has an endowment of fiat money m, which can be positive, negative or zero.

The production function is

$$f(k_t, \ell_t) = \theta k_t^\alpha \ell_t^{1-\alpha},$$

and capital depreciates at the rate δ per period, $0 \leq \delta \leq 1$.

a) Define a sequential market equilibrium for this economy.

SOLUTION: ■

b) Define an Arrow–Debreu equilibrium for this economy.

Solution: ■

c) State and prove two theorems that establish the essential equivalence between a sequential market equilibrium and an Arrow–Debreu equilibrium.

Solution: ■

d) Define a steady state for this economy. Write down an equation or equations that characterize a steady state.

Solution: ■

e) Suppose now that consumers born in period 1 and afterwards live for four, rather than two, periods. Specify a stationary environment analogous to that above by specifying utility functions, endowments, and a production technology. Define a sequential markets equilibrium for this economy.

Solution: ■

f) Define a steady state for the economy in part e. Write down an equation or equations that characterize a steady state.

Solution: ■

2.6.6 Prelim: Fall 2015 (Overlapping Generations)

Consider an overlapping generations economy in which there is one good in each period and each generation, except the initial one, lives for two periods. The representative consumer in generation t, $t = 1, 2, \ldots$, has the utility function

$$\log c_t^t + \log c_{t+1}^t$$

and the endowment $(w_t^t, w_{t+1}^t) = (4, 2)$. The representative consumer in generation 0 lives only in period 1, prefers more consumption to less, and has the endowment $w_1^0 = 2$. There is no fiat money.

a) Define an Arrow–Debreu equilibrium for this economy. Calculate the unique Arrow–Debreu equilibrium.

Solution: ■

b) Define a sequential markets equilibrium for this economy. Calculate the unique sequential markets equilibrium.

SOLUTION: ■

c) Define a Pareto efficient allocation. Prove either that the equilibrium allocation in part a is Pareto efficient or prove that it is not.

SOLUTION: ■

d) Suppose now that there is a continuum of measure 1 of two types of consumers in each generation t, $t = 1, 2, \ldots$. Both types of consumers have the utility function

$$\log c_t^{it} + \log c_{t+1}^{it}, \quad i = 1, 2.$$

Consumers of type 1 have the endowment $(w_t^{1t}, w_{t+1}^{1t}) = (4, 2)$, while consumers of type 2 have the endowment $(w_t^{2t}, w_{t+1}^{2t}) = (1, 2)$. The two representative consumers in generation 0 live only in period 1, prefer more to less, and have the endowment $w_1^{i0} = 2$, $i = 1, 2$. There is no fiat money. Define an Arrow–Debreu equilibrium for this economy.

SOLUTION: ■

d) In the equilibrium allocation is $c_t^{1t} = 4$? Explain carefully why or why not.

SOLUTION: ■

3 ECON 8106 (Chari)

3.1 Dynamic Programming (Deterministic and Stochastic)

Dynamic programming is one of the fundamental tools that we use in macroeconomics. In my experience, getting the basic hang of it is not too hard at all but there is a sophisticated set of mathematical machinery that goes on in the background to ensure that it works. The Stokey, Lucas, Prescott textbook is basically all you need when it comes to building up the rigour behind dynamic programming. I will obviously not go through everything at the same level of depth as SLP, but I'll write down some key notes from the book.

The two main themes, as with most solution concepts, is existence and uniqueness. We need to search for conditions such that the

3.2 Recursive Competitive Equilibria

This is perhaps the concept that I have had the most trouble with in this class. Not because it encapsulates a particularly difficult concept, but more so because it is a lot to remember without the right intuition. The idea is, now that we can represent problems with respect to functional equations and using the tools of dynamic programming, what is a corresponding equilibrium concept. As is the case with equilibrium concepts that we have studied before, the definition changes based on how the economy is structured so I'll discuss three main types of models that Chari covers: the Neo-Classical Growth, heterogenous agents in a complete market setting, heterogenous agents in an incomplete markets setting.

First, let's remember the setup for the stochastic neoclassical growth model. The household faces the following **sequence problem**

$$\begin{aligned}
\max_{c_t(s^t),\ell_t(s^t),k_{t+1}(s^t)} \quad & \sum_{t=0}^{\infty}\sum_{s^t}\beta^t\pi(s^t)u(c_t(s^t),\ell_t(s^t)) \\
s.t. \quad (1) \quad & c_t(s^t)+k_{t+1}(s^t)\le w_t(s^t)\ell_t(s^t)+r_s(s^t)k_t(s^{t-1})+(1-\delta)k_t(s^{t-1}), \quad \forall t,s^t, \\
(2) \quad & c_t(s^t),k_{t+1}(s^t)\ge 0, \quad \forall t,s^t, \\
(3) \quad & 0\le \ell_t(s^t)\le 1 \quad \forall t,s^t.
\end{aligned}$$

There is a firm with some production technology $F(\cdot)$ that exists in a competitive environment. As such, they minimize costs but also make zero profits which yields the two **factor price** expressions

$$\begin{aligned}
r_t(s^t) &= F_K(K_t(s^t),L_t(s^t)), \\
w_t(s^t) &= F_L(K_t(s^t),L_t(s^t)).
\end{aligned}$$

This is the basic setup, let's impose a little more structure with respect to the stochasicity and write it up in a functional form so that we can build some intuition for the final RCE statement. Specifically, consider the states s^t to follow a Markov process and the randomness appears in firm productivity $A(s^t)$. Specifically, we write the bellman equation that represents the

Definition 10 (RCE for Neo-Classical Growth Model)**.**

3.2.1 Stationary Competitive Equilibrium

3.3 Cash-Credit-Good Models

All the models that we have discussed so far deals with buying and selling objects that priced with respect to the consumption good.

3.4 Sticky Prices

3.5 Exercises

There is a very specific set of questions that Chari tends to pull from for the prelim and final/midterm. Here's a quick list

- Cash-Credit Goods Models (with and without Production)
- Recursive Competitive Equilibria
- Search and Unemployment Models
- Sticky Price Models
- Island Economy Models
- Sustainable Commitment

3.5.1 Prelim: Spring 2025 (Inflation and Welfare)

Consider a stochastic cash credit goods economy. in which households have preferences over deterministic sequences of the form

$$\sum_{t=0}^{\infty} \beta^t \log c_{1t} + \alpha \log c_{2t}$$

where c_{1t} and c_{2t} denote consumption of cash and credit goods respectively, and $0 < \beta < 1$ is the discount factor. Households are endowed with y units of a composite good which can be converted into cash and credit goods at a 1-1 rate. The resource constraint for deterministic sequences is given by

$$c_{1t} + c_{2t} = y.$$

The endowment y follows a first order Markov process and is the only source of uncertainty in the economy. The securities market meets at the beginning of the period. The household's securities market constraint (for a deterministic version of the economy) is

$$M_t + B_t = (M_{t-1} - p_{t-1}c_{1t-1}) - p_{t-1}c_{2t-1} + p_{t-1}y + R_{t-1}B_{t-1} + T_{t-1}$$

where M_t denotes cash balances, B_t denotes holdings of one-period debt, p_t denotes the price level, R_t denotes the (gross) interest rate on debt and T_t denotes lump-sum transfers by the government. The cash in advance constraint (for a deterministic version of the economy) is

$$p_t c_{1t} \leq M_t$$

Assume real debt holdings are bounded below by a large negative number.

(a) Define a competitive equilibrium. In particular, be precise about what allocations depend on.

SOLUTION: A competitive equilibrium in this economy (assuming complete markets) is a collection of household allocations $\{\hat{c}_{1t}(s^t), \hat{c}_{2t}(s^t), \hat{M}_t(s^t), \hat{B}_t(s^t)\}_{t,s^t}$, government allocations $\{\hat{M}^g_t(s^t), \hat{B}^g_t(s^t), \hat{T}_t(s^t)\}_{t,s^t}$, and prices $\{\hat{p}_t(s^t), \hat{R}_t(s^t)\}_{t,s^t}$ such that:

- Household Maximization: Given prices, endowment y, initial money holdings $\bar{M}_0(s^0)$, initial bond holdings $\bar{B}_0(s^0)$, government transfers, and probabilities of state the household allocations solve the maximization problem

$$\max_{c_{1t}(s^t), c_{2t}(s^t), M_t(s^t), B_t(s^t)} \sum_{t=0}^{\infty} \sum_{s^t} \beta^t (\log c_{1t}(s^t) + \alpha \log c_{2t}(s^t))$$

$$\begin{aligned} s.t. \quad (1) \quad & M_{t+1}(s^{t+1}) + B_{t+1}(s^{t+1}) = (M_t(s^t) - \hat{p}_t(s^t) c_{1t}(s^t)) \\ & -\hat{p}_t(s^t) c_{2t}(s^t) + \hat{p}_t(s^t) y_t(s^t) \\ & +\hat{R}_t(s^t) B_t(s^t) + \hat{T}_t(s^t), \\ (2) \quad & \hat{p}_t(s^t) c_{1t}(s^t) \leq M_t(s^t), \\ (3) \quad & c_{1t}(s^t), c_{2t}(s^t), M_t(s^t) \geq 0, \\ (4) \quad & B_t(s^t) \geq -B. \end{aligned}$$

- Government Budget Constraint: Given prices, the government allocations satisfy the following budget clearing constraint for all $t \geq 1$ and s^t

$$\hat{M}^g_t(s^t) + B^g_t(s^t) = \hat{M}^g_{t-1}(s^t) + \hat{R}_{t-1}(s^t) B^g_{t-1}(s^t) + \hat{T}_{t-1}(s^t).$$

- Market Clearing: The allocations also satisfy the following market clearing conditions for all $t \geq 0$ and s^t

$$\begin{aligned} [Goods]: \quad & \hat{c}_{1t}(s^t) + \hat{c}_{2t}(s^t) = y_t(s^t), \\ [Bonds]: \quad & \hat{B}^g_t(s^t) = \hat{B}_t(s^t), \\ [Money]: \quad & \hat{M}^g_t(s^t) = \hat{M}_t(s^t). \end{aligned}$$

■

(b) Assume government policy is characterized by a sequence of constant interest rates, $R_t = R > 1$ for all t and for all realizations of the exogenous uncertainty. Characterize the stationary competitive equilibrium.

Solution: Let's first take a look at the household's optimization problem and corresponding first order conditions, given that the utility function satisfies the inada conditions, we can write the lagrangian as

$$\begin{aligned}\mathcal{L}(c_{1t}(s^t), c_{2t}(s^t), M_t(s^t), B_t(s^t), \lambda_t(s^t), \mu_t(s^t)) &= \sum_{t=0}^{\infty} \beta^t (\log c_{1t}(s^t) + \alpha \log c_{2t}(s^t)) \\ &+ \sum_{t=0}^{\infty} \lambda_t(s^t)(\text{long BC}) \\ &+ \sum_{t=0}^{\infty} \mu_t(s^t)(M_t(s^t) - p_t(s^t)c_{1t}(s^t)).\end{aligned}$$

The corresponding FOCs are then:

$$\begin{aligned}[c_{1t}(s^t)] &: \frac{\beta^t}{c_{1t}(s^t)} - \lambda_t(s^t)\hat{p}_t(s^t) - \mu_t(s^t)\hat{p}_t(s^t) = 0, \\ [c_{2t}(s^t)] &: \frac{\alpha\beta^t}{c_{2t}(s^t)} - \lambda_t(s^t)\hat{p}_t(s^t) = 0, \\ [M_{t+1}(s_{t+1}, s^t)] &: -\lambda_t(s^t) + \lambda_{t+1}(s_{t+1}, s^t) + \mu_{t+1}(s_{t+1}, s^t) = 0, \\ [B_{t+1}(s_{t+1}, s^t)] &: -\lambda_t(s^t) + R_{t+1}(s_{t+1}, s^t)\lambda_{t+1}(s_{t+1}, s^t) = 0.\end{aligned}$$

If the cash-in-advance constraint is not binding, then we know that $\mu_{t+1}(s_{t+1}, s^t) = 0$ for all $t \geq 0$ and (s_{t+1}, s^t). From the third FOC, this gives us that $\lambda_t(s^t) = \lambda_{t+1}(s_{t+1}, s^t)$ which further implies (from the fourth FOC) that $R_{t+1}(s_{t+1}, s^t) = R = 1$. This, however, is a contradiction with the government policy that $R > 1$.

Now let's handle the case in which the money in advance constraint is indeed binding, specifically that $M_t(s^t) = p_t(s^t)c_{1t}(s^t)$. Using this, we can rewrite the first FOC as

$$\beta^t - \lambda_t(s^t)M_t(s^t) - \mu_t(s^t)M_t(s^t) = 0 \implies \lambda_t(s^t) + \mu_t(s^t) = \frac{\beta^t}{M_t(s^t)}.$$

Using the third FOC, we then get that

$$\begin{aligned}-\lambda_t(s^t) + \lambda_{t+1}(s_{t+1}, s^t) + \mu_{t+1}(s_{t+1}, s^t) = 0 &\implies \lambda_t(s^t) = \frac{\beta^{t+1}}{M_{t+1}(s_{t+1}, s^t)}, \\ &\implies \beta^t = \lambda_{t-1}(s^{t-1})p_t(s_t, s^{t-1})c_{1t}(s_t, s^{t-1}).\end{aligned}$$

Next from the second FOC, we get

$$\beta^t = \frac{\lambda_t(s^t)p_t(s^t)c_{2t}(s^t)}{\alpha}.$$

Dividing these two expressions we get that

$$1 = \frac{\alpha\lambda_{t-1}(s^{t-1})c_{1t}(s_t, s^{t-1}}{\lambda_t(s_t, s^{t-1})c_{2t}(s_t, s^{t-1})} \implies c_{2t}(s^t) = \alpha R c_{1t}(s^t).$$

To get consumption, we can finally plug this into the market clearing for goods.

$$c_{1t}(s^t)\,(1+\alpha R) = y_t(s^t) \implies \boxed{c_{1t}(s^t) = \frac{1}{1+\alpha R}y_t(s^t),\ c_{2t}(s^t) = \frac{\alpha R}{1+\alpha R}y_t(s^t).}$$

Next, to pin down prices, we know from the last FOC that

$$\begin{aligned}
R &= \frac{\lambda_t(s^t)}{\lambda_{t+1}(s_{t+1}, s^t)}, \\
R &= \frac{\alpha\beta^t}{p_t(s^t)c_{2t}(s^t)}\frac{p_{t+1}(s_{t+1}, s^t)c_{2t+1}(s_{t+1}, s^t)}{\alpha\beta^{t+1}}, \\
\beta R &= \frac{p_{t+1}(s_{t+1}, s^t)y_{t+1}(s_{t+1}, s^t)}{p_t(s^t)y_t(s^t)}, \\
p_t(s_t, s^{t-1}) &= \beta R p_{t-1}(s^{t-1})\frac{y_{t-1}(s^{t-1})}{y_t(s_t, s^{t-1})}.
\end{aligned}$$

Now finally, setting $p_0(s_0) = 1$ as the numeraire, we get that prices are

$$\boxed{p_t(s^t) = p_t(s_t, s^{t-1}) = (\beta R)^t \prod_{s=0}^{\tau-1} \frac{y_\tau(s^\tau)}{y_{\tau+1}(s_{\tau+1}, s^\tau)}.}$$

Money holdings follows very clearly

$$\boxed{M_t^g(s^t) = M_t(s^t) = p_t(s^t)c_{1t}(s^t).}$$

Finally, we can't really pin down $B_t(s^t)$ and $T_t(s^t)$ (we don't have enough equations to do so) other than noting that $T_t(s^t) = B_{t+1}(s_{t+1}, s^t) - RB_t(s^t)$ and $B_t^g(s^t) = B_t(s^t)$.

Keep in mind, everything I have done above assumes complete markets for bonds and money holdings! ■

(c) Compare stationary equilibrium outcomes and welfare for two economies with different constant interest rates, but are otherwise identical.

Solution: Welfare in this economy is simply the sum

$$\sum_{t=0}^{\infty}\sum_{s^t}\beta^t(\log c_{1t}(s^t) + \alpha\log c_{2t}(s^t))$$

$$= \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \left(\log \left(\frac{y_t(s^t)}{1+\alpha R} \right) + \alpha \log c_{2t} \left(\frac{\alpha R y_t(s^t)}{1+\alpha R} \right) \right)$$
$$= \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \left[\alpha \log(\alpha R) - (1+\alpha) \log(1+\alpha R) + (1+\alpha) \log(y_s(s^t)) \right].$$

Since both $\alpha < 1+\alpha$ and $\log(\alpha R) < \log(1+\alpha R)$ we can say that increases in R decreases welfare. For prices, we can see clearly than an increase in R also increases prices and similarly for money holdings. ■

3.5.2 Prelim: Fall 2025 (A Cash-Credit Goods Model with Private Money)

Consider a cash-credit goods model in which households have preferences of the form $\sum_{t=0}^{\infty} \beta^t (\log c_{1t} + \phi \log c_{2t})$, where c_{1t} and c_{2t} denote consumption of cash and credit goods respectively, ϕ is a parameter, and $0 < \beta < 1$ is the discount factor. Households supply labor inelastically. The resource constraint is given by

$$c_{1t} + c_{2t} + k_{t+1} = k_t^{\alpha} + (1-\delta) k_t$$

where k_t denotes the capital stock, the capital share parameter α and the depreciation rate δ are both positive and less than 1.

It is convenient to let q_t denote the price of money (or the inverse of the price level). The representative household's budget constraint is given by

$$q_t M_{t+1} + c_{1t} + c_{2t} + k_{t+1} \leq q_t M_t + R_{kt} k_t + w_t l_t + q_t T_t$$

where M_t denotes cash balances, R_{kt} denotes the return to capital and T_t denotes lump sum transfers of cash by the government. The household can use fraction θ of its capital stock to purchase cash goods. The cash in advance constraint is

$$c_{1t} \leq q_t M_t + \theta k_t.$$

(a) Define a competitive equilibrium.

Solution: A competitive equilibrium consists of:

- a household allocation $z_H = \{(\hat{c}_{1t}, \hat{c}_{2t}, \hat{l}_t, \hat{k}_{t+1}, \hat{M}_{t+1})\}_{t=0}^{\infty}$,
- a firm allocation $z_F = \{(\hat{k}_t^F, \hat{l}_t^F)\}_{t=0}^{\infty}$,
- a government policy $z_G = \{(\hat{M}_t^g, \hat{T}_t^g)\}_{t=0}^{\infty}$,
- prices $P = \{(\hat{w}_t, \hat{R}_{kt}, \hat{q}_t)\}_{t=0}^{\infty}$,

such that:

- Given P, M_0, k_0, the household allocation z_H solves

$$\max \sum_{t=0}^{\infty} \beta^t \left[\log(c_{1t}) + \phi \log(c_{2t})\right]$$

subject to

$$\begin{aligned} c_{1t} + c_{2t} + q_t M_{t+1} + k_{t+1} &\leq q_t M_t + R_{kt} k_t + w_t l_t + q_t T_t, \\ c_{1t} &\leq q_t M_t + \theta k_t, \\ c_{1t}, c_{2t} \geq 0, \quad & 0 \leq l_t \leq 1. \end{aligned}$$

- Given P, the firm allocation z_F satisfies the firm's optimality conditions:

$$R_{kt} = \alpha (k_t^F)^{\alpha-1} (l_t^F)^{1-\alpha} + (1-\delta), \qquad w_t = (1-\alpha)(k_t^F)^{\alpha} (l_t^F)^{-\alpha}.$$

- The government policy z_G satisfies the government's budget constraint:

$$q_t M_{t+1}^g = q_t M_t^g + T_t^g.$$

- All markets clear:

$$c_{1t} + c_{2t} + k_{t+1} = (k_t^F)^{\alpha} (l_t^F)^{1-\alpha} + (1-\delta) k_t^F, \qquad k_t^F = k_t, \quad l_t^F = l_t, \quad M_t^g = M_t.$$

■

(b) Assume that the aggregate stock of money grows at a constant rate, π. Characterize the steady state.

Solution: **Labor supply.** Since labor does not enter preferences, households supply labor inelastically: $l_t = 1$ for all t.

First-order conditions. Let $\lambda_t \geq 0$ be the multiplier on the budget constraint and $\mu_t \geq 0$ be the multiplier on the CIA constraint. The household FOCs are:

$$\begin{aligned} [c_{1t}] &: \quad \frac{\beta^t}{c_{1t}} = \mu_t + \lambda_t, \\ [c_{2t}] &: \quad \frac{\phi \beta^t}{c_{2t}} = \lambda_t, \\ [k_{t+1}] &: \quad \lambda_t = R_{k,t+1} \lambda_{t+1} + \theta \mu_{t+1}, \end{aligned}$$

$$[M_{t+1}]: \quad q_t\lambda_t = q_{t+1}\lambda_{t+1} + q_{t+1}\mu_{t+1} \;\Rightarrow\; \frac{q_t}{q_{t+1}}\lambda_t = \lambda_{t+1} + \mu_{t+1}.$$

The first two FOCs immediately give

$$\lambda_t = \frac{\phi\beta^t}{c_{2t}}, \qquad \mu_t = \frac{\beta^t}{c_{1t}} - \frac{\phi\beta^t}{c_{2t}}.$$

Money supply. Since $M_t = (1+\pi)^t M_0$, we have $q_{t+1}/q_t = 1/(1+\pi)$, so the $[M_{t+1}]$ condition becomes

$$(1+\pi)\lambda_t = \lambda_{t+1} + \mu_{t+1}.$$

Steady-state ratio c_1^*/c_2^*. In a steady state, all real quantities are constant. Substituting the expressions for λ_t and μ_t into the $[M_{t+1}]$ condition:

$$(1+\pi)\frac{\phi\beta^t}{c_2^*} = \frac{\phi\beta^{t+1}}{c_2^*} + \beta^{t+1}\left(\frac{1}{c_1^*} - \frac{\phi}{c_2^*}\right) = \frac{\beta^{t+1}}{c_1^*}.$$

Cancelling β^t and solving:

$$\boxed{\frac{c_1^*}{c_2^*} = \frac{\beta}{\phi(1+\pi)}.}$$

Steady-state return to capital R_k^*. Substituting into the $[k_{t+1}]$ condition:

$$\frac{\phi\beta^t}{c_2^*} = R_k^*\frac{\phi\beta^{t+1}}{c_2^*} + \theta\beta^{t+1}\left(\frac{1}{c_1^*} - \frac{\phi}{c_2^*}\right).$$

Dividing through by β^t and substituting $1/c_1^* = \phi(1+\pi)/(\beta c_2^*)$:

$$\frac{\phi}{c_2^*} = R_k^*\frac{\phi\beta}{c_2^*} + \frac{\theta\phi}{c_2^*}(1+\pi-\beta).$$

Dividing by ϕ/c_2^*:

$$1 = R_k^*\beta + \theta(1+\pi-\beta),$$

which gives

$$\boxed{R_k^* = \frac{1-\theta(1+\pi-\beta)}{\beta}.}$$

Steady-state capital k^*. From the firm's optimality condition with $l = 1$:

$$R_k^* = \alpha(k^*)^{\alpha-1} + (1-\delta) \;\Rightarrow\; \boxed{k^* = \left(\frac{R_k^* + \delta - 1}{\alpha}\right)^{\frac{1}{\alpha-1}}.}$$

Price of money. The CIA constraint binds in equilibrium: $c_1^* = q_t M_t + \theta k^*$, so $q_t M_t = c_1^* - \theta k^*$ is constant. Since $M_t = (1+\pi)^t M_0$:

$$\boxed{q_t = \frac{q_0}{(1+\pi)^t}, \quad \text{where } q_0 = \frac{c_1^* - \theta k^*}{M_0}.}$$

Consumption levels. The resource constraint in steady state gives $c_1^* + c_2^* = (k^*)^\alpha - \delta k^*$. Using $c_2^* = c_1^* \phi(1+\pi)/\beta$:

$$c_1^* = \frac{\beta}{\beta + \phi(1+\pi)}\left[(k^*)^\alpha - \delta k^*\right],$$
$$c_2^* = \frac{\phi(1+\pi)}{\beta + \phi(1+\pi)}\left[(k^*)^\alpha - \delta k^*\right].$$

Wage. $w^* = (1-\alpha)(k^*)^\alpha$. ■

(c) Show that if π is above a critical threshold, the economy has a nonmonetary steady state in which the price of money is zero.

Solution: The monetary equilibrium requires $q_0 > 0$, i.e., $c_1^* > \theta k^*$. The economy has a nonmonetary steady state when $q_0 \leq 0$. Tracing through the equivalences:

$$\begin{aligned} q_0 \leq 0 \;&\Leftrightarrow\; c_1^* - \theta k^* \leq 0 \\ &\Leftrightarrow\; \frac{\beta}{\beta + \phi(1+\pi)}\left[(k^*)^\alpha - \delta k^*\right] \leq \theta k^* \\ &\Leftrightarrow\; \frac{\beta}{\beta + \phi(1+\pi)}\left[(k^*)^{\alpha-1} - \delta\right] \leq \theta. \end{aligned}$$

Substituting $(k^*)^{\alpha-1} = (R_k^* + \delta - 1)/\alpha$:

$$\frac{\beta}{\beta + \phi(1+\pi)} \cdot \frac{R_k^* + \delta - 1 - \delta\alpha}{\alpha} \leq \theta.$$

Substituting $R_k^* = [1 - \theta(1+\pi-\beta)]/\beta$ and rearranging:

$$1 - \beta - \delta\beta(\alpha - 1) \leq \theta\big[\alpha(\beta + \phi(1+\pi)) + (1+\pi-\beta)\big].$$

Collecting $(1+\pi)$ terms on the right:

$$1 - \beta - \beta(\alpha-1)\delta - \theta\beta(\alpha-1) \leq \theta(1+\pi)(1+\alpha\phi),$$

so the nonmonetary steady state exists when

$$1 + \pi \geq A \;\equiv\; 1 + \frac{1 - \beta - \beta(\alpha-1)(\delta+\theta)}{\theta(1+\alpha\phi)}.$$

■

(d) Show that below the critical threshold, comparing steady states, per capita output is higher if the growth rate of money is higher.

SOLUTION: Suppose $A > 1 + \pi^* > 1 + \pi$ (both economies are in the monetary equilibrium region). Then:

$$R_k^{*\prime} = \frac{1 - \theta(1+\pi^*-\beta)}{\beta} < \frac{1-\theta(1+\pi-\beta)}{\beta} = R_k^*,$$

since $\pi^* > \pi$. Because $\alpha - 1 < 0$, the function $x \mapsto (x/\alpha)^{1/(\alpha-1)}$ is decreasing in x, so:

$$k^{*\prime} = \left(\frac{R_k^{*\prime} + \delta - 1}{\alpha}\right)^{\frac{1}{\alpha-1}} > \left(\frac{R_k^* + \delta - 1}{\alpha}\right)^{\frac{1}{\alpha-1}} = k^*.$$

Hence per capita output $(k^{*\prime})^\alpha > (k^*)^\alpha$, so output is strictly increasing in π throughout the monetary equilibrium region $1 + \pi < A$. Note that both expressions for $k^{*\prime}$ and k^* are well defined and positive since $A > 1 + \pi^* > 1 + \pi$ ensures $R_k^{*\prime} + \delta - 1 > 0$.

■

(e) Do your results in part (d) imply that the Friedman Rule is not optimal in this economy?

SOLUTION: Friedman rule says that we should ensure certain value of R_{kt} that maximizes a welfare, which is not just about an inflation rate. Thus, (d) does not necessarily imply that the Friedman Rule is not optimal in this economy. ■

3.5.3 Prelim: Spring 2024 (Search and Unemployment)

Consider the behavior of a worker who seeks to maximize the expected present discounted value of utility given by

$$\sum \beta^t u(c_t).$$

The discount factor is $\beta \in (0,1)$. In this economy, jobs are destroyed at rate $1-\theta$, and the worker becomes unemployed. If an individual is unemployed he/she receives $b > 0$ units of consumption (unemployment compensation) if he/she has been unemployed for at most $n-1$ periods. From the nth period on, an unemployed worker receives $b = 0$. Suppose first that the worker cannot borrow or lend. The worker receives wage draws from a fixed distribution $F(w)$.

A. Consider the situation of an unemployed worker who is in his/her nth period of unemployment (that is, he/she is not collecting unemployment benefits). Argue that the optimal strategy is of the reservation wage variety, and display a formula for the reservation wage. Denote this reservation wage by $w^*(0)$.

SOLUTION: We can write out a bellman equation to describe this unemployed worker's optimization problem,

$$V_{unemp}(w,0) = \max\left\{u(0) + \beta\mathbb{E}\left[V_{unemp}(w',0)\right], V_{emp}(w)\right\}.$$

Here $V_{emp}(w)$ represents the value of worker who is employed at wage w, which we can represent as

$$V_{emp}(w) = u(w) + \beta\left[\theta V_{emp}(w) + (1-\theta)\mathbb{E}\left[V_{unemp}(w',n)\right]\right],$$
$$\implies V_{emp}(w) = \frac{1}{1-\beta\theta}\left[u(w) + \beta(1-\theta)\mathbb{E}[V_{unemp}(w',n)]\right].$$

In $V_{unemp}(w,0)$, we know that the left hand term in the expression is a constant and the right hand side is increasing w (and also contains the LHS in the range of the RHS for proper distributions $F(\cdot)$). As such, by intermediate value theorem, we know there must exist some $w^*(0)$ such that

$$\beta\mathbb{E}\left[V_{unemp}(w',0)\right] = V_{emp}(w^*(0)),$$

such that if $w > w^*(0)$ then the worker accepts the wage offer and if $w < w^*(0)$ then the worker declines the offer. To more explicitly define $w^*(0)$, we can write

$$\frac{1}{1-\beta\theta}\left[u(w^*(0)) + \beta(1-\theta)\mathbb{E}[V_{unemp}(w',n)]\right] = u(0) + \beta\mathbb{E}\left[V_{unemp}(w',0)\right],$$
$$\implies u(w^*(0)) + \beta(1-\theta)\mathbb{E}[V_{unemp}(w',n)] = (1-\beta\theta)\left[u(0) + \beta\mathbb{E}\left[V_{unemp}(w',0)\right]\right].$$

This gives us the expression

$$\boxed{w^*(0) = u^{-1}\left((1-\beta\theta)\left[u(0) + \beta\mathbb{E}\left[V_{unemp}(w',0)\right] - \beta(1-\theta)\mathbb{E}[V_{unemp}(w',n)]\right]\right).}$$

■

B. Consider next the situation of an unemployed worker who is in his/her $(n-1)$th period of unemployment (that is, this is the last period that the worker will collect unemployment compensation). Let the reservation wage of such a worker be $w^*(1)$. Show that $w^*(1) > w^*(0)$ (if you cannot prove it, a serious argument will receive partial credit).

Solution: We can once again write out the bellman equation that represents the problem that this worker faces,

$$V_{unemp}(w,1) = \max\left\{u(b) + \beta\mathbb{E}[V_{unemp}(w',0), V_{emp}(w)]\right\},$$

where $V_{emp}(w)$ is as written above. Once again, we can write

$$\begin{aligned} u(b) + \beta\mathbb{E}[V_{unemp}(w',0) &= V_{emp}(w^*(1)), \\ u(b) + \beta\mathbb{E}\left[V_{unemp}(w',0)\right] &= \frac{1}{1-\beta\theta}\left[u(w^*(1)) + \beta(1-\theta)\mathbb{E}[V_{unemp}(w',n)]\right], \\ u(w^*(1)) + \beta(1-\theta)\mathbb{E}[V_{unemp}(w',n)] &= (1-\beta\theta)\left[u(b) + \beta\mathbb{E}\left[V_{unemp}(w',0)\right]\right]. \end{aligned}$$

This finally gives us

$$w^*(1) = u^{-1}\left((1-\beta\theta)\left[u(b) + \beta\mathbb{E}\left[V_{unemp}(w',0)\right]\right] - \beta(1-\theta)\mathbb{E}[V_{unemp}(w',n)]\right).$$

Since $b > 0$, we know that $u(b) > u(0)$; since we know $u(\cdot)$ is strictly increasing, we know that $u^{-1}(\cdot)$ is also strictly increasing and therefore

$$w^*(1) > w^*(0).$$

■

C. What does the result from (b) say about the probability of leaving unemployment as a function of the duration of the unemployment spell?

Solution: It suggests that the probability of **leaving unemployment increases as the duration of the unemployment spell increases**. We will show later that $w^*(T) > w^*(T-1)$ for all $T < n$. We know that the strategy of the worker boils down to accepting any wage below the reservation wage. Since we know the cdf $F(\cdot)$, we can claim that the probability than an unemployment spell ends is the probability $1 - F(w^*)$. As such

$$
\begin{aligned}
& w^*(T) > w^*(T-1), \\
\implies & F(w^*(T)) > F(w^*(T-1)), \\
\implies & 1 - F(w^*(T)) < 1 - F(w^*(T-1)).
\end{aligned}
$$

■

D. Let $w^*(2)$ be the reservation wage of a worker who can collect unemployment compensation for two periods before his/her eligibility expires. Show that $w^*(2) > w^*(1)$.

Solution: We can follow a similar train of logic as we did in previous parts. First let's calculate $w^*(2)$ and then go from there

$$
\begin{aligned}
u(b) + \beta \mathbb{E}[V_{unemp}(w', 1)] &= V_{emp}(w^*(2)), \\
\frac{1}{1-\beta\theta}\left[u(w^*(2)) + \beta(1-\theta)\mathbb{E}[V_{unemp}(w', n)]\right] &= u(b) + \beta \mathbb{E}\left[V_{unemp}(w', 1)\right], \\
(1-\beta\theta)\left[u(b) + \beta \mathbb{E}\left[V_{unemp}(w', 1)\right]\right] &= u(w^*(2)) + \beta(1-\theta)\mathbb{E}[V_{unemp}(w', n)].
\end{aligned}
$$

From there we can then write

$$
w^*(2) = u^{-1}\left((1-\beta\theta)\left[u(b) + \beta \mathbb{E}\left[V_{unemp}(w', 1)\right]\right] - \beta(1-\theta)\mathbb{E}[V_{unemp}(w', n)]\right).
$$

We can clearly see the similarities in the formulation between $w^*(2)$ and $w^*(1)$ and as such, proving $w^*(2) > w^*(1)$ boils down (after doing some algebra) to proving $\mathbb{E}[V_{unemp}(w', 1)] > \mathbb{E}[V_{unemp}(w', 0)]$. Furthermore we know that $V_{unemp}(w, 1) > V_{unemp}(w, 0)$ for any w would imply $\mathbb{E}[V_{unemp}(w', 1)] > \mathbb{E}[V_{unemp}(w', 0)]$. As such, pick any arbitrary w in the domain of possible w. There are then four cases

- Case 1: If $V_{unemp}(w, 1) = V_{emp}(w)$ and $V_{unemp}(w, 0) = V_{emp}(w)$ then $V_{unemp}(w, 1) = V_{unemp}(w, 0)$
- Case 2: If $V_{unemp}(w, 1) = V_{emp}(w)$ and $V_{unemp}(w, 0) = u(0) + \beta \mathbb{E}[V_{unemp}(w', 0)]$ then

$$
V_{unemp}(w, 1) = V_{emp}(w) \geq u(b) + \beta \mathbb{E}[V_{unemp}(w', 0)] > u(0) + \beta \mathbb{E}[V_{unemp}(w', 0)].
$$

- Case 3: If $V_{unemp}(w, 1) = u(b) + \beta \mathbb{E}[V_{unemp}(w', 0)]$ and $V_{unemp}(w, 0) =$

$\beta\mathbb{E}[V_{unemp}(w',0)]$ then very simply

$$V_{unemp}(w,1) = u(b) + \beta\mathbb{E}[V_{unemp}(w',0)] > u(0) + \beta\mathbb{E}[V_{unemp}(w',0)].$$

- Case 4: If $V_{unemp}(w,1) = u(b) + \beta\mathbb{E}[V_{unemp}(w',0)]$ and $V_{unemp}(w,0) = V_{emp}(w)$ then again very simply

$$V_{unemp}(w,1) = u(b) + \beta\mathbb{E}[V_{unemp}(w',0)] > V_{emp}(w) = V_{unemp}(w,0).$$

For proper distributions $F(\cdot)$, we know that the measure of w's that lie in cases 2,3, and 4 are non-zero and as such we can conclude that

$$V_{unemp}(w,1) > V_{unemp}(w,0) \implies w^*(2) > w^*(1).$$

■

E. Use an induction argument to show that $w^*(T) > w^*(T-1)$ for all $T \leq n$.

SOLUTION: Let's do a proof by induction

- Base Case: Let $n = 1$, we have shown above that $w^*(1) > w^*(0)$.
- Induction Hypothesis: Assume that for any $k < n$ we have that $w^*(k) > w^*(k-1)$ we want to show that $w^*(k+1) > w^*(k)$.
- Induction Step: Pick any arbitrary $k < n$. As evidenced in the previous problem, we know that

$$w^*(k+1) = (1-\beta\theta)\left[u(b) + \beta\mathbb{E}\left[V_{unemp}(w',k)\right]\right] - \beta(1-\theta)\mathbb{E}[V_{unemp}(w',n)].$$

As such, proving $w^*(k+1) > w^*(k)$ is equivalent to proving that $\mathbb{E}\left[V_{unemp}(w',k)\right] > \mathbb{E}\left[V_{unemp}(w',k-1)\right]$. Fix any arbitrary w in the domain. Again, there are three cases

  - Case 1: If $V_{unemp}(w,k) = V_{emp}(w)$ and $V_{unemp}(w,k-1) = V_{emp}(w)$ then $V_{unemp}(w,k) = V_{unemp}(w,k-1)$
  - Case 2: If $V_{unemp}(w,k) = V_{emp}(w)$ and $V_{unemp}(w,k-1) = u(b) + \beta\mathbb{E}[V_{unemp}(w',k-2)]$ then

$$V_{unemp}(w,k) = V_{emp}(w) \geq u(b) + \beta\mathbb{E}[V_{unemp}(w',k-2)] > V_{unemp}(w,k-1).$$

  - Case 3: If $V_{unemp}(w,k) = u(b) + \beta\mathbb{E}[V_{unemp}(w',k-1)]$ and $V_{unemp}(w,k-$

$1) = u(b) + \beta\mathbb{E}[V_{unemp}(w', k-2)]$. From the induction hypothesis we know that $w^*(k) > w^*(k-1)$ which also implies

$$\mathbb{E}[V_{unemp}(w', k-1)] > \mathbb{E}[V_{unemp}(w', k-2)].$$

This concludes this case.

– Case 4: If $V_{unemp}(w,k) = u(b) + \beta\mathbb{E}[V_{unemp}(w', k-1)]$ and $V_{unemp}(w, k-1) = V_{emp}(w)$ then again very simply

$$V_{unemp}(w,k) = u(b) + \beta\mathbb{E}[V_{unemp}(w', k-1)] > V_{emp}(w) = V_{unemp}(w, k-1).$$

For proper distributions $F(\cdot)$, we know that the measure of w's that lie in cases 2,3, and 4 are non-zero and as such we can conclude that

$$V_{unemp}(w,k) > V_{unemp}(w,k-1) \implies w^*(k+1) > w^*(k).$$

Thus concludes the proof. ■

F. Suppose now that the worker can save (but not borrow) at an exogenous interest rate R. Set up the dynamic programming problem of the worker. Assume the worker starts with A_0 units of assets and is unemployed. Show (for extra credit) that a worker with larger assets has a higher reservation wage.

SOLUTION: The new dynamic programming problem of an unemployed worker, who has $1 \leq k \leq n$ many periods of unemployment benefits left, a many assets is now, and faces a wage draw w is

$$V_{unemp}(w,a,k) = \max_{0 \leq a' \leq b+a} \left\{ u(b + a - a')) + \beta\mathbb{E}[V_{unemp}(w', Ra', k-1)], V_{emp}(w,a) \right\},$$

with

$$V_{unemp}(w,a,0) = \max_{0 \leq a' \leq a} \left\{ u(a - a') + \beta\mathbb{E}[V_{unemp}(w', Ra', k-1)], V_{emp}(w,a) \right\},$$

and

$$V_{emp}(w,a) = \max_{0 \leq a' \leq a+w} \left\{ u(w + a - a') + \beta \left[\theta V_{emp}(w, Ra') + (1-\theta)\mathbb{E}[V_{unemp}(w', Ra', n)] \right] \right\}.$$

■

3.5.4 Prelim: Fall 2024 (Mobility, Unemployment, and Equilibrium)

Consider an infinite-horizon economy with a continuum of islands. At the beginning of each period, each island receives a shock z which takes on a finite set of values. The shock is identically, independently distributed across islands, and follows a Markov process with transition probabilities $\pi(z', z)$. In each island, competitive firms operate a technology. Let $f(x, z)$ denote the marginal product of labor on an island with workforce x and current shock z. Assume that f is strictly decreasing in x and strictly increasing in z, and that $f(0, z)$ is strictly positive. Also assume that there is some $\bar{x}$ such that $f(\bar{x}, z) = 0$ for all z. Workers are risk neutral and have preferences given by

$$\sum_{t=0}^{\infty} \beta^t c_t.$$

At the beginning of each period, after observing the realization of z on all the islands, workers in an island indexed by, say, x, z, can choose either to continue working in that island, receiving a wage of $f(x, z)$, or move. If they choose to move, they receive a wage of 0 in the current period, and can move to any island of their choice at the beginning of the next period, after observing the shocks z' in all islands.

Your main task is to define a stationary competitive equilibrium. To do so, conjecture that there are three types of islands in this economy.

A. One type of island has a positive flow of immigrants into it in the current period. Also all current workers stay. The wage must be the same in all these islands. Why? Let θ denote this wage.

B. In the second type of island, all current workers stay and no workers arrive. Why might this happen?

C. The third type of island has some or all current workers leave and no immigrants arrive.

With this conjecture, what is the Bellman equation of workers in the three types of islands? Write a common Bellman equation for all workers.

Solution: Let's carefully go through each of the cases

(IA) In this situation, the wage in the island (x, z) is high enough such that it is higher than the value of leaving and seaching. Crirically the wage θ must be equal in all such islands because if there is some island with wage θ' then no one would choose to go to any island with wage θ and these islands would be of type B instead of type A.

(IB) In this situation, the wage in the island (x, z) is high enough such that it is higher than the value of leaving and seaching but not high enough such people in other islands would do the same. The wage in this island is high but not high enough that others would forgo a period of wages in their own island to come over.

(IC) In this situation the wage in island (x, z) is low enough such that the value of leaving the island, forgoing one time period of wage, and searching for another island is higher than staying in the island and accepting the wage. In theory, it is possible for the wage to be lower than the search value but we will note that in equilibrium it must be that the wage is exaclty the value of searching (which is something that comes from the fact that $f(x, z)$ is strictly decreasing as x increases and visa-versa).

Since islands of Type A all have the same wage, we know that the value of searching for such an island must be equal for a household in any state (x, z). Also note that the wage in a given island must equal the marginal product of labor on that same island. Given θ, the corresponding bellman equation that all workers solve is

$$V(x, z) = \max \left\{ f(x, z) + \beta \sum_{z'} \pi(z', z) V(x', z'), \beta V(x_\theta, z_\theta) \right\}$$

where x_θ, z_θ are any state such that $\theta = f(x_\theta, z_\theta)$.

■

Define a stationary competitive equilibrium. The key object here is a distribution. A distribution over what objects?

SOLUTION: ■

Sketch how you might prove that such an equilibrium exists.

SOLUTION: ■

3.5.5 Prelim: Fall 2023 (The New Classical Phillips Curve)

Consider an infinite-horizon economy cash in advance economy. In each period the economy is affected by a shock s which takes on a finite set of values, and is i.i.d. over time. Consumers have preferences over the single final consumption good, c_t, in each period given by

$$E \sum_{t=0}^{\infty} \beta^t \log c_t$$

and supply one unit of labor inelastically. The final consumption good is an aggregate of intermediate goods c_{it} given by

$$c_t = \left[\int_0^1 c_{it}^{\theta} \, di \right]^{1/\theta}. \tag{1}$$

The technology for each intermediate good is given by

$$c_{it} = l_{it}, \tag{2}$$

where l_{it} is the amount of labor allocated to intermediate good i in period t. Competitive retail firms use (1) to produce the final good using as inputs the intermediate goods. Intermediate goods firms are monopolists of their particular good. A fraction α of intermediate goods is produced by *sticky price* firms which must set prices for their good in period t before the current shock s_t is realized. These firms are required to meet all the demand at their posted prices. These prices are sticky only for one period at a time. The remaining $1-\alpha$ of intermediate goods firms set prices after the shock is realized.

Consumers face a cash in advance constraint of the form

$$P_t c_t \leq M_t + T_t \tag{3}$$

where M_t denotes the amount of money brought into period t and T_t denotes lump sum transfers by the government. Consumers have budget constraints of the form

$$P_t c_t + M_{t+1} \leq W_t l_t + M_t + T_t + \Pi_t$$

where P_t is the nominal price of the final consumption good, W_t is the wage rate, and Π_t is profits from intermediate goods prices.

The growth rate of the money supply depends on the shock s and is given by an increasing function $\mu(s)$.

A. Define a *recursive equilibrium* for this economy. Specifically, this is an equilibrium in which the continuation equilibrium for any history depends only on the current shock s.

> Solution: A recursive equilibrium for this economy is a collection of a value function $V(\cdot)$, policy function $C(\cdot)$, ■

B. To do so, set up the Bellman equation for consumers. What are the state variables for each consumer? What are the state variables for the economy as a whole? Set up the problems for the three types of firms. What are the market clearing conditions?

> Solution: Consumers face the following bellman equation
>
> $$\begin{aligned} V(M,s) = \max_{c,l,M'} & \{\log(c) + \beta\mathbb{E}[V(M',s')]\} \\ s.t. \quad & (1) \quad P(M,s)C \leq M + T(M,s), \\ & (2) \quad P(M,s) + M' \leq W(M,s)\cdot l + M + T(M,s) + \Pi(s), \end{aligned}$$

$$(3) \quad c, M' \geq 0,$$

$$(4) \quad 0 \leq l \leq 1.$$

■

C. Consider a version of this economy with no shocks. Characterize the equilibrium.

SOLUTION: ■

D. Now suppose there is a shock only in period 1. Characterize the equilibrium for this economy. (You may assume the equilibrium is as in part 3 from period 1 onwards.)

SOLUTION: ■

E. (Extra Credit.) Characterize the equilibrium with i.i.d. shocks in each period.

SOLUTION: ■

4 ECON 8107 (Amador)

4.1 Hugget Model

4.2 Aiyagari Model

4.3 Exercises

4.3.1 Prelim: Spring 2025 (Incomplete Markets, Labor Supply, and Aggregate Risk)

This question is related to the 2025 8108 final. There is a mass one of households. The household's discount factor is $\beta \in (0, 1)$ and their period utility function is given by

$$u(c, n) = \log(c) - \psi n,$$

where c is consumption, n is labor supply, and $\psi > 0$.

A household supplies zn units of effective labor to the market at a wage w_t and where z is stochastic. It can take one of two values: z_H and z_L, with $z_H > z_L > 0$. The probability of being in state z_H is π_z and the probability of being in state z_L is $1 - \pi_z$. The realization is idiosyncratic, independent across households and time. Households also have access to a one-period risk-free uncontingent asset that pays a gross interest rate R_t. A household faces the following budget constraint:

$$c + a' = R_{t-1}a + w_t zn,$$

where a are assets, and has the following (ad-hoc) borrowing constraint:

$$a' \geq -\phi,$$

where $\phi > 0$. In what follows we allow the household's choice of n to be in $(-\infty, \infty)$.

Firms operate in a competitive environment and have a linear production function to produce the final consumption good:

$$Y = A_t N,$$

where N are the *total effective units of labor* and A_t is aggregate productivity. The aggregate productivity A_t is realized at the beginning of period t, taking value A_H with probability π_A and A_L with probability $1 - \pi_A$, with $A_H > A_L > 0$. This realization is iid.

We will discuss the competitive equilibrium of this incomplete markets economy.

(a) Show that the consumption of a household with state z when productivity is A is:

$$c(z, A) = \frac{zA}{\psi}.$$

SOLUTION: First, we know that since firms operate in a competitve environment, it must be that wages are equal to marginal returns to labor. Specifically, we know that $w = Y_n = A_t$. With this in mind, we can then write the following bellman equation

$$\begin{aligned} V(z, A, a) &= \max_{c,n,a'} \{\log(c) - \psi n + \beta \mathbb{E}[V(z', A', a')]\} \\ s.t. \quad & c + a' = Ra + Azn. \end{aligned}$$

The budget constraint allows us to remove n which let's us rewrite

$$V(z, A, a) = \max_{c,a'} \left\{ \log(c) - \frac{\psi}{Az}(c + a' - Ra) + \beta \mathbb{E}[V(z', A', a')] \right\}$$

Now, let's take a look at the first order condition for c. We get

$$\frac{1}{c} - \frac{\psi}{Az} = 0 \implies c(z, A) = \frac{zA}{\psi}.$$

■

(b) The equilibrium interest rate R_t is stochastic, but only a function of the realized aggregate productivity level A. Find $R(A)$. **Hint:** *Use that households with the lowest z, z_L, are always borrowing constrained.*

Solution: Let's first derive the Euler equation. Starting with the envelope condition with respect to assets and then plugging this into the FOC with respect to a', we can write.

$$
\begin{aligned}
V_a(z', A', a') &= \frac{\psi R}{Az}, \\
\implies \frac{\psi}{Az} &= \beta \mathbb{E}[V_a(z', A', a')], \\
\implies \frac{\psi}{Az} &= \beta \mathbb{E}\left[\frac{\psi R'}{A'z'}\right], \\
\implies \frac{1}{Az} &= \beta \mathbb{E}\left[\frac{R'}{A'z'}\right], \\
\implies \frac{1}{R'} &= \beta Az \mathbb{E}\left[\frac{1}{A'z'}\right], \\
\implies \frac{1}{R'} &= \beta Az \left(\frac{\pi_z \pi_A}{A_H z_H} + \frac{\pi_z(1-\pi_A)}{A_L z_H} + \frac{(1-\pi_z)\pi_A}{A_H z_L} + \frac{(1-\pi_z)(1-\pi_A)}{A_L z_L}\right).
\end{aligned}
$$

We have now boiled down R as an expression of z and A (note also that I have conjectured that R is time invariant which I will later confirm). Then, using the hint, we know that households that have been hit with z_L are borrowing constrained and therefore may not necessarily have a binding Euler condition (given that they may want to borrow more). On the other hand, individuals who have been hit with the high productivity shock are guaranteed to have a binding Euler condition (no matter their previous asset holdings) by the asset market clearing condition. As such, we can then continue to write

$$
\begin{aligned}
\frac{1}{R'} &= \beta Az \left(\frac{\pi_z \pi_A}{A_H z_H} + \frac{\pi_z(1-\pi_A)}{A_L z_H} + \frac{(1-\pi_z)\pi_A}{A_H z_L} + \frac{(1-\pi_z)(1-\pi_A)}{A_L z_L}\right), \\
\implies \frac{1}{R'} &= \beta Az_H \left(\frac{\pi_z \pi_A}{A_H z_H} + \frac{\pi_z(1-\pi_A)}{A_L z_H} + \frac{(1-\pi_z)\pi_A}{A_H z_L} + \frac{(1-\pi_z)(1-\pi_A)}{A_L z_L}\right), \\
\implies R' &= \frac{1}{\beta Az_H} \left(\frac{\pi_z \pi_A}{A_H z_H} + \frac{\pi_z(1-\pi_A)}{A_L z_H} + \frac{(1-\pi_z)\pi_A}{A_H z_L} + \frac{(1-\pi_z)(1-\pi_A)}{A_L z_L}\right)^{-1}.
\end{aligned}
$$

Since this expression is independent of time period (which confirms the conjecture), we can write

$$
\boxed{R(A) = \frac{1}{\beta Az_H} \left(\frac{\pi_z \pi_A}{A_H z_H} + \frac{\pi_z(1-\pi_A)}{A_L z_H} + \frac{(1-\pi_z)\pi_A}{A_H z_L} + \frac{(1-\pi_z)(1-\pi_A)}{A_L z_L}\right)^{-1}.}
$$

■

(c) Let's introduce a government that issues a constant amount of bonds $B > 0$ in every period and uses lump-sum transfers/taxes to balance its budget. What happens to the equilibrium interest rate after this policy?

Solution: The **equilibrium interest rate would stay the same**. To show this, we can look to see what changes and what stays the same.

- Changes - BC and Assets Market Clearing: Now households are faced with two types of savings mehcanisms and also are hit with a transfer/tax. Specifically their budget constraint now becomes (we know that the rate of return for asset holdings and government bonds must be the same by no arbitrage conditions)

$$c + a' + b' = R_{t-1}a + R_{t-1}b + A_t zn + T_t,$$

where the government budget constraint is now instead

$$B = R_{t-1}B + T_t.$$

Next, the asset's market clearing is

$$B = \sum_{z,a} b'(z,a) + a'(z,a).$$

These conditions, however, don't enter the euler condition and therefore will not have an effect on the value of R.

- Same - Euler Condition: We derived this in a manner that was independent of asset holdings. Specifically we know that since both assets a' and b' earn the same return, their handling in the first order conditions and envelope condition are the same and we will still get an expression for R that is solely dependent on A.

■

(d) Under what conditions is such a policy certain to deliver a Pareto improvement to all households? How does your answer here differ from the case with no aggregate risk? Explain your answers.

Solution: We know that $B > 0$ implies that these government bonds are a savings mechanism for households, rather than a borrowing mechanism. Additionally, we know from the government budget constraint that

$$B = R(A)B + T_t \implies T_t = (1 - R(A))B.$$

If $R(A) > 1$ then $T_t < 0$ and if $R(A) < 1$ then $T_t > 0$. This is important since we know that there are some households that are borrowing constraint. A positive transfer would guaranteed lessen the number of borrowing constrained individuals since their in-period income increases whereas a negative transfer would cause more people to become borrowing constrained. More and less people being borrowing constraint is important for welfare because it implies that a given household is not optimally consumed according to their preferences. As such, this government bond policy would certainly deliver a pareto improvement if $\boldsymbol{R(A) < 1}$ for all $A \in \{A_H, A_L\}$.

Since this is an incomplete markets model, risk incentivizes precautionary savings and the interest rate reflects this fact. We know that removing the aggregate risk from this economy would reduce overall risk and decrease the need for precationary savings, thereby increasing $R(A)$ and making it less likely that $R(A) < 1$ holds. Removing aggregate risk makes it **less likely that the government policy is a pareto improvement**. ■

4.3.2 Prelim: Fall 2025 (Incomplete Markets, Labor Supply, and Aggregate Risk)

This question is based on the 2025 8108 final and the May prelim. There is a mass one of households. The household's discount factor is $\beta \in (0, 1)$ and their period utility function is given by

$$u(c, n) = \log(c) - \psi n$$

where c is consumption, n is labor supply, and $\psi > 0$.

A household supplies zn units of effective labor to the market at a wage w_t and where z is stochastic. It can take one of two values: z_H and z_L, with $z_H > z_L > 0$. The probability of being in state z_H is π_z and the probability of being in state z_L is $1 - \pi_z$. The realization is idiosyncratic, independent across households and time. Households also have access to a one-period risk-free uncontingent asset that pays a gross interest rate R_t. A household faces the following budget constraint:

$$c + a' = R_{t-1}a + w_t zn,$$

where a are assets, and has the following (ad-hoc) borrowing constraint $a' \geq -\phi$, where $\phi > 0$. In what follows we allow the household's choice of n to be in $(-\infty, \infty)$.

Firms operate in a competitive environment and have a linear production function to produce the final consumption good:

$$Y = A_t N$$

where N are the *total effective units of labor* and A_t is aggregate productivity. The aggregate productivity A_t is realized at the beginning of period t, taking value A_H with probability π_A and A_L with probability $1 - \pi_A$, with $A_H > A_L > 0$. This realization is iid.

We will discuss the competitive equilibrium of this incomplete markets economy. In such an equilibrium, the consumption of a household with z units of labor when aggregate productivity is A is $c(z, A) = zA/\psi$.

(a) Why is this an incomplete markets economy?

SOLUTION: This is an incomplete markets economy simply because although there are different potential realizations (multiple states) across aggregate and idiosyncratic utility, the household can only trade bonds that are uncontingent of these realizations. ■

(b) Show that the equilibrium interest rate $R_t = R(A_t)$ where the function R is given by:

$$R(A) = \frac{1}{\beta}\left(\frac{1}{\pi_A \frac{A}{A_H}\left(\pi_z \frac{z_H}{z_H} + (1-\pi_z)\frac{z_H}{z_L}\right) + (1-\pi_A)\frac{A}{A_L}\left(\pi_z \frac{z_H}{z_H} + (1-\pi_z)\frac{z_H}{z_L}\right)}\right).$$

Hint: Use that the household with the lowest z, z_L, is borrowing constrained, and thus the Euler equation of the household with the highest z must hold.

SOLUTION: Let's first derive the Euler equation. Starting with the envelope condition with respect to assets and then plugging this into the FOC with respect to a', we can write.

$$\begin{aligned}
V_a(z', A', a') &= \frac{\psi R}{Az}, \\
\implies \frac{\psi}{Az} &= \beta \mathbb{E}[V_a(z', A', a')], \\
\implies \frac{\psi}{Az} &= \beta \mathbb{E}\left[\frac{\psi R'}{A'z'}\right], \\
\implies \frac{1}{Az} &= \beta \mathbb{E}\left[\frac{R'}{A'z'}\right], \\
\implies \frac{1}{R'} &= \beta Az \mathbb{E}\left[\frac{1}{A'z'}\right], \\
\implies \frac{1}{R'} &= \beta Az\left(\frac{\pi_z \pi_A}{A_H z_H} + \frac{\pi_z(1-\pi_A)}{A_L z_H} + \frac{(1-\pi_z)\pi_A}{A_H z_L} + \frac{(1-\pi_z)(1-\pi_A)}{A_L z_L}\right).
\end{aligned}$$

We have now boiled down R as an expression of z and A (note also that I have conjectured that R is time invariant which I will later confirm). Then, using the hint, we know that households that have been hit with z_L are borrowing constrained and therefore may not necessarily have a binding Euler condition (given that they may want to borrow more). On the other hand, individuals who have been hit with the high productivity shock are guaranteed to have a binding Euler condition (no matter their previous asset holdings) by the asset market clearing condition. As such, we can then continue to write

$$\begin{aligned}
\frac{1}{R'} &= \beta A z \left(\frac{\pi_z \pi_A}{A_H z_H} + \frac{\pi_z (1-\pi_A)}{A_L z_H} + \frac{(1-\pi_z)\pi_A}{A_H z_L} + \frac{(1-\pi_z)(1-\pi_A)}{A_L z_L} \right), \\
\implies \frac{1}{R'} &= \beta A z_H \left(\frac{\pi_z \pi_A}{A_H z_H} + \frac{\pi_z (1-\pi_A)}{A_L z_H} + \frac{(1-\pi_z)\pi_A}{A_H z_L} + \frac{(1-\pi_z)(1-\pi_A)}{A_L z_L} \right), \\
\implies R' &= \frac{1}{\beta A z_H} \left(\frac{\pi_z \pi_A}{A_H z_H} + \frac{\pi_z (1-\pi_A)}{A_L z_H} + \frac{(1-\pi_z)\pi_A}{A_H z_L} + \frac{(1-\pi_z)(1-\pi_A)}{A_L z_L} \right)^{-1}, \\
\implies R' &= \frac{1}{\beta A z_H} \left(\frac{\pi_A}{A_H} \left(\frac{\pi_z}{z_H} + \frac{1-\pi_z}{z_L} \right) + \frac{1-\pi_A}{A_L} \left(\frac{\pi_z}{z_H} + \frac{1-\pi_z}{z_L} \right) \right)^{-1},
\end{aligned}$$

Since this expression is independent of time period (which confirms the conjecture), we can distribute the A and z_H to write

$$\boxed{R(A) = \frac{1}{\beta} \left(\frac{1}{\pi_A \frac{A}{A_H} \left(\pi_z \frac{z_H}{z_H} + (1-\pi_z) \frac{z_H}{z_L} \right) + (1-\pi_A) \frac{A}{A_L} \left(\pi_z \frac{z_H}{z_H} + (1-\pi_z) \frac{z_H}{z_L} \right)} \right).}$$

■

(c) Suppose that $R(A) < 1$ for all $A \in \{A_L, A_H\}$. Suppose that a government issues a constant level of debt B financed by lump sum taxes/transfers. This policy generates a *Pareto improvement* over the initial equilibrium. Explain why.

SOLUTION: We know that the bonds are constant over time and also that, conditioned on A, the interest rate is time invariant. The resulting government budget constriant is

$$B = R(A)B + T \implies (1 - R(A))B = T.$$

Since $R(A) < 1$, we know that $1 - R(A)$ is positive. As such, there are then three cases.

- $\underline{B = 0}$: The trivial case, nothing changes.
- $\underline{B > 0}$: If bonds are positive, then transfers will also be positive. This pulls some people away from the borrowing constraint and is a pareto improvement.
- $\underline{B < 0}$: This is not a pareto improvement for the opposite reasoning as above. As such the policy is only a pareto improvement when $B > 0$.

■

4.3.3 Final: Spring 2025 (Aiyagari economy with a storage technology)

Consider a standard Aiyagari economy. There is a continuum of mass one of households with utility $u(c)$ (strictly increasing, strictly concave) and discount factor $\beta \in (0,1)$. Each household receives an idiosyncratic labor productivity shock $z \in S = \{z_1, z_2, \dots, z_N\}$, with all $z_i > 0$, which follows a Markov chain with transition matrix Π that has full support (all entries strictly positive). Shocks are independent across households and the law of large numbers applies.

A representative firm operates a neoclassical production function $F(K, L)$ with constant returns to scale, where K is capital and $L = \bar{z}$ is aggregate effective labor (constant in the stationary equilibrium). Capital depreciates at rate $\delta \in (0,1)$. Competitive factor prices are:

$$R(K) = F_K(K, L) + 1 - \delta, \qquad w(K) = F_L(K, L).$$

Households can save in capital ($a' \geq 0$) but cannot borrow. The household's budget constraint is:

$$c + a' = Ra + wz, \qquad a' \geq 0, \; c \geq 0.$$

Part I: The standard economy.

Denote the stationary equilibrium of this economy by $(R^\circ, w^\circ, K^\circ)$. $\boxed{\textbf{Assume that } R^\circ < 1.}$

(a) Write down the Bellman equation for a household in the stationary equilibrium. What are the state and choice variables?

SOLUTION: The state variables are assets a and the labor productivity shock z. The choice variable is future assets a'. Here is the corresponding bellman equation

$$\begin{aligned} V(a,z) = &\max_{a'} \left\{ c(Ra + wz - a') + \beta \mathbb{E}[V(a', z')] \right\}, \\ s.t. \quad &(1) \quad Ra + wz - a' \geq 0, \\ &(2) \quad a' \geq 0. \end{aligned}$$

■

(b) Is the borrowing constraint $a' \geq 0$ an ad-hoc or a natural borrowing limit? Explain.

SOLUTION: We know that we can calculate the natural borrowing constraint by looking at the consumer's budget constraint and thinking about a scenario in which they have taken on the maximum amount of debt and dedicate all their resources (labor income and future debt) to only repaying debt given that they have experienced the worse possible income shock. Specifically, we can write

$$\begin{aligned} c - \phi &= -R\phi + wz_{low}, \\ -\phi &= -R\phi + wz_{low}, \\ -\phi &= \frac{wz_{low}}{1-R}. \end{aligned}$$

Since we know that every possible income shock is strictly positive and $R^\circ < 1$, we can conclude that the expression above is also strictly positive. As such $a' \geq 0$ is an ad-hoc borrowing constraint. ∎

(c) Briefly explain why $R^\circ < 1$ is possible in the Aiyagari model but not in the representative-agent neoclassical model.

SOLUTION: In the representative-agent neoclassical growth model (assuming the same firm and consumer relationship where firms buy capital in entirety instead of renting it out from consumers as inticated by the factor prices above), consumers face the following euler condition

$$u'(c_t) = \beta \mathbb{E}[u'(c_{t+1})(R_{t+1})].$$

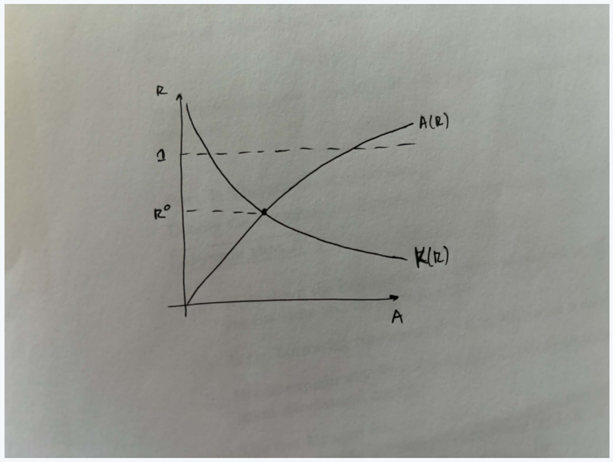
In a stationary equilibrium, however, we can simplify this expression into

$$\begin{aligned} u'(c_t) &= \beta \mathbb{E}[u'(c_{t+1})(R_{t+1})], \\ u'(c) &= \beta \mathbb{E}[u'(c)(R)], \\ 1 &= \beta(R), \\ R &= \frac{1}{\beta} > 1. \end{aligned}$$

In the Aiyagari model on the other hand, consumers are faced with uninsurable risk in the form of ad-hoc borrowing constraints and idiosyncratic shocks without state contingent asset claims. This incentivizes precautionary savings more than in the neoclassical model and as such, consumers are more tolerant of a lower interest rate. ∎

(d) Use the Aiyagari diagram (assets A on the x-axis, gross interest rate R on the y-axis) to illustrate the stationary equilibrium with $R^\circ < 1$. Label the asset supply curve $A(R)$ and the capital demand curve $K(R)$.

Solution: Here is the figure



■

Part II: Introducing storage.

Suppose households now gain access to a **storage technology**: one unit of the consumption good stored today yields one unit tomorrow. That is, storage pays a gross return of 1. Storage is available in perfectly elastic supply (households can store any non-negative amount).

Let k denote a household's holdings of capital and s its holdings of storage. The budget constraint becomes:

$$c + k' + s' = Rk + s + wz,$$

with $k' \geq 0$, $s' \geq 0$, and $c \geq 0$. Neither asset can be held in negative amounts.

(e) Write down the Bellman equation for a household with access to both capital and storage.

Solution: The state variables are capital k, labor productivity shock z, and stored goods s. The choice variables are future capital k' and future storage s'. Here is the corresponding bellman equation

$$V(k,s,z) = \max_{k',s'} \left\{ c(Rk + s + wz - k' - s') + \beta \mathbb{E}[V(k',s',z')] \right\},$$

$$\begin{aligned} s.t. \quad (1) \quad & Rk + s + wz - k' - s' \geq 0, \\ (2) \quad & k' \geq 0, \\ (3) \quad & s' \geq 0. \end{aligned}$$

■

(f) Define a stationary equilibrium for this economy. Be precise about the market clearing conditions.

Solution: A stationary equilibrium is a collection of a value function $V(k,s,z)$, policy functions $S(k,s,z)$ and $K(k,s,z)$, price R, and distribution $\lambda(k,s,z)$ such that

- The value function and policy functions $(V(\cdot), S(\cdot), K(\cdot))$ solve the household's problem given R.
- $\lambda(\cdot)$ is the stationary distribution implied by S, A, and transition matrix Π.
- Given R, The following markets clear

$$[Capital] \quad K = K(R) \equiv \sum_{(k,s,z)\in(\mathcal{K},\mathcal{S},\mathcal{Z})} k * \lambda(k,s,z),$$

$$[Goods] \quad \sum_{(k,s,z)\in(\mathcal{K},\mathcal{S},\mathcal{Z})} S * \lambda(k,s,z) + K * \lambda(k,s,z) + C = F(K,L) + (1-\delta)K.$$

Where $K = K(k,s,z)$, $S = S(k,s,z)$, and $C = Rk + s + wz - K - S$.

■

(g) Show that when $R > 1$, no household holds storage in equilibrium (i.e., $s' = 0$ for all). When $R = 1$, argue that households are indifferent between capital and storage, so only the *total* saving $a' = k' + s'$ matters.

Solution: When $R > 1$, households have the option between storing b units of consumption in storage and getting b units of consumption back in the next period or investing b units of consumption into assets and then getting back $(1+R)b > b$ units of consumption back in the next period. Since they get a strictly larger return from assets, they have no incentive to use storage technology.

When $R = 1$, as seen above, they get the same return from both savings channels and therefore would be ambivalent between the two. From their perspective, they decide some aggregate amount of total savings and randomly distribute this across storage and assets and recieve in return the same level of savings back in the next period.

■

(h) Argue that in any equilibrium, $R \geq 1$. What would happen if $R < 1$?

Solution: If $R < 1$, then consumers would be incentivized to use only the storage facility for intertemporal savings since this offers a strictly higher return. This, however, is incompatible with equilibrium conditions because then aggregate capital used by firms would be 0 and

$$F(K,L) + (1-\delta)K = F(0,L) + (1-\delta) * 0 = 0$$

which means that there would be no stock of goods that consumers would use for either consumption or saving; as such the Euler condition would then imply that R is driven higher towards infinity but we have already claimed that $R < 1$, thus we have reached a contradiction. ■

(i) Let $A(R,w)$ denote the total asset supply (capital plus storage) as a function of the gross return R and wage w. In the standard Aiyagari diagram, w is determined by R through the factor price frontier: given R, capital K is determined by $R = F_K(K,L) + 1 - \delta$, and the wage is $w(R) = F_L(K,L)$. Write $A(R) \equiv A(R, w(R))$.

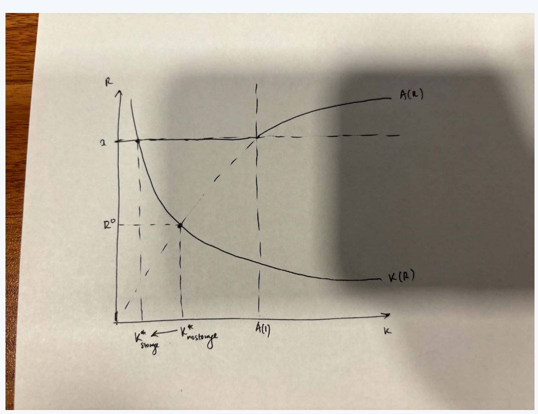
Show how the storage technology modifies the capital supply curve in this diagram. In particular, explain why the modified curve has the following shape (with K on the horizontal axis and R on the vertical):

(i) a *flat* segment at $R = 1$ running from $K = 0$ to $K = A(1)$, and

(ii) the standard upward-sloping $A(R)$ for $R > 1$ (to the right of $K = A(1)$).

Draw the modified diagram, add the capital demand curve $K(R)$, and identify the new equilibrium. Where on the supply curve does the equilibrium occur given $R^\circ < 1$?

Solution: When we intorduce storage, the supply curve has a flate segment at $R = 1$ from $K = 0$ to $K = A(1)$ because, as explained above, storage forces the equilibrium interest rate to be at least 1 so that consumers are ambivalent between choosing capital or storage as their savings method. The supply curve then reverts back to the standard upwards sloping supply curve for $R > 1$ because now consumer's don't use the storage technology anymore and the consumer's problem is the same as in part I. Here is the figure



■

(j) How does the introduction of storage affect the equilibrium capital stock and wages compared to the original economy (where $R^\circ < 1$)? What happens to the excess savings that would have driven R below 1?

SOLUTION: As we can see in the diagram above, $K_{\text{storage}}(R^\circ) < K_{\text{vanilla}}(R^\circ)$ for $R^\circ < 1$. Since we know that

$$w(K) = F_L(K, L)$$

and $F_L(K, L)$ is increasing in K we can claim that the equilibrium wage decreases when storage is introduced:

$$w(K_{\text{storage}}(R^\circ)) = F_L(K_{\text{storage}}(R^\circ), L) < F_L(K_{\text{vanilla}}(R^\circ), L) = w(K_{\text{vanilla}}(R^\circ)).$$

Now, instead of $R^\circ < 1$ being driven by excess asset/capital savings, these excess savings are going into the storage technology. ■

(k) How does the new steady-state capital stock compare to the golden rule level?

Hint: The golden rule capital stock K_{gold} is defined by $F_K(K_{\text{gold}}, L) = \delta$.

SOLUTION: The new steady-state capital stock is exactly the golden rule level. Specifically, we can start from the factor price equation

$$\begin{aligned} R(K) &= F_K(K, L) + 1 - \delta, \\ 1 &= F_K(K, L) + 1 - \delta, \\ F_K(K, L) &= \delta = F_K(K_{\text{gold}}, L), \\ \implies K &= K_{\text{gold}}. \end{aligned}$$

■

Part III: Welfare and policy.

We now consider the welfare and policy implications of the arrival of the storage technology.

(l) Would introducing the storage technology generate a Pareto improvement relative to the original $R^\circ < 1$ equilibrium? Discuss. *Hint:* Think of the wage.

SOLUTION: To be a pareto improvement, we must have that every consumer is weakly better off. Although it is not directly clear if households who have non-zero savings are better or worse off we know for sure that households who have zero assets are worse off in the world with storage. As explained in part j, we know that the wage decreases as storage is introduced, as such, effective income for household with zero savings satisfies

$$\begin{aligned} Ra + w_{\text{vanilla}} z &> Rk + s + w_{\text{storage}} z, \\ \Leftrightarrow w_{\text{vanilla}} z &> w_{\text{storage}} z, \\ \Leftrightarrow w_{\text{vanilla}} &> w_{\text{storage}}. \end{aligned}$$

Since at least one individual is worse off, we know that introducing the storage technology is not a Pareto improvement. ■

(m) Consider a policy where the government, upon the arrival of the storage technology:

(i) taxes both capital returns and storage returns so the household faces a common after-tax gross return $R' = R^\circ$,

(ii) lets capital fall to a new level $K' < K^\circ$,

(iii) uses a wage subsidy to maintain the effective wage at w°.

Argue that the government's net revenues from the three instruments take the form:

$$\begin{aligned} \text{Capital tax revenue} &= \big[R(K_t) - R^\circ\big] K_t, \\ \text{Storage tax revenue} &= (1 - R^\circ)\, S_t, \\ \text{Wage subsidy cost} &= \big[w^\circ - w(K_t)\big] L, \end{aligned}$$

where K_t is the capital stock and S_t is aggregate storage at date t.

Using the firm's first-order conditions, compute each of these three quantities separately for $t = 0$ and for $t > 0$ (the new steady state). For $t > 0$, first argue how much storage S the economy holds.

Note: Because capital is a **predetermined variable**, treat $t = 0$ and $t > 0$ separately. At $t = 0$ the capital stock is still K°; only from $t = 1$ onward does capital equal K'.

Hint: At $t = 0$: what is K_0? How much storage S_0 is there? For $t > 0$: why are total household savings still K°, and what does that imply for $S = K^\circ - K'$?

Solution: At $t = 0$, households face the same budget constraint that they did in previous periods when there was no storage technology and government policy since these states are remnants of household choices under a different model. As such, $K_0 = K^\circ$, $S_0 = 0$ and we can write that at $t = 0$,

$$\text{Capital tax revenue} = \big[R(K_0) - R^\circ\big] K_0 = \big[R^\circ - R^\circ\big] K_0 = 0,$$

$$\text{Storage tax revenue} = (1 - R^\circ)\, S_0 = (1 - R^\circ) * 0 = 0,$$
$$\text{Wage subsidy cost} = \big[w^\circ - w(K_0)\big] L = \big[w^\circ - w^\circ\big] L = 0.$$

Now at $t > 0$, households have access to storage technology but the effective return rate they face between storage and capital holdings is still R° from the government policy. As such, they are indifferent between storage and capital holdings and are faced with a savings problem that is identical to before. Specifically, we have that $S + K' = K^\circ$. We can then write that for $t > 0$,

$$\text{Capital tax revenue} = \big[R(K_t) - R^\circ\big] K_t = \big[R(K') - R^\circ\big] K' = \big[F_K(K', L) + 1 - \delta - R^\circ\big] K',$$
$$\text{Storage tax revenue} = (1 - R^\circ)\, S_t = (1 - R^\circ)(K^\circ - K'),$$
$$\text{Wage subsidy cost} = \big[w^\circ - w(K_t)\big] L = \big[w^\circ - F_L(K', L)\big] L.$$

■

(n) **[Diamond-style policy: fiscal surplus.]**

Using the revenues and costs from (m), write the government's net fiscal surplus at $t > 0$. Show that it simplifies to

$$\text{Surplus} = \big[F(K', L) - \delta K'\big] - \big[F(K^\circ, L) - \delta K^\circ\big],$$

and argue that this is *positive* for any $K' \in [K_{\text{gold}}, K^\circ)$.

Hint: Use the CRS identity $F(K, L) = F_K(K, L)\, K + F_L(K, L)\, L$ to simplify.

Explain why this policy can lead to a Pareto improvement.

Solution: To write out the net fiscal surplus, we just have to sum all the terms up together.

$$\begin{aligned}
\text{Surplus} &= \big[F_K(K', L) + 1 - \delta - R^\circ\big] K' + (1 - R^\circ)(K^\circ - K') - \big[w^\circ - F_L(K', L)\big] L,\\
&= K' F_K(K', L) + K' - \delta K' - R^\circ K' + K^\circ - K' - R^\circ K^\circ + R^\circ K' - w^\circ L + L F_L(K', L),\\
&= (K' F_K(K', L) + L F_L(K', L) - \delta K') + K^\circ - R^\circ K^\circ - w^\circ L,\\
&= [F(K', L) - \delta K'] - [(F_K(K^\circ, L) + 1 - \delta) K^\circ + F(K^\circ, L) L - K^\circ],\\
&= [F(K', L) - \delta K'] - [F(K^\circ, L) - \delta K^\circ].
\end{aligned}$$

Pick any $K' \in [K_{\text{gold}}, K^\circ)$ we can write

$$\begin{aligned}
\text{Surplus} &= [F(K', L) - \delta K'] - [F(K^\circ, L) - \delta K^\circ],\\
&= [F(K', L) - F(K^\circ, L)] + \delta [K^\circ - K'].
\end{aligned}$$

From concavity of F, we know that $F'(K', L) < \delta$. Since F is increasing in K we know that $F(K', L) - F(K^\circ, L)$ is negative. We need to show that

$$F(K^\circ, L) - F(K', L) < \delta[K^\circ - K'].$$

From properties of concave functions though we know that

$$F(K^\circ, L) - F(K', L) < F'(K', L)[K^\circ - K'] \leq \delta[K^\circ - K'].$$

As such, surplus is positive. This policy can lead to a pareto improvement because in this world, consumers experience the same wage and return on savings, but the government has a surplus of resources which they can then transfer back to consumer's which would generate a Pareto improvement because no individual faces a worse situation than before. ■

5 ECON 8108 (Luttmer)

I've never done any continuous time macro before this class and so getting intuition about what is going on is taking me some time. These notes will likely be very detailed as I'm writing them in order to learn myself.

5.1 The canonical continuous time consumer problem

Here, **preferences** are

$$\int_0^\infty e^{-\rho t} U(C_t)$$

for some well behaving $U(C_t)$. Off the get go, the main thing to note that instead of the β temporal discounting that we had before, the continuous time equivalent is $e^{-\rho t}$ where we should note that $\int_0^\infty e^{-\rho t}\, dt = \frac{1}{\rho}$ for $\rho > 0$. Next, the present-value **budget constraint** is (think an Arrow-Debreu world) is

$$\int_0^\infty \pi_t C_t \, dt \leq \pi_0 W_0.$$

We are abstracting right now from firms or income processes. We are looking at lifetime wealth W_0 and prices π_t as some exogenous feature. So with this in mind, let's write down the lagrangian

$$\mathcal{L}(C_t, \lambda) = \int_0^\infty e^{-\rho t} U(C_t)\, dt - \lambda \left(\int_0^\infty \pi_t C_t \, dt - \pi_0 W_0 \right).$$

Which leads to the following two first order conditions

$$[C_t]: \quad e^{-\rho t} DU(C_t) = \lambda \pi_t$$

$$[\lambda]: \quad \pi_0 W_0 = \int_0^\infty \pi_t C_t \, dt.$$

Taking the first FOC, we can write

$$\begin{aligned} e^{-\rho t} DU(C_t) &= \lambda \pi_t, \\ DU(C_t) &= e^{\rho t} \lambda \pi_t, \\ C_t &= [DU]^{-1} \left(\lambda \pi_t e^{\rho t} \right). \end{aligned}$$

Putting this into the second first order conditions, we get

$$\pi_0 W_0 = \int_0^\infty \pi_t [DU]^{-1} \left(\lambda \pi_t e^{\rho t} \right) \, dt$$

with which we can solve for $\lambda > 0$. To do a concrete example, take CES utility

$$U(C) = \frac{C^{1-1/\epsilon}}{1 - 1/\epsilon} \implies DU(C) = C^{-1/\epsilon} \implies [DU]^{-1}(A) = A^{-\epsilon}.$$

Putting this into the expression above, we get that

$$\begin{aligned} \int_0^\infty \pi_t [DU]^{-1} \left(\lambda \pi_t e^{\rho t} \right) \, dt &= \pi_0 W_0 \\ \int_0^\infty \pi_t \left(\lambda \pi_t e^{\rho t} \right)^{-\epsilon} \, dt &= \pi_0 W_0 \\ \int_0^\infty \lambda^{-\epsilon} \pi_t^{1-\epsilon} e^{-\epsilon \rho t} \, dt &= \pi_0 W_0 \\ \lambda^{-\epsilon} \int_0^\infty \pi_t^{1-\epsilon} e^{-\epsilon \rho t} \, dt &= \pi_0 W_0 \\ \lambda^{-\epsilon} &= \frac{\pi_0 W_0}{\int_0^\infty \pi_t^{1-\epsilon} e^{-\epsilon \rho t} \, dt} \\ \lambda &= \left[\frac{\pi_0 W_0}{\int_0^\infty \pi_t^{1-\epsilon} e^{-\epsilon \rho t} \, dt} \right]^{-1/\epsilon} \end{aligned}$$

We can plug in the original expression that we derived for C_t above and we get that

$$\pi_t C_t = \frac{e^{\epsilon \rho t}}{\int_0^\infty \pi_t^{1-\epsilon} e^{-\epsilon \rho t} \, dt} \times \pi_0 W_0 \xrightarrow{\epsilon \to 1} \pi_t C_t = \rho^{-\rho t} \pi_0 W_0.$$

This finally implies that $C_0 = \rho W_0$.

All the discussion thus far has been for the present value budget constraint case. We could also, however, consider the same maximization problem but now with a sequential markets budget constraint

$$dW_t = (r_t W_t - C_t) \, dt \iff DW_t = r_t W_t - C_t,$$

where r_t represents some measure of an interest rate. Specifically, we would have

$$\pi_t = \pi_0 \exp\left(-\int_0^t r_s\, ds\right).$$

5.2 Kass-Coopmans

Now, let's introduce some production and investment dynamics into the model. This time, consider some utility function $U(\cdot)$ that is strictly increasing, concave, and sufficiently smooth. This economy contains one representative agent who has **preferences**, for some $\rho > 0$,

$$\int_0^\infty e^{-\rho t} U(C_t)\, dt.$$

We will have production, but we abstract from having a firm that manages production. Specifically, households will have the chance to choose between dedicating their available resources between consumption and capital investment. There is a production function $F(k, l)$ that is strictly increasing, concave, sufficiently smooth, and homogeneous of degree 1. We will also denote $f(k) = F(k, 1)$. Finally let X_t be the amount of investment at time t and δ be the depreciation rate. These definition all help define the **resource contraints** that the consumer faces

$$\begin{aligned} C_t + X_t &\leq f(K_t), \\ DK_t &\leq -\delta K_t + X_t, \\ C_t, K_t &\geq 0. \end{aligned}$$

Combining all of these constraints gives us the one collapsed constraint

$$DK_t \leq f(K_t) - \delta K_t - C_t.$$

Now let's look at the lagrangian now and derive the first order constraints that can help us solve this model.

$$\mathcal{L}(C_t, K_t, \lambda_t) = \int_0^\infty e^{-\rho t} U(C_t)\, dt + \int_0^\infty e^{-\rho t} \lambda_t (f(K_t) - \delta K_t - C_t - DK_t)\, dt.$$

Since we don't like to work with DK_t, let's do integration by parts.

$$\begin{aligned} \int_0^\infty e^{-\rho t} \lambda_t DK_t\, dt &= e^{-\rho T} \lambda_T K_T \Big|_{T=0}^{\infty} - \int_0^\infty e^{-\rho t} K_t (D\lambda_t - \rho \lambda_t)\, dt, \\ &= \lim_{T \to \infty} e^{-\rho T} \lambda_T K_T - \lambda_0 K_0 - \int_0^\infty e^{-\rho t} K_t (D\lambda_t - \rho \lambda_t)\, dt. \end{aligned}$$

Using this, we can then rewrite the lagrangian as

$$\mathcal{L}(C_t, K_t, \lambda_t) = \int_0^\infty e^{-\rho t} U(C_t)\, dt + \int_0^\infty e^{-\rho t} \lambda_t (f(K_t) - \delta K_t - C_t - DK_t)\, dt,$$

$$\begin{aligned}
&= \int_0^\infty e^{-\rho t}\left[U(C_t) + \lambda_t(f(K_t) - \delta K_t - C_t)\right] dt - \int_0^\infty e^{-\rho t}\lambda_t DK_t\, dt,\\
&= \lambda_0 K_0 - \lim_{T\to\infty} e^{-\rho T}\lambda_T K_T + \int_0^\infty e^{-\rho t}\left[U(C_t) + \lambda_t(f(K_t) - (\delta+\rho)K_t - C_t) + K_t D\lambda_t\right] dt,\\
&= \lambda_0 K_0 + \int_0^\infty e^{-\rho t}\left[U(C_t) + \lambda_t(f(K_t) - (\delta+\rho)K_t - C_t) + K_t D\lambda_t\right] dt.
\end{aligned}$$

Note, we use the transversality condition to claim $\lim_{T\to\infty} e^{-\rho T}\lambda_T K_T = 0$. Now let's take the first order conditions:

$$\begin{aligned}
&[C_t] : e^{-\rho t}[U'(C_t) - \lambda_t] = 0 \implies DU(C_t) = \lambda_t \implies C_t = [DU]^{-1}(\lambda_t),\\
&[K_t] : e^{-\rho t}[\lambda_t f'(K_t) - (\delta+\rho)\lambda_t + D\lambda_t] = 0 \implies D\lambda_t = \lambda_t(\rho + \delta - Df(K_t)).
\end{aligned}$$

Right now, this is a system of two equations has three unkowns in C_t, λ_t and K_t. We can, however, collapse this into a system of two equations of two unknowns using the condensed budget constraint. Specifically we plug in the first FOC condition. This gives us the following system of two differential equations

$$\begin{aligned}
DK_t &= f(K_t) - \delta K_t - [DU]^{-1}(\lambda_t),\\
D\lambda_t &= \lambda_t(\rho + \delta - Df(K_t)).
\end{aligned}$$

5.2.1 Solving the System

If you have done some Partial Differential Equations before then you know that a solution to the system above is not some simple concept of neat expressions for K_t and λ_t but rather a dynamic object in itself that we often try to represent using things such as phase diagrams.

The idea is that given an initial condition of capital K_0 and λ_0 (which we can recover from the envelope condition $\lambda_0 = DV(K_0)$ an interpret as the shadow price of capital) we trace out the curve of K_t and λ_t depending on the direction of DK_t and $D\lambda_t$ which changes as K_t and λ_t evolves. This traced out curve is the solution and we can use phase diagrams to help build out this curve (1).

A **steady state solution** is, however, something that we can think about and back out easily. We know that a steady state implies that the equilibrium allocations don't change over time. As such, by definition, we can very directly claim that a steady state implies that $DK_t = 0$, $D\lambda_t = 0$ and we get the new system

$$\begin{aligned}
0 &= f(K) - \delta K - [DU]^{-1}(\lambda),\\
0 &= \lambda(\rho + \delta - Df(K)).
\end{aligned}$$

Disregarding the edge case of $\lambda = 0$, we can then back out

$$K = [Df]^{-1}(\rho + \delta),\ C = f(K) - \delta K.$$

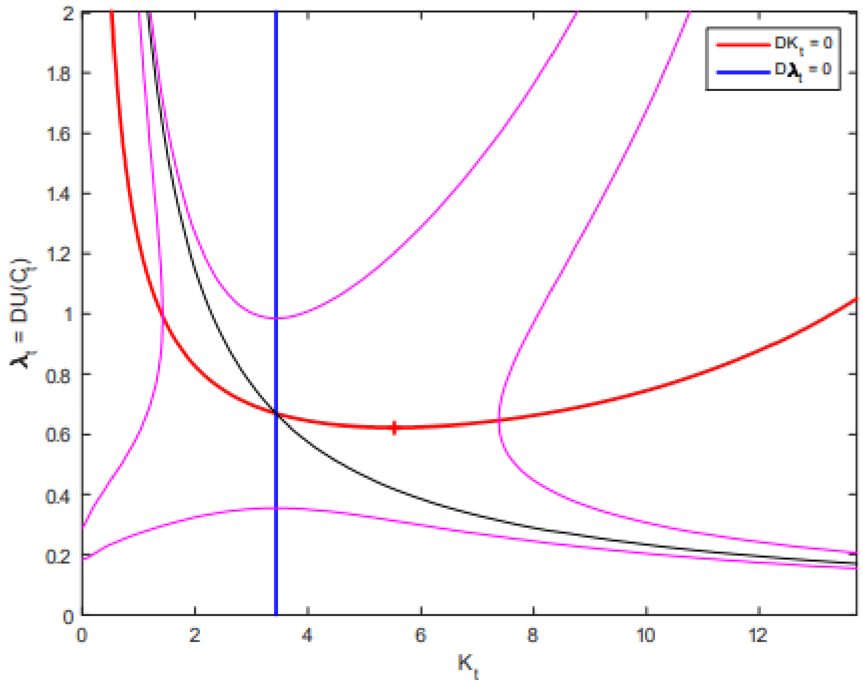


Figure 1: The phase diagram for (K_t, λ_t).

One alternate simple solution class that we can think about is the **golden rule**. That is, choosing a capital stock such that change in capital is offset by depreciation. Specifically, picking some K_G such that

$$Df(K_G) = \delta \iff K_G = [Df]^{-1}(\delta).$$

Since $\rho > 0$ and Df is a decreasing function, we know that the steady state level of capital that we described in the system of equations above must be smaller than the golden rule capital. As such, we know that this steady state level of capital is not optimal.

5.2.2 Irreversible Investment

Before we dive into competitive eqiulibrium, let's consider the same version of the model above but also require that capital investment decisions must be non-negative. Specifically, preferences are the same but the budget constraints now change to

$$\begin{aligned} C_t + X_t &\leq f(K_t), \\ DK_t &\leq -\delta K_t + X_t, \\ C_t, K_t, X_t &\geq 0. \end{aligned}$$

Starting from the beginning, let's write the lagrangian

$$\mathcal{L}(C_t, X_t, \mu_t, \lambda_t) = \int_0^\infty e^{-\rho t} U(C_t)\, dt + \int_0^\infty e^{-\rho t} \lambda_t (f(K_t) - X_t - C_t) + \int_0^\infty e^{-\rho t} \mu_t (X_t - \delta K_t - DK_t).$$

Once again, since we don't like to work with DK_t, let's do integration by parts.

$$\begin{aligned}\int_0^\infty e^{-\rho t}\mu_t DK_t\,dt &= e^{-\rho T}\mu_T K_T\Big|_{T=0}^{\infty} - \int_0^\infty e^{-\rho t}K_t(D\mu_t - \rho\mu_t)\,dt, \\ &= \lim_{T\to\infty} e^{-\rho T}\mu_T K_T - \mu_0 K_0 - \int_0^\infty e^{-\rho t}K_t(D\mu_t - \rho\mu_t)\,dt.\end{aligned}$$

Using this, we can then rewrite the lagrangian as

$$\begin{aligned}\mathcal{L}(C_t, X_t, \mu_t, \lambda_t) = -\lim_{T\to\infty} e^{-\rho T}\mu_T K_T + \mu_0 K_0 + \int_0^\infty e^{-\rho t}\,[U(C_t) + \lambda_t(f(K_t) - X_t - C_t) \\ +\mu_t(X_t - \delta K_t) + K_t(D\mu_t - \rho\mu_t)]\,dt.\end{aligned}$$

Again, with the tranversality condition, we can rewrite

$$\mathcal{L}(C_t, X_t, \mu_t, \lambda_t) = \mu_0 K_0 + \int_0^\infty e^{-\rho t}\,[U(C_t) + \lambda_t(f(K_t) - X_t - C_t) + \mu_t(X_t - \delta K_t) + K_t(D\mu_t - \rho\mu_t)]\,dt.$$

Assuming that the non-negativity conditions don't apply and the utility function satisfies the inada conditions (implying that the other constraints bind as well), we can write the following FOCs

$$\begin{aligned}&[C_t] : e^{-\rho t}[U'(C_t) - \lambda] = 0 \implies U'(C_t) = \lambda_t, \\ &[K_t] : e^{-\rho t}[\lambda_t f'(K_t) - \mu_t\delta + D\mu_t - \rho\mu_t] = 0 \implies \lambda_t Df(K_t) = \mu_t(\delta+\rho) - D\mu_t.\end{aligned}$$

Again, we plug in the first FOC into the second FOC and also write down the condensed household budget constraint to get the following system of equations

$$\begin{aligned}D\mu_t &= \mu_t(\delta+\rho) - DU(C_t)Df(K_t), \\ DK_t &= f(k_t) - C_t - \delta K_t.\end{aligned}$$

This along with the transversality condition, an initial value for K_0, and the following conditions classifies the phase diagram (2) and any equilibria:

$$\begin{aligned}DU(C_t) &\geq \mu_t, \\ f(K_t) &\geq C_t, \\ (DU(C_t) - \mu_t)(f(K_t) - C_t) &= 0.\end{aligned}$$

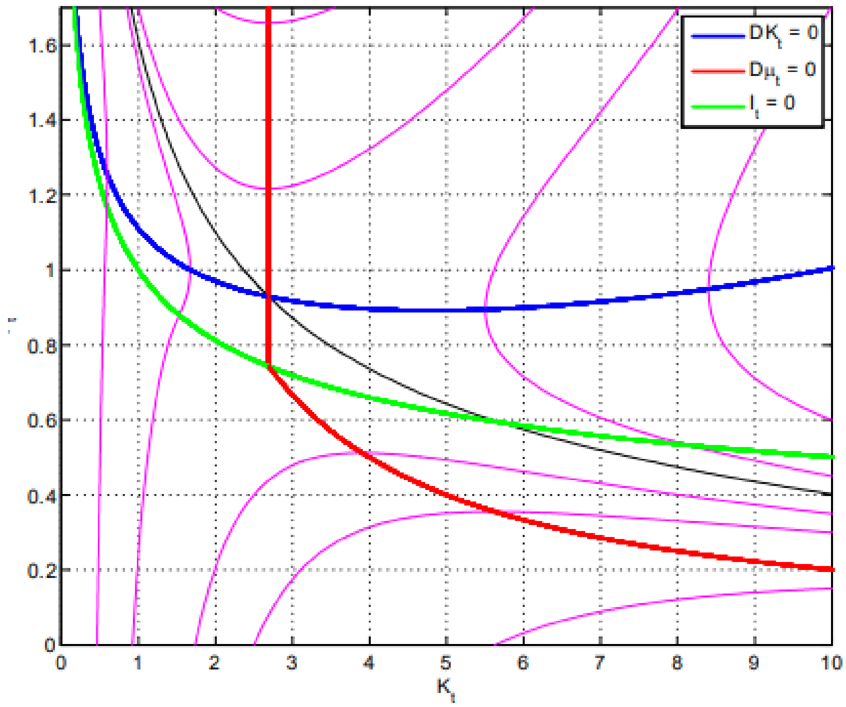


Figure 2: The phase diagram for (K_t, μ_t) with irreversible investment.

5.2.3 Competitive Equilibrium

So far, we have looked at this economy from a social planner's perspective without prices. Let's now try to add firms and prices.

5.2.4 Producers

I will preface, I feel like Erzo's treatment of the intermediate and final good producing firms does not seem very intuitively parallel to the discrete time versions that we have learned about in Chari. The Erzo firm is actually a quite simpler beast with simpler technology, but I think it is the skipping straight to final and intermediate good producers instead of first having them as one block and the reason for doing so that is tripping me up. I'll go really slow.

The **final good producer** takes a homogenous intermediate good g as an input and linearly transforms this into either capital, x, or the final consumption good c. Let q_t be the price of capital, p_t be the price of the intermediate good, and normalize the price of the consumption good to 1. This setup yields the following maximization problem

$$\max_{c,x,g}\{c + q_t x - p_t g\} \quad s.t. \quad g = c + x.$$

Which we can more simply just write as

$$\max_{c,x}\{c + q_t x - p_t(c+x)\}.$$

We know that there can't be unbounded profits, which means that we must have $p_t \geq 1$ and $p_t \geq q_t$ because otherwise we could just buy infinite amount of the intermediate good and produce an infinite amount of either capital or the consumption good. As such, we can clearly state

$$p_t = \max\{1, q_t\}.$$

Now consider, if $q_t > 1$ then $p_t = q_t$ and the final goods producer would have no incentive to produce any consumption good. Since, from the inada conditions, we know that there cannot be an equilibrium with 0 final good consumption, it must be then that an equilibrium has $p_t = 1 \geq q_t$ where there is zero capital production if $q_t < 1$ for the same argument as just stated but visa-versa.

The **intermediate good producer** takes in capital and labor as an input and produces the homogenous intermediate good. Given wages w_t, define

$$v_t = \max_{l}\{F(1,t) - w_t l\}.$$

Since we know $p_t = 1$ and given that capital input here is is non-zero which implies that $q_t = 1$, we can claim that v_t represents the income earned by an intermediate firm who owns one unit of capital. We know that the aggregate labor supply is 1 and the aggregate capital stock is K_t so taking the first order condition for the expression above, we get

$$D_2 F(K_t, 1) = w_t.$$

Then, because F has constant returns to scale, we can claim that $v_t = D_1 F(K_t, 1) = Df(K_t)$ and therefore

$$w_t = f(K_t) - Df(K)K_t.$$

This final good producer may seem quite trivial but the reason of defining one is because of the irreversible investment. Specifically, in these sorts of models with constant returns to scale, the price of capital q_t can be less than 1 unit of the consumption good. This leads to arbitrage scenarios because one can scale production of the consumption good linearly for less than the cost of the consumption good. This would lead to some equilibrium where all resources would be funneled into investment and yield no consumption. There needs to be a mechanism that stops capital prices from being less than one.

5.2.5 Consumers

Preferences are, of course, the same as in planner's problem but the budget constraint does change. Now, the consumer earns labor income w_t, consumes C_t, and can borrow or lend at interest rate r_t for assets A_t. Assets evolve according to

$$DA_t = r_t A_t + w_t - C_t.$$

Erzo doesn't write this derivation out explicitly but I tend to get confused if every step is not spelled out, so we can then rewrite the asset evolution expression as

$$\begin{aligned}
DA_t &= r_t A_t + w_t - C_t, \\
DA_t - r_t A_t &= w_t - C_t, \\
\left(-\int_0^t r_u \, du\right) [DA_t - r_t A_t] &= \left(-\int_0^t r_u \, du\right) [w_t - C_t], \\
\left(-\int_0^t r_u \, du\right) DA_t - \left(-\int_0^t r_u \, du\right) r_t A_t &= \left(-\int_0^t r_u \, du\right) [w_t - C_t], \\
D \left[\exp\left(-\int_0^t r_u \, du\right) A_t\right] &= \left(-\int_0^t r_u \, du\right) [w_t - C_t].
\end{aligned}$$

Integrating this differential equation from t to $T > t$ we then get

$$\begin{aligned}
\int_t^T D \left[\exp\left(-\int_0^s r_u \, du\right) A_s\right] ds &= \int_t^T \exp\left(-\int_0^s r_u \, du\right) [w_s - C_s] \, ds, \\
\exp\left(-\int_0^T r_u \, du\right) A_T - \exp\left(-\int_0^0 r_u \, du\right) A_t &= \int_t^T \exp\left(-\int_0^s r_u \, du\right) [w_s - C_s] \, ds, \\
\int_t^T \exp\left(-\int_0^s r_u \, du\right) C_s \, ds + \exp\left(-\int_0^T r_u \, du\right) A_T &= A_t + \int_t^T \exp\left(-\int_0^s r_u \, du\right) w_s \, ds.
\end{aligned}$$

Then express everything relative to time t consumption, as opposed to time 0 consumption, we get

$$\int_t^T \exp\left(-\int_t^s r_u \, du\right) C_s \, ds + \exp\left(-\int_t^T r_u \, du\right) A_T = A_t + \int_t^T \exp\left(-\int_t^s r_u \, du\right) w_s \, ds.$$

This gives us a map between a consumer's asset level A_t and their consumption choices from t to T, $\{C_s\}_{s\in[t,T]}$, into A_T. We are getting there slowly, but we have an expression to tell us how consumption plans and current asset holsings can give us future asset holdings. We can also use this expression to classify the borrowing constraint, as $T \to \infty$

$$\liminf_{T\to\infty} \exp\left(-\int_t^T rs \, ds\right) A_T \geq 0.$$

Define wealth (aka current assets plus the present value of future income flows) as

$$W_t = A_t + \int_t^\infty \exp\left(-\int_t^s r_u \, du\right) w_s \, ds$$

and we can then claim from the long expression above that

$$\int_t^\infty \exp\left(-\int_t^s r_u\, du\right) C_s\, ds \leq W_t.$$

Qualitatively, this is not too crazy, our time t consumption plans are limited by our time t wealth. But... we have gotten our consumer budget constraint now! Here is the consumer's problem so far

$$\begin{aligned} \max_{C_t} \quad & \int_0^\infty e^{-\rho t} U(C_t) \\ s.t. \quad & \int_0^\infty \exp\left(-\int_0^s r_u\, du\right) C_s\, ds \leq W_0. \end{aligned}$$

with the corresponding lagrangian

$$\mathcal{L}(C_t, \lambda_0) = \int_0^\infty e^{-\rho t} U(C_t) + \lambda_0 \left[W_0 - \int_0^\infty \exp\left(-\int_0^s r_u\, du\right) C_s\, ds \right].$$

Finally we can write the following FOC (wrt to consumption

$$e^{-\rho t} DU(C_t) = \lambda_0 \exp\left(-\int_0^t r_s\, ds\right).$$

Differentiating this will give us our euler condition $r_t = \rho + \theta(C_t) DC_t / C_t$. Next, FINALLY! We can solve for consumption

$$C_t = [DU]^{-1} \left(e^{\rho t} \lambda_0 \exp\left(-\int_0^t r_s\, ds\right)\right).$$

Which let's us plug C_t back into the budget constraint

$$\int_0^\infty \exp\left(-\int_0^s r_u\, du\right) C_s\, ds \leq W_0$$

to solve for λ_0 and plug this back in to fully pin down consumption.

5.2.6 Putting everything together - Equilibrium

Let's work on prices. From the consumer's side, we know that the rate of return on assets is r_t and on the firm's side we know a unit of capital generates factor income equal to v_t. Also capital depreciates at a rate δ which implies a value loss of δq_t per unit of time. This gives us the asset pricing equation

$$r_t q_t = v_t + D q_t - \delta q_t.$$

There are some derivations that I don't quite understand yet but we ultimately get $v_t = r_t + \delta$. I'll come back to this later. Also something else that I will come back to later is the derivation for the expression that will let us compute the trajectory of capital holdings, $\{K_s\}_{s>t}$,

$$A_t + \int_t^\infty \exp\left(-\int_t^s r_u\, du\right) w_s\, ds = \int_t^\infty \exp\left(-\int_t^s r_u\, du\right) (f(K_s) - X_s)\, ds.$$

Otherwise, combining a lot of the work above, we get

$$r_t = \rho - \frac{D\lambda_t}{\lambda_t}, \quad q_t = \frac{\mu_t}{\lambda_t}, \quad v_t = Df(K_t), \quad w_t = f(K_t) - Df(K_t)K_t.$$

5.3 Hamiltonians

So far, we have done optimization via writing down the lagrangian and then considering the first order conditions. Hamiltonians offer a more direct means of getting to a system of ODEs that give us our solution. It was incredibly groundbreaking for me when I came to the realization of what was going on. It is a set of UNIVERSAL ODEs (as long as the consumer problem can be written in a specific form that I will get to) where to get the final system, all we need to do is look at our formulation of the maximization problem and create this hamiltonian equation that I will get to and voila, we are done. One proof to show that it works needed and then you can go all willy-nilly with it. If you don't get why this is so cool and useful right now (I didn't either until I talked to Claude a lot about why it was cool and useful), then hopefully you will at the end of this.

I'm going to add some context here that I think is woefully missing from Erzo's notes. Let's start with a simple Cass-Koopmans type model that we dealt with in the section above. The social planner solves the problem (given some initial level of capital K_0)

$$\begin{aligned}
\max_{C_t, X_t, L_t, K_t} \quad & \int_0^\infty e^{-\rho t} u(C_t, L_t)\, dt \\
s.t. \quad (1) \quad & C_t + X_t \leq F(K_t, L_t), \\
(2) \quad & DK_t = X_t - \delta K_t, \\
(3) \quad & C_t \geq 0,\ 0 \leq L_t \leq 1.
\end{aligned}$$

Define $U(K, X) = \max_{C,L} \{u(C_t, L_t) : C + X \leq F(K, L)\}$ and note how we can write the equivalent social planner problem

$$\begin{aligned}
\max_{K_t, X_t} \quad & \int_0^\infty e^{-\rho t} U(K_t, X_t)\, dt \\
s.t. \quad & DK_t = X_t - \delta K_t.
\end{aligned}$$

The following work we will do works for any problem that can be written in the form above such that U is concave, increasing in K, and decreasing in X. Define $e^{-\rho t} q_t$ as the shadow price of captial (and the lagrange multiplier for the constraint on the motion of capital). The corresponding lagrangian is then

$$\mathcal{L}(K_t, X_t, q_t) = \int_0^\infty e^{-\rho t} \left[U(K_t, X_t) + q_t(X_t - \delta K_t - DK_t)\right] dt.$$

Once again, let's do integration by parts on DK_t.

$$\begin{aligned}\int_0^\infty e^{-\rho t} q_t DK_t &= e^{-\rho t} q_t K_t \Big|_{t=0}^\infty - \int_0^\infty e^{-\rho t} K_t (Dq_t - \rho q_t)\, dt, \\ &= \lim_{T\to\infty} e^{-\rho T} q_T K_T - q_0 K_0 - \int_0^\infty e^{-\rho t} K_t (Dq_t - \rho q_t)\, dt.\end{aligned}$$

With this, we can rewrite the lagrangian as

$$\mathcal{L}(K_t, X_t, q_t) = \int_0^\infty e^{-\rho t} \left[U(K_t, X_t) + q_t(X_t - \delta K_t) + K_t(Dq_t - \rho q_t)\right] dt + q_0 K_0 - \lim_{T\to\infty} e^{-\rho T} q_T K_T.$$

From here, define the **Hamiltonian**

$$H(K, q) = \max_X \{U(K, X) + q(X - \delta K)\}.$$

Just for some basic intuition, this construction is simply the utility represented as a function of investment decisions X and capital holdings K plus the value (given that q is the shadow price of capital) of the flow of capital (aka the value of the change of capital). Using this, we can reformulate the lagrangian as

$$\mathcal{L}(K_t, q_t) = \int_0^\infty e^{-\rho t} \left[H(K_t, q_t) + K_t(Dq_t - \rho q_t)\right] dt + q_0 K_0 - \lim_{T\to\infty} e^{-\rho T} q_T K_T.$$

This yields the following first order condition

$$[K_t]: \quad H_1(K_t, q_t) + (Dq_t - \rho q_t) = 0 \implies Dq_t = \rho q_t - DH_1(K_t, q_t).$$

Then from the envelope condition for H with respect to q, we get that

$$DH_2(K, q) = X - \delta K = DK_t.$$

Using these two statements, we can get to the following system of differential equations

$$\begin{aligned} Dq_t &= \rho q_t - DH_1(K_t, q_t), \\ DK_t &= DH_2(K_t, q_t). \end{aligned}$$

And we are done! This system defines the solution to the social planner's problem. H will of course change depending on U and u but we will cover some examples in the exercises section to solidify this concretely.

The crazy part is that we were very general with this derivation. Any model where the planning problem can be writen as

$$\max_{K_t, X_t} \quad \int_0^\infty e^{-\rho t} U(K_t, X_t)\, dt$$
$$s.t. \quad DK_t = X_t - \delta K_t.$$

yields the corresponding solution system of differential equations above.

5.3.1 From Hamiltonians to Competitive Equilibrium

There is a lot more than Luttmer talks about in his notes such as the slopes of the resulting differential equations, what more we can milk from the hamiltonians with the Jacobians, and identifying the stable manifold. You can refer to Erzo's notes for that information but I did want to touch up on how to move from the social planners problem that we just spent some time on towards competitive equilibrium as the Hamilonian method makes this easy as well. Critically, the lagrange multiplier q_t is the shadow price of capital (remember back to Kehoe where we studied how the lagrange multipliers of the social planners problem correspond to the prices in competitive equilibrium). Next, the system of ODEs above gives us solutions for K_t and q_t. The rest of the equilibrium prices and allocations will of course depend on the specific problem and what else is going on beyond the household's problem but consider the Cass-Koopman's model. We can recover everything else from just knowing K_t and q_t.

- $\mathbf{X_t}$, we can recover from the hamiltonian FOC as the solution to $DU_2(K_t, X_t) + q_t = 0$.
- $\mathbf{L_t}$ and $\mathbf{C_t}$, since we already know X_t and K_t, we can recover these from the inner maximization problem of $U(K, X) = \max_{C,L} \{u(C_t, L_t) \,:\, C + X \leq F(K, L)\}$.
- $\mathbf{w_t} = f(K_t) - Df(K_t)K_t$, this is from the firm's problem.
- $\mathbf{v_t} = Df(K_t)$, this is also from the firm's problem.
- $\mathbf{r_t} = \frac{v_t - \delta q_t + Dq_t}{q_t}$, this comes from the identities that we derived in the Cass-Koopman's section.

5.4 The Uzawa two-sector economy

Alright now let's consider a type of economy where the factors of production have a bit more nuance to them. Qualitatively, this will be a scenario where we can split our capital and labor resources into capital/labor used to generate more capital and capital/labor used to generate the consumption good. In other words, we can freely transfer production between consumption and capital; there is different technology that produces either of them.

I'll first set up the model and corresponding dynamics from the social planner's perspective, then look at the competitive equilibrium problem (the formulation of what is going on with the firms is what is interesting), and then talk about relevant steady states. To begin the household has the following **preferences**.

$$\mathcal{U}(C) = \int_0^\infty e^{-\rho t} U(C_t)\, dt,$$

where $\rho > 0$ and $U(\cdot)$ is strictly increasing and concave. They also are given initial capital $K_0 > 0$ and a fixed supply of labor over time $L > 0$. Here is the **technology** for producing consumption and capital paths,

$$\begin{aligned} K_{c,t} + K_{k,t} &= K_t, \\ L_{c,t} + L_{k,t} &= L_t, \\ C_t &\leq F(K_{c,t}, L_{c,t}), \\ X_t &\leq G(K_{k,t}, L_{k,t}), \\ DK_t &= -\delta K_t + X_t. \end{aligned}$$

Where both G and F exhibit constant returns to scale and all $K_{c,t}, K_{k,t}, L_{c,t}, L_{k,t}$ are non-negative. A normal one sector economy would have $F = G$ but differentiating the production functions creates these two consumption and capital sectors. There are some more assumptions placed on F and G and their relationship but you can refer to the notes for that, I'll try to stick to the intuition and deriving the intuition.

5.4.1 The competitive equilibrium environment

There are prices: v_t factor price of capital services, w_t factor price of labor services, p_t output price of consumption, q_t output price of capital, and r_t discount bond prices. On the production side, since there are constant returns to scale and it is competitive markets, the firms choose capital and labor to minimize costs (and also make 0 profits). Their profit functions looks like

$$\begin{aligned} p_t F(k,l) - v_t k - w_t l \leq 0, \\ p_t G(k,l) - v_t k - w_t l \leq 0. \end{aligned}$$

Which lead to the following set of firm problems

$$\begin{aligned} \min_{k,l} \{v_t k + w_t l \,:\, F(k,l) \geq 1\} \geq p_t, \\ \min_{k,l} \{v_t k + w_t l \,:\, G(k,l) \geq 1\} \geq q_t \end{aligned}$$

If we write $k_{c,t}$ and $k_{k,t}$as the cost-minimizing capital-labor ratios, the allocation of labor can be determined by

$$\begin{bmatrix} k_{k,t} & k_{c,t} \\ 1 & 1 \end{bmatrix} \begin{bmatrix} L_{k,t} \\ L_{c,t} \end{bmatrix} = \begin{bmatrix} K \\ L \end{bmatrix}.$$

Consumption and investment are then

$$C_t = F(k_{c,t}, 1)L_{c,t} > 0, \quad I_t = G(k_{k,t}, 1)L_{k,t} \geq 0.$$

Once again, the price of capital satisfies the asset pricing equations

$$r_t q_t = v_t - \delta q_t + Dq_t,$$

which jointly with the Euler condition implies

$$D\mu_t = (\rho + \delta)\mu_t - \lambda_t v_t.$$

As always, the capital stock evolves according to

$$DK_t = -\delta K_t + G(k_{k,t}, 1)L_{k,t}.$$

Together they jointly describe how the state (K_t, μ_t) evolves.

5.4.2 Writing down the hamiltonian

The corresponding hamiltonian for this problem is

$$H(K, \mu) = \max_{C,I} \{U(C) + \mu(I - \delta K)\},$$

where C and I must satisfy the constraints $C \leq F(K_c, L_c)$, $I \leq G(K_k, L_k)$, $K_c + K_k \leq K$, and $L_c + L_k \leq L$. Critically, this formulation abstracts a little from basing the hamiltonian on specific values of K_k, K_c, L_k, L_c but more so possible values of K and L that can be achieved by any combination of the sectoral specific counterparts. Since we know that $U(\cdot)$ satisfies the inada conditions, we can equivalently write

$$H(K, \mu) = \max_{K_c, L_c, I} \{U(F(K_c, L_c)) + \mu(I - \delta K)\}.$$

We know then that the system of differential equations that dictate the solution to this model are

$$\begin{aligned}
D\mu_t &= \rho\mu_t - DH_1(K_t, \mu_t) = \rho\mu_t + \delta\mu_t - DU(C_t)D_1F(K_{c,t}, L_{c,t})), \\
DK_t &= DH_2(K_t, \mu_t) = I_t - \delta K_t.
\end{aligned}$$

Note the parallels between backing these back from the hamiltonian versus the the asset pricing above (it is just a matter of substituting $G(\cdot)$ and $F(\cdot)$ into the hamiltonian).

The final thing that I wanted to talk about is backing out the sectoral capital and labor allocations. In steady state $DK_t = 0$ and $D\mu_t = 0$. Using equality of the constraints and the system of equations above, we can write that this economy has a unique steady state, characterized by steady-state capital-labor ratios, k_k^* and k_c^* that are defined by,

$$\delta + \rho = D_1 G(k_x, 1), \quad \frac{D_1 G(k_x, 1)}{D_2 G(k_x, 1)} = \frac{D_1 F(k_c, 1)}{D_2 F(k_c, 1)}.$$

For the golden rule, the conditions are slightly different

$$\delta = D_1 G(k_x, 1), \quad \frac{D_1 G(k_x, 1)}{D_2 G(k_x, 1)} = \frac{D_1 F(k_c, 1)}{D_2 F(k_c, 1)}.$$

5.5 Capital Accumulation and Adjustment Costs

To motivate why we would consider the models I will talk about soon, consider the general growth model in discrete time or the Cass-Koopmans that we have talked about in this Luttmer section. The model is constructed in a such a manner that investment today converts into capital tomorrow one-to-one. There are no frictions or delays in capital formulation. Reality is a bit different, simply investing in the construction of a building today does not mean that the building is added into our capital stock overnight. There are additional costs and things take time.

The idea in this section is to model that by introducing an additional cost to investment flow. Specifically, we introduce function $A(K_t, L_t)$ that affects the resource constraint in the following manner by replacing $C_t + I_t \leq f(K_t, L_t)$ with

$$C_t + A(K_t, I_t) \leq F(K_t, L),$$

where

$$A(K_t, I_t) = a\left(\frac{I_t}{K_t}\right) L_t.$$

Here $a(\cdot)$ is convex, which means that there are increasing returns to scale in I_t/K_t, or in other words, the resource cost increases at I_t increases but get smaller as K_t is large: capital heavy firms require fewer resource costs to raise investment. To pin everything down with a specific setup, let's consider the following model. Households have **preferences**

$$\mathcal{U}(C) = \int_0^\infty e^{-\rho t} U(C_t)\, dt.$$

Labor is taken to be **inelastic** and equal to L. The **constraints for the social planner's** problem are

$$\begin{aligned} C_t + A(K_t, I_t) &\leq F(K_t, L), \\ DK_t &\leq I_t - \delta K_t. \end{aligned}$$

As mentioned before $A(K_t, I_t) = a\left(I_t/K_t\right) K_t$ where $a(0) = 0$, $a(\cdot)$ is increasing and continuous, and $a(\cdot)$ is frequently taken to be quadratic. We can also see how we can recover the Cass-Koopman model if we let $a(\cdot)$ be linear (or better yet the identity. This is not the only way of modeling adjustment costs but is pretty popular. Alternatively, we could also equivalentely write the same formulation with the following constraints

$$
\begin{aligned}
C_t + X_t &\leq F(K_t, L), \\
DK_t &\leq G(K_t, X_t) - \delta K_t,
\end{aligned}
$$

where $G(\cdot)$ is defined as $A(1, G(1, x)) = x$. That being said, let's look to solve the problem from the planners perspective. We can start with writing down the **hamiltonian**.

$$
\mathcal{H}(K, \mu) = \max_{C, I \geq 0} \{U(C) + \mu(I - \delta K) \,:\, C \leq F(K, L) - A(K, I)\}.
$$

The corresponding **first order conditions** for this hamiltonian (given that λ is the lagrange multiplier for the resource constraint, also assuming that investment is non-zero) are

$$
\begin{aligned}
[C] &: DU(C) - \lambda = 0, \\
[I] &: \mu - \lambda D_2 A(K, I) = 0
\end{aligned}
$$

The corresponding **envelope conditions** are

$$
\begin{aligned}
D_1 \mathcal{H}(K, \mu) &= -\mu\delta + \lambda(D_1 F(K, L) - D_1 A(K, I)), \\
D_2 \mathcal{H}(K, \mu) &= I - \delta K.
\end{aligned}
$$

Combining statements all together, we can write out the system of ODEs that classify the solution to this economy

$$
\begin{aligned}
D\mu_t &= \rho\mu_t - D_1 \mathcal{H}(K, \mu) = \mu_t(\rho + \delta) - DU(C_t)(D_1 F(K_t, L_t) - D_1 A(K_t, I_t)), \\
DK_t &= D_2 \mathcal{H}(K_t, \mu_t) = I_t - \delta K_t.
\end{aligned}
$$

We can then also pin down consumption and investment from the resource constraint and FOC,

$$
\begin{aligned}
C_t + A(K_t, I_t) &= F(K_t, L), \\
\mu_t &= DU(C_t) D_2 A(K_t, I_t).
\end{aligned}
$$

Now, moving onto competitive equilibrium

5.5.1 Using Labor to Produce New Capital

5.6 Vintage Capital

5.7 Unemployment, Search, and Matching

Consider a model with

5.8 Perpetual Youth

This model has a gounding in demographics where the key idea is trying to capture the birth and death of consumers. Specifically, there is a unit measure of consumers, new consumers are born at a rate of θ and also consumers also die randomly at a rate of θ (hence the name perpetual youth because one's mortality is independent of age). One way to interpret this is an OLG with some different structure placed on when consumers are born and die.

Let's consider the discrete case, compute some probabilities, and then take some limits to identify the continuous case parallel. Let Δ be a time interval, and let the probability that a consumer survives one time period be $1-\theta\Delta$. We know then that the probability that a consumer survives to age (or equivalently, the fraction of a cohort that survives to age) $n\Delta$ is

$$PR[N > n] = (1-\theta\Delta)^n.$$

Now let $a = n\Delta$ represent the total amount of time and we can write $a/\Delta = n$,

$$PR[N > a/\Delta] = (1-\theta\Delta)^{a/\Delta}.$$

Let's shoot Δ (aka the duration of the time period) down to 0 to get the continuous time equivalents of what we have done so far

$$\begin{aligned}
\lim_{\Delta\to 0}\left(1-\theta\frac{a}{a/\Delta}\right)^{a/\Delta} &= \lim_{\Delta\to 0}\left(1-\theta\frac{a}{a/\Delta}\right)^{a/\Delta}, \\
&= \lim x \to \infty \left(1-\frac{\theta a}{x}\right)^{x}, \\
&= e^{-\theta a}.
\end{aligned}$$

This also gives us that the average life span in a stationary population is $\frac{1}{\theta}$ and distribution of life spans being exponential with density $\theta e^{-\theta a}$. Now let's get to the economics. Consumer preferences are as follows

$$\mathcal{U}(C) = \int_0^\infty \theta e^{-\theta t}\left(\int_0^t e^{-\rho a} U(C_a)\,da\right)dt.$$

We can interpret this as the expected utility.

5.9 Fiscal Theory of the Price Level

5.10 Semi-Endogenous Growth

The main idea here is to try to write down a model that has per-capita growth and we do so with preferences for variety, returns to scale, and population growth. Let's start with the dynastic household. It is called as such because it increases in size every time period (representing population growth) whose aggregate consumption is C_t. Specifically, we have **preferences**

$$\mathcal{U} = \int_0^\infty e^{-\rho t} H_t \ln\left(\frac{C_t}{H_t}\right) dt,$$

where C_t is a CES composite good ($\epsilon > 1$ and dictates the "love of variety")

$$C_t = \left(\int_0^{\Omega_t} C_{\omega,t}^{1-1/\epsilon}\, d\omega\right)^{1/(1-1/\epsilon)}.$$

The dynastic present-value **budget constraint** is

$$\int_0^\infty \pi_t C_t \, dt \leq wealth$$

5.11 Exercises

5.11.1 Durable Consumption

Consider an economy in which K_t represents the stock of durable consumption goods at time t, and C_t measures the flow of consumption services generated by the stock of durable consumption goods. Preferences are

$$\int_0^\infty e^{-\rho t} \left(\ln(C_t) - J(L_t)\right) dt.$$

The technology is

$$\begin{aligned} C_t &\leq AK_t \\ DK_t &= -\delta K_t + zL_t \end{aligned}$$

Here, ρ, δ, A, and z are all positive parameters. The function $J : \mathbb{R}_+ \to \mathbb{R}$ is smooth, strictly increasing, and strictly convex.

Consider the competitive equilibrium of this economy. Write w_t for the wage, and q_t for the price of a unit of the durable consumption good, both measured in units of consumption services at time t. The real interest rate (in terms of consumption services) is r_t. A consumer with wealth W_t can consume subject to the sequence of constraints $dW_t = (r_t W_t - C_t)dt$ together with $W_t \geq 0$ at all times.

(a) Explain why aggregate consumption services and the real interest rate must satisfy $r_t = \rho + DC_t/C_t$.

> Solution: Let's write down the household's hamiltonian
>
> $$\mathcal{H}(W, \mu) = \max_{C,L} \left\{\ln(C) - J(L) + \mu(rW - C)\right\}.$$

Note here that the choice of L does indeed intrinsically factor into W but the question writes the household budget constraint differently. The corresponding FOC for consumption is

$$\frac{1}{C_t} - \mu_t = 0 \implies \mu_t = \frac{1}{C_t} \implies D\mu_t = -\frac{DC_t}{C_t^2},$$

and the resulting ODE for μ_t is

$$D\mu_t = \rho\mu_t - r_t\mu_t.$$

Combining these two expressions, we get

$$\begin{aligned}
D\mu_t &= -\frac{DC_t}{C_t^2}, \\
\frac{D\mu_t}{\mu_t} &= -\frac{DC_t}{C_t}, \\
\frac{\rho\mu_t - r_t\mu_t}{\mu_t} &= -\frac{DC_t}{C_t}, \\
\rho - r_t &= -\frac{DC_t}{C_t}, \\
r_t &= p + \frac{DC_t}{C_t}.
\end{aligned}$$

■

(b) There will be an asset pricing equation of the form $r_t q_t = v_t + Dq_t - \delta q_t$. What is v_t? What will be the wage w_t?

Solution: In general, v_t is flow income earned by owning one unit of the durable good (which is capital in this case). There are two ways that capital is used, to produce the consumption good and to produce more capital. Specifically, we can write the flow value of capital to be

$$v_t = A + \max_{l_t}\{q_t z l_t - w_t l_t\}.$$

For the latter maximization problem, the foc very clearly gives us that

$$\boxed{w_t = q_t z.}$$

which then implies that

$$\boxed{v_t = A.}$$

(c) Give the differential equation for K_t and q_t/C_t that must hold in any equilibrium. Construct the phase diagram for this differential equation.

SOLUTION: I won't construct the phase diagram for now, but may come back to it later. ■

(d) Suppose the economy is in its steady state and there is an unforeseen permanent increase in A at $t = 0$. Describe what happens to $\{K_t, q_t/C_t, C_t, L_t, w_t, q_t, r_t\}_{t\geq 0}$.

SOLUTION: ■

(e) Suppose the economy is in its steady state and there is an unforeseen permanent increase in z at $t = 0$. Describe what happens to $\{K_t, q_t/C_t, C_t, L_t, w_t, q_t, r_t\}_{t\geq 0}$. **Note:** If you want to compute an example to check your logic, use $J(L) = L^{1+1/\psi}/(1+1/\psi)$, for some $\psi > 0$.

SOLUTION: ■

5.11.2 A Simple Consumption-Savings Problem

Define

$$U(W) = \max_{C_t \geq 0} \left(\int_0^\infty \rho e^{-\rho t} C_t^{1-1/\varepsilon} \, dt \right)^{1/(1-1/\varepsilon)}$$

subject to

$$DW_t = rW_t - C_t$$

and $W_t \geq 0$ for all $t \geq 0$, starting from $W_0 = W$ for some given $W > 0$.

(a) Find a candidate solution for $\{C_t\}_{t\geq 0}$ and verify that it is well defined if

$$\rho - \left(1 - \frac{1}{\varepsilon}\right) r > 0$$

Compute $U(W)$.

SOLUTION: We know that consumer preferences are

$$U(W) = \max_{C_t \geq 0} \left(\int_0^\infty \rho e^{-\rho t} C_t^{1-1/\varepsilon} \, dt \right)^{1/(1-1/\varepsilon)}.$$

Since $f(x) = x^{1/(1-1/\epsilon)}$ is a monotone transformation of x, we know that we can find the consumption maximizing allocation $\{C_t\}_{t\geq 0}$ by looking at the alternate problem

$$\max_{C_t \geq 0} \int_0^\infty \rho e^{-\rho t} C_t^{1-1/\varepsilon} \, dt.$$

Now with this in mind, let's write down the hamiltonian. We know the flow of consumer's wealth is $DW_t = rW_t - C_t$. Letting μ be the costate of wealth, we can write

$$\mathcal{H}(W, \mu) = \max_C \left\{ \rho C^{1-1/\epsilon} + \mu(rW - C) \right\}.$$

Taking the first order condition, we have

$$[C_t]: \quad \rho(1-1/\epsilon)C_t^{-1/\epsilon} - \mu_t = 0 \implies C_t = \left(\frac{\mu_t}{\rho(1-1/\epsilon)}\right)^{-\epsilon}.$$

The corresponding system of ODE's is

$$\begin{aligned} DW_t &= rW_t - C_t, \\ D\mu_t &= \rho\mu_t - \mu_t r \implies \mu^{-1} D\mu_t = \rho - r. \end{aligned}$$

Taking this system of equations, lets continue writing

$$\begin{aligned} DC_t &= D\left[\left(\frac{\mu_t}{\rho(1-1/\epsilon)}\right)^{-\epsilon}\right], \\ &= D\left[\mu_t^{-\epsilon} \left(\rho(1-1/\epsilon)\right)^{\epsilon}\right], \\ &= \left(\rho(1-1/\epsilon)\right)^{\epsilon} (D\mu_t)(-\epsilon)\mu_t^{-\epsilon-1}. \\ \frac{DC_t}{C_t} &= -\epsilon D\mu_t \mu_t^{-1}, \\ &= -\epsilon(\rho - r), \\ &= \epsilon(r - \rho). \end{aligned}$$

This last expression is our euler equation for consumption. Since the right hand side is a constant, we know from the get go that

$$C_t = C_0 e^{\epsilon(r-\rho)t}.$$

We still, however, need to pin down C_0. To do so, we can conjecture that $C_t = \alpha W_t$; that consumption is always a constant fraction of wealth. To confirm this conjecture, we jointly look at the budget constraint ODE and the transverality condition.

Specifically we have that

$$DW_t = rW_t - C_t = (r-\alpha)W_t \implies W_t = W_0 e^{(r-\alpha)t}.$$

To confirm this conjecture, we need to spell out α and then put some constraints on it through the transversality condition. We know that consumption grows at a rate of $e^{\epsilon(r-\rho)t}$ and wealth grows at a rate of $e^{(r-\alpha)t}$, to make these elements mesh together well we need

$$e^{(r-\alpha)t} = e^{\epsilon(r-\rho)t} \implies r - \alpha = \epsilon(r-\rho) \implies \alpha = r(1-\epsilon) + \epsilon\rho.$$

Plugging this into the transversality condition, we get

$$\begin{aligned}
\lim_{T\to\infty} e^{-\rho T}\mu_T W_T &= \lim_{T\to\infty} e^{-\rho T}\mu_T W_0 e^{(r-r(1-\epsilon)-\epsilon\rho)T}, \\
&= \lim_{T\to\infty} e^{-\rho T}\mu_T W_0 e^{(r\epsilon-\epsilon\rho)T}, \\
&= \lim_{T\to\infty} \mu_T W_0 e^{(r\epsilon-\epsilon\rho-\rho)T}, \\
&= \lim_{T\to\infty} \mu_0 e^{(\rho-r)T} W_0 e^{(r\epsilon-\epsilon\rho-\rho)T}, \\
&= \lim_{T\to\infty} \mu_0 W_0 e^{(r\epsilon-\epsilon\rho-r)T}.
\end{aligned}$$

In order for this limit to equal zero, we need

$$\begin{aligned}
r\epsilon - \epsilon\rho - r &< 0, \\
r - \rho - \frac{r}{\epsilon} &< 0, \\
-r + \rho + \frac{r}{\epsilon} &> 0, \\
\rho - \left(1 - \frac{1}{\epsilon}\right) r &> 0.
\end{aligned}$$

As such, along as this condition holds (which is the condition in the problem), we know that our conjecture holds. With this confirmed, we can finally claim

$$\boxed{C_t = (r(1-\epsilon)+\epsilon\rho)We^{\epsilon(r-\rho)t}.}$$

Finally, to compute $U(W)$ we write

$$\begin{aligned}
U(W) &= \max_{C_t \geq 0} \left(\int_0^\infty \rho e^{-\rho t} C_t^{1-1/\varepsilon}\, dt \right)^{1/(1-1/\varepsilon)}, \\
&= \left(\int_0^\infty \rho e^{-\rho t} (r(1-\epsilon)+\epsilon\rho)^{1-1/\epsilon} W^{1-1/\epsilon} e^{(\epsilon-1)(r-\rho)t}\, dt \right)^{1/(1-1/\epsilon)},
\end{aligned}$$

$$= (r(1-\epsilon)+\epsilon\rho)W\left(\int_0^{\infty}\rho e^{-\rho t}e^{(\epsilon-1)(r-\rho)t}\,dt\right)^{1/(1-1/\epsilon)},$$

$$= (r(1-\epsilon)+\epsilon\rho)W\left(\int_0^{\infty}\rho e^{(\epsilon(r-\rho)-r)t}\,dt\right)^{1/(1-1/\epsilon)},$$

$$= \boxed{(r(1-\epsilon)+\epsilon\rho)W\left(\frac{\rho}{r-\epsilon(r-\rho)}\right)^{1/(1-1/\epsilon)}.}$$

■

Now suppose labor income is $zL > 0$ and assets on hand evolve subject to

$$DA_t = rA_t + zL - C_t$$

and $A_t \geq 0$ for all $t \geq 0$, starting from $A_0 = A$ for some given $A > 0$.

(b) Suppose that $r \geq \rho > (1-1/\varepsilon)r$. What is the optimal consumption trajectory?

SOLUTION: The difference now is that each period, the household now gets a fixed amount of wealth increases.

$$\mathcal{H}(A,\mu) = \max_C\left\{\rho C^{1-1/\epsilon} + \mu(rA + zL - C)\right\}.$$

Critically, note how the first order condition for C_t and differential equation for μ_t stays the same. This means that we will eventually get to

$$\frac{DC_t}{C_t} = \epsilon(r-\rho),$$

which implies that $C_t = C_0 e^{-\epsilon(r-\rho)}$ and although C_0 may be different, the consumption trajectory is indeed the same. ■

(c) Suppose that $r < \rho$. Use the Lagrangian for this problem. Conjecture that there will be some optimal time $T \in (0,\infty)$ such that $A_t = 0$ if and only if $t \in [T,\infty)$. Find the equation that defines T and show that it must have a solution in $(0,\infty)$.

SOLUTION: Qualitatively, the idea is that if the rate of return on assets is less that the temporal preference discount factor then it is optimal behavior to consume all of the aggregate assets at some point in time. Specifically here, we have the same condition

$$\frac{DC_t}{C_t} = \epsilon(r-\rho),$$

which combined with $r < \rho$ and $C_t \geq 0$ means that consumption is decreasing over time and the household would at some point want to borrow to fuel consumption. We are told, however, that $A_t \geq 0$ which means that households can't borrow which means that from the conjecture $A_t = 0$ at some point and at which point the household becomes hand-to-mouth and consumes exactly their labor income $C_t = zL$. Like the question suggests, lets write down the lagrangian

$$\mathcal{L}(C_t, \lambda_t, A_t) = \max_{C_t, A_t} \int_0^\infty \rho e^{-\rho t}$$

■

5.11.3 An Euler Equation with Consumption Risk

Take some $\Delta \in \mathbb{R}_{++}$ and consider an economy with preferences over consumption sequences $\{C_{n\Delta}\}_{n=0}^\infty$ determined by

$$\mathcal{U}(\{C_{n\Delta}\}_{n=0}^\infty) = \mathrm{E}_0 \left[\sum_{n=0}^\infty e^{-\rho \Delta n} \left(\frac{C_{n\Delta}^{1-1/\varepsilon} - 1}{1 - 1/\varepsilon} \right) \right].$$

Suppose the aggregate consumption process follows

$$C_{(n+1)\Delta} = C_{n\Delta}\, e^{\mu\Delta} \times \begin{cases} 1, & \text{w.p. } 1 - \alpha\Delta \\ \delta, & \text{w.p. } \alpha\Delta \end{cases}$$

Markets are complete. What is the price $q_{n\Delta}$ at time $n\Delta$ of a one-period discount bond? Pick some time $t \in \mathbb{R}_{++}$ and calculate the limits

$$g_t = \lim_{\Delta \downarrow 0} \frac{1}{\Delta} \mathrm{E}_{\lfloor t/\Delta \rfloor} \left[\frac{C_{(\lfloor t/\Delta \rfloor + 1)\Delta} - C_{\lfloor t/\Delta \rfloor \Delta}}{C_{\lfloor t/\Delta \rfloor \Delta}} \right], \qquad r_t = \lim_{\Delta \downarrow 0} \frac{1 - q_{\lfloor t/\Delta \rfloor \Delta}}{\Delta}.$$

Show that $r_t < \rho + g_t/\varepsilon$. The notation here is: $\lfloor x \rfloor$ is the integer part (floor) of the real number x.

Solution: ■

5.11.4 The Log Utility Special Case

Consider a consumer with preferences over possibly stochastic consumption trajectories $\{C_t\}_{t\geq 0}$ determined by

$$\mathcal{U}(\{C_t\}_{t\geq 0}) = \exp\left(\mathrm{E}_0 \left[\int_0^\infty \rho e^{-\rho t} \ln(C_t)\, dt \right] \right),$$

where $\rho \in \mathbb{R}_{++}$.

4.1 The Consumption-Savings Problem

This consumer has initial wealth $W_0 = W > 0$ and can accumulate wealth subject to $DW_t = rW_t - C_t$ and $W_t \geq 0$ for all $t \in [0, \infty)$. The consumer takes $r \in \mathbb{R}$ as given.

a. Write down the Hamiltonian for log utility. Write q_t for the co-state and find the differential equation for $[W_t, q_t]$. What is the transversality condition?

> SOLUTION: ■

b. Solve the differential equation for $q_t W_t$. Use the transversality condition to explain why $q_0 W_0 = 1$.

> SOLUTION: ■

c. Calculate $V(W) = \ln(\mathcal{U}(\{C_t\}_{t\geq 0}))$ for this consumer.

> SOLUTION: ■

d. Suppose the labor income of this consumer is $z_t L$, where $z_t = z_0 e^{-\lambda t}$ and $\lambda \geq 0$. Assume $r + \lambda > 0$. The initial assets of this consumer are $A_0 = A \geq 0$. What is W? Describe how the trajectory of A_t depends on the sign of $r - \rho$.

> SOLUTION: ■

4.2 A Poisson Shock

Suppose this consumer faces the stochastic consumption trajectory

$$\tilde{C}_t = \begin{cases} C_t & t \in [0, \tau) \\ C'_t & t \in [\tau, \infty) \end{cases}$$

Here $\{C_t\}_{t\geq 0}$ and $\{C'_t\}_{t\geq 0}$ are deterministic consumption trajectories. And τ is a random time with a probability distribution $\Pr[\tau \geq T] = e^{-\lambda T}$, where $\lambda > 0$. Note that $S > T$ gives $\Pr[\tau \geq S \mid \tau \geq T] = e^{-\lambda(S-T)}$. Define V_0 and $\{U_T\}_{T\geq 0}$ by

$$\ln(V_0) = \mathrm{E}_0\left[\int_0^\infty \rho e^{-\rho t} \ln(\tilde{C}_t)\, dt\right], \qquad e^{-\rho T}\ln(U_T) = \int_T^\infty \rho e^{-\rho t} \ln(C'_t)\, dt, \quad T \in [0, \infty).$$

Show that

$$\ln(V_0) = \int_0^\infty (\rho + \lambda) e^{-(\rho+\lambda)t} \left(\frac{\rho}{\rho + \lambda} \times \ln(C_t) + \frac{\lambda}{\rho + \lambda} \times \ln(U_t)\right) dt.$$

Note the distinction between V_0 and U_0.

SOLUTION: ■

4.3 Actuarially Fair Insurance

Now suppose the labor income of this consumer is $\tilde{z}_t L$ and that $\{\tilde{z}_t\}_{t\geq 0}$ is a stochastic process defined by

$$\tilde{z}_t = \begin{cases} z & t \in [0, \tau) \\ 0 & t \in [\tau, \infty), \end{cases}$$

where $\Pr[\tau \geq T] = e^{-\lambda T}$ and $\lambda > 0$. Assume $r \in \mathbb{R}$ and $r + \lambda \in \mathbb{R}_{++}$. Write A_t for assets on hand. The consumer can trade subject to

$$DA_t = rA_t + zL - (C_t + \lambda G_t)$$

for all $t \in [0, \tau)$. Here, C_t is flow consumption and λG_t is an insurance premium paid by the consumer. At $t = \tau$, assets on hand will jump to

$$A_\tau = A_{\tau^-} + G_{\tau^-}.$$

Here, A_{τ^-} and G_{τ^-} are the left limits $A_{\tau^-} = \lim_{s\uparrow\tau} A_s$ and $G_{\tau^-} = \lim_{s\uparrow\tau} G_s$, respectively. In addition, there is a borrowing constraint that says that $A_t + zL/(r + \lambda) \geq 0$ for all $t \in [0, \tau)$ and $A_t \geq 0$ for all $t \in [\tau, \infty)$.

a. Take for granted that the borrowing constraints will not bind. Construct the Hamiltonian $\mathcal{H}(A, q)$ for this problem and find the optimality conditions. In particular, what is the transversality condition?

SOLUTION: ■

b. Define $W_t = A_t + zL/(r + \lambda)$ for all $t \in [0, \tau)$ and $W_t = A_t$ for all $t \in [\tau, \infty)$. Show that the optimal decision rule for this consumer is $C_t = \rho W_t$ at all times. What happens to C_t between τ^- and τ?

SOLUTION: ■

4.4 No Insurance

Consider the same labor income shock as before, but now suppose the consumer must trade subject to

$$DA_t = rA_t + zL - C_t$$

for all $t \in [0, \tau)$ and

$$DA_t = rA_t - C_t$$

for all $t \in [\tau, \infty)$, and $A_t \geq 0$ for all $t \in [0, \infty)$.

a. Set up the Hamiltonian $\mathcal{H}(A, q)$ for the problem faced by this consumer whenever $t < \tau$. What is the resulting differential equation for $[A_t, q_t]$? What is the transversality condition?

SOLUTION: ■

b. Assume $r < \rho$. Show that there is a well-defined steady state conditional on $t < \tau$. Show the phase diagrams for $r < 0$ and for $r \in (0, \rho)$, taking for granted that the optimal trajectory does in fact converge to the steady state. What happens to consumption at $t = \tau$?

SOLUTION: ■

c. More generally, find the differential equation for $q_t A_t$, integrate it over the interval $[t, T]$, and use the transversality condition to show that

$$1 = q_t \left(A_t + \frac{1}{\pi_t} \int_t^\infty e^{-(\rho+\lambda)(s-t)} \pi_s zL \, ds \right)$$

for all $t \in [0, \infty)$ conditional on $t < \tau$, where π_t is the "shadow date-0 price" $\pi_t = e^{-(\rho+\lambda)t} q_t$. Conclude that $C_t = \rho W_t$, where W_t is "shadow wealth".

SOLUTION: ■

d. Note that $q_t A_t \in (0, 1)$. Explain why $Dq_t < -(r - \rho) q_t$. Use this to show that

$$0 < \frac{1}{q_t} \int_t^\infty e^{-(\rho+\lambda)(s-t)} q_s zL \, ds < \int_t^\infty e^{-(r+\lambda)(s-t)} zL \, ds = \frac{zL}{r+\lambda}$$

as long as $r + \lambda > 0$.

SOLUTION: ■

e. Assume that $r > \rho$. Combine your results for **c** and **d** to show that $\lim_{t\to\infty} q_t A_t = 1$. What does this imply for Dq_t/q_t and DA_t/A_t as $t < \tau$ becomes large? Set up the phase diagram for the case $r \in (\rho, \rho + \lambda)$ and make an informed guess about the optimal trajectory for $[A_t, q_t]$.

SOLUTION: ■

5.11.5 The AK Economy

Consider an economy with preferences

$$\mathcal{U}(C) = \left(\int_0^\infty e^{-\rho t} C_t^{1-1/\varepsilon} \, dt \right)^{1/(1-1/\varepsilon)},$$

where $\rho > 0$ and $\varepsilon \in (0, \infty)$. The technology for accumulating capital given by

$$DK_t = \mu K_t - C_t,$$

starting from some positive K_0.

a. Consider consumption trajectories of the form $C_t = Ce^{gt}$, for some feasible C and g. If $\varepsilon \in (0, 1)$, what do you have to assume about parameters to ensure that utility is positive? If $\varepsilon \in (1, \infty)$, what do you have to assume to ensure that utility is finite? What if $\varepsilon = 1$?

SOLUTION: ■

Consider competitive equilibria for this economy, taking for granted that it will not be an equilibrium to consume all of the capital stock in finite time.

b. What is the Euler condition for consumers?

SOLUTION: ■

c. What is the price of capital in units of consumption? What is the real interest rate in this economy?

SOLUTION: ■

d. Conjecture that $C_t/K_t > 0$ is constant and determine a candidate equilibrium.

SOLUTION: ■

e. What do you have to assume about parameters to ensure that this equilibrium is well defined? What do you have to assume to ensure that the economy grows over time?

SOLUTION: ■

5.11.6 An AK Economy with a Fixed Factor

Consider an economy with preferences

$$u(C) = \int_0^\infty e^{-\rho t} \left(\frac{C_t^{1-1/\varepsilon} - 1}{1 - 1/\varepsilon} \right) dt,$$

and a technology for accumulating capital given by

$$DK_t = \mu K_t - X_t, \qquad C_t = X_t^{1-\alpha} L^\alpha, \qquad X_t \in [0, \infty),$$

starting from some positive K_0. Observe that the production of consumption requires labor services and a flow X_t of capital that depletes the capital stock.

Assume the parameters are such that this economy has a well defined competitive equilibrium. Let v_t and w_t be the factor prices of capital and labor in the consumption sector. Let q_t be the price of capital in units of consumption.

a. What is the Euler equation for this economy?

SOLUTION: ■

b. Given a wage w_t, what must v_t be? What is the relation between v_t and q_t?

SOLUTION: ■

c. Conjecture that there is an equilibrium with $X_t/K_t = x > 0$. Construct the candidate equilibrium.

SOLUTION: ■

d. What do you have to assume about parameters to ensure that this equilibrium is well defined? What do you have to assume to ensure that the economy grows over time?

SOLUTION: ■

e. Suppose the technology is, instead, $DK_t = \mu \times (K_t - H_t)$ and $C_t = H_t^{1-\alpha} L^\alpha$ with $H_t \in [0, K_t]$. Show how you can translate the equilibrium constructed in **a–d** into a candidate equilibrium for this economy by ignoring the constraint $H_t \leq K_t$. Given an assumption on parameters that ensures this constraint does not bind.

SOLUTION: ■

5.11.7 An Exhaustible Resource

There is a representative consumer whose preferences over consumption trajectories $\{C_t\}_{t\geq 0}$ are given by

$$\mathcal{U}(C) = \left(\int_0^\infty e^{-\rho t} C_t^{1-1/\varepsilon} \, dt \right)^{1/(1-1/\varepsilon)},$$

where $\rho > 0$ and $\varepsilon \in (0, \infty)$. The aggregate supply of labor is $L > 0$. Output can be produced using capital, labor, and an exhaustible resource. The initial supply of capital is $K_0 > 0$ and the initial supply of the exhaustible resource is $A_0 > 0$. Output at time t is

$$Y_t = K_t^{1-\alpha-\beta} (Z_t L)^\alpha M_t^\beta,$$

where K_t is the capital stock at time t, $Z_t = Z e^{\theta t}$ is an exogenously determined productivity trajectory, and M_t is a flow of materials that depletes the exhaustible resource. The factor share parameters α, β, and $1 - \alpha - \beta$ are all in $(0, 1)$. The capital stock and the stock of the exhaustible resource evolve according to

$$\begin{aligned} DK_t &= -\delta K_t + Y_t - C_t \\ DA_t &= -M_t \end{aligned}$$

and K_t and A_t are required to be positive at all times. Write w_t and u_t for the factor prices of labor and materials, respectively. The market value of the remaining stock A_t of the exhaustible resource at time t is $s_t A_t$. All these prices are in units of consumption. The risk-free rate is r_t, also in units of consumption.

a. What is the Euler equation for this economy? Explain why $u_t = s_t$. What is the relation between s_t and r_t?

Solution: ■

Conjecture that the economy has a balanced growth path with an interest rate $r_t = r$, consumption growing at some rate g, and the stock of the exhaustible resource evolving according to $A_t = A_0 e^{-\lambda t}$, for some $\lambda > 0$ to be determined.

b. Determine a candidate balanced growth path for this economy.

Solution: ■

c. Give the parameter restrictions needed to ensure that this candidate equilibrium is well defined. What is needed for consumption to grow at a positive rate?

Solution: ■

You should get pretty clean formulas that allow you to give interpretable answers in **c**.

5.11.8 An Interpretation of Cobb-Douglas

Consider an economy with a measure K of machines and a measure L of workers who each supply one unit of labor. Assume $K > L$. Machines are heterogeneous. The output that can be produced with a type-z machine is

$$y = z \min\{l, 1\},$$

where l is the amount of labor employed with this machine. The distribution of types is $P(z)$. This distribution is continuous, with a support $[x, \infty)$ and a finite mean.

Consider competitive equilibria. Write $v(z)$ for the price of a type-z machine and w for the wage, both in units of output.

a. Determine $v(z)$ and w, as well as aggregate output.

> SOLUTION: ■

b. Suppose you add a type-z machine to this economy. This is an infinitesimal change in the capital stock of this economy. What is the infinitesimal change in aggregate output?

> SOLUTION: ■

c. Suppose you add a worker to this economy. This is an infinitesimal change in the aggregate supply of labor of this economy. What is the infinitesimal change in aggregate output?

> SOLUTION: ■

Now suppose that $P(z) = 1 - (z/x)^{-\theta}$, where $z \in [x, \infty)$ and $x > 0$.

d. Show that aggregate output is

$$Y = \frac{x}{\alpha} \times K^{1-\alpha} L^{\alpha}$$

where $\alpha \in (0, 1)$. What is the interpretation of the marginal product of K in this economy?

> SOLUTION: ■

5.11.9 A Vintage Capital Economy

Consumer preferences are

$$\mathcal{U}(C) = \int_0^\infty e^{-\rho t} \ln(C_t)\, dt.$$

Consumption is produced using an intermediate good. At time t, a flow $x \geq 0$ of the intermediate good produces a flow $c = e^{\theta t} x$ of consumption, where $\theta \geq 0$. The intermediate good can be produced using labor and machines of different vintages. Labor can also be used to produce new machines. The technology for producing the intermediate good at time t with machines of a vintage $v \leq t$ is

$$x = Z(v, t-v) \min\left\{k, \frac{l}{B}\right\},$$

where k is the number of vintage-v machines used, and l is the amount of labor used. In this economy, the vintage of a machine is the date the machine is produced. The technology for producing machines is linear in labor. It takes A units of labor to produce a unit flow of machines of the latest vintage. The function $Z(v, a)$ is given by

$$Z(v, a) = Ze^{\mu v - \Delta(a)}, \qquad v \in (-\infty, \infty), \quad a \in [0, \infty)$$

where $\Delta : \mathbb{R}_+ \to \mathbb{R}_+$ is a continuous and strictly increasing function that satisfies $\Delta(0) = 0$ and $\lim_{a\to\infty} \Delta(a) = \infty$. The parameter μ is strictly positive.

Write M_t for the flow of new machines and X_t for the flow of the intermediate good produced at time t. Write s_t for the price of the intermediate, w_t for the wage, and $p_t(v)$ for the factor price and $q_t(v)$ for the price of a machine of vintage $v \in (-\infty, t]$, all at time t. These prices are in units of consumption.

a. Show that this economy has a balanced growth path. Give the steady state equilibrium conditions. What is the balanced growth rate of consumption? Also, simplify the equilibrium conditions for the special case of $\Delta(a) = \delta a$.

SOLUTION: ■

b. Assume the economy is on a balanced growth path. Suppose you can observe the trajectories of C_t, w_t, s_t, and $\{p_t(v) : v \in (-\infty, t]\}$. Show how you can identify the parameters of this economy, with the understanding that we will never know the shape of $\Delta(\cdot)$ for machines that are obsolete.

SOLUTION: ■

c. How does your answer change if you cannot observe the trajectory of s_t?

SOLUTION: ■

5.11.10 A Messed-Up Model of the Firm Size Distribution

Consumer preferences are

$$\mathcal{U}(C) = \int_0^\infty e^{-\rho t} \ln(C_t)\, dt$$

for some $\rho > 0$. The aggregate supply of labor is inelastic and given by $\mathcal{L}$. Final output is produced by heterogeneous firms, using labor.

A firm is a unit of capital, and the quality of that capital varies across firms. A firm j with capital quality $z_{j,t}$ using $l_{j,t}$ units of labor produces $y_{j,t} = F(z_{j,t}, l_{j,t})$ units of final output. The quality of the capital of a firm of age $a \in [0, \infty)$ is $z_a = e^{\theta a}$, for some $\theta > 0$. Firms are destroyed randomly at a rate $\delta > 0$.

Final output can be consumed and used to create new firms. A flow $X_t \geq 0$ of final output generates a flow βX_t of new firms, where β is a positive parameter. In everything that follows assume that this economy has steady state and consider only steady state outcomes. Write w for the wage and let $v = \max_{m \geq 0}\{F(1, m) - wm\}$ and write l for the labor input that attains this maximum. At any given point in time, the price of a new firm is q. These prices are all in units of consumption. The risk-free interest rate in units of consumption is r. The measure of firms at time t is N_t.

a. Outside a steady state, how does the measure N_t of firms evolve? What is the steady state measure of firms? What is the age distribution of firms in that steady state?

> SOLUTION: ■

b. What is the distribution of capital quality in a steady state? What is the average capital quality?

> SOLUTION: ■

c. What do you know about q/v? What does the technology for creating new firms tell you about q itself?

> SOLUTION: ■

d. Give the steady state equilibrium condition that pins down l in terms of F and parameters. Construct the rest of the steady state and be sure to verify that X_t is less than final output.

> SOLUTION: ■

So far, so good. But now suppose the technology for a firm with capital quality $z_{j,t}$ is $y_{j,t} = F(z_{j,t}, B_t l_{j,t})$, where $B_t = Be^{gt}$ for some parameters $B > 0$ and $g > 0$. And now assume $\theta \in (0, \delta + g)$.

e. Construct the balanced growth path for this economy and determine two firm size distributions: one for firm employment, and the other for firm output.

> SOLUTION: ■

5.11.11 Long-Term Employment Contracts

There is a unit measure of consumer-workers whose preferences over consumption trajectories are

$$\mathcal{U} = \int_0^\infty e^{-\rho t} u(C_t)\, dt.$$

Workers can be matched with a project or they can be unmatched. Unmatched workers produce a flow of $x > 0$ units of consumption. Matched workers produce a flow of $y > x$ units of consumption. While producing $x > 0$, unmatched workers can also use a flow $a_t \geq 0$ of output to search for a new project that will match their personal characteristics. The arrival time of such a project is random and exponentially distributed with mean $M(1, a_t)$. Here, $M : \mathbb{R}_+^2 \to \mathbb{R}_+$ is a strictly increasing production function that is continuous and homogeneous of degree 1. Assume $M(0,1) = 0$, $M(1,0) = 0$, and $\lim_{a \to 0} D_2 M(0, a) = \infty$. Once the worker has been matched with a new project, that worker must stay with the project or the project disappears. The project will also disappear randomly at a rate $\delta > 0$.

Write N_t for the measure of matched workers and let $A_t = (1 - N_t)a_t$. So the technology for aggregate consumption is

$$C_t = (1 - N_t)x + N_t y - A_t.$$

Markets are complete and the risk-free interest rate is r_t in units of consumption. Write U_t for the present value of all output produced now and into the indefinite future by a worker who is currently unmatched. Write V_t for the present value of all output produced now and into the indefinite future by a worker who is currently matched with a project.

a. What is the differential equation for N_t?

SOLUTION: ■

b. Give the asset pricing equations for U_t and V_t. What is the first-order condition for a_t? Construct the asset pricing equation for $S_t = V_t - U_t$.

SOLUTION: ■

c. Write $\lambda_t = Du(C_t)$ and use the Euler equation to construct a differential equation for $s_t = \lambda_t S_t$.

SOLUTION: ■

d. Show that this equilibrium solves the planner's problem for a representative consumer.

SOLUTION: ■

e. Can you think of a decentralization in which workers work for employers who keep all the output produced and always pay a going wage w_t, whether the worker is matched with a project or not? Consider a setting in which firms can commit to long-term contracts that take this form. Determine the equilibrium conditions for the wage w_t and the value of a firm with a single matched worker employed under such a contract.

SOLUTION: ■

5.11.12 Permanent Primary Deficits

Consider the Blanchard [1985] economy as presented in class. In particular, take labor endowments at age a to be $L_a = L_y e^{-\lambda a}$ where $\lambda > \rho$. Assume throughout that this economy has a no-bubble steady state with a real interest rate $r < 0$.

Suppose there is a government that gives a flow of newly printed useless pieces of paper to newborn consumers. Suppose that it so happens that, in equilibrium, the time-t prices s_t of these useless pieces of paper are strictly positive. The government regulates the flow of transfers to newborn consumers so that the aggregate market value of this flow is $T \geq 0$ units of consumption. Consumers alive at $t = 0$ own a strictly positive supply $D_0 > 0$ of useless pieces of paper. The aggregate supply at time t is $D_t = s_t D_t$.

a. Given a trajectory $\{s_t\}_{t\geq 0} \subset \mathbb{R}_{++}$, how does D_t evolve over time? Given real interest rates $\{r_t\}_{t\geq 0}$, how does B_t evolve over time? What do you know about steady state real interest rates if there is a steady state with $T > 0$?

SOLUTION: ■

Write $W_{L,t}$ for the present value of the labor income of a newborn consumer at time t. Write $W_{y,t}$ for the initial wealth of a newborn consumer at time t.

b. Explain why $\theta(W_{y,t} - W_{L,t}) = T$.

SOLUTION: ■

c. Assuming that there is a steady state in which the government can transfer a flow $T \geq 0$ to newborn consumers, show that the steady state conditions for $K > 0$, $B \geq 0$, and $W > 0$ are

$$(\rho + \theta)W = F(K, L) - \delta K, \tag{4}$$

$$(\lambda + \theta)(K + B) = (\lambda + D_1 F(K, L) - (\rho + \delta))\, W + T, \tag{5}$$

$$0 = (D_1 F(K, L) - \delta)\, B + T. \tag{6}$$

SOLUTION: ■

d. Given the assumption that the $T = 0$ version of this economy has a no-bubble steady state with $r < 0$, show that the economy also has a steady state for all $T > 0$ close enough to zero.

SOLUTION: ■

e. What happens to $P_t = 1/s_t$ over time in a steady state in which $T > 0$? What would your answer be if the government was also paying a nominal rate of interest $i \in \mathbb{R}_{++}$ on useless pieces of paper (by printing more of those pieces of paper)?

SOLUTION: ■

5.11.13 Prelim: Spring 2024 (Firm Size Distributions)

Consider an economy with a representative household whose preferences over trajectories $\{C_t\}_{t\geq 0}$ are given by $\int_0^\infty e^{-\rho t}U(C_t)\,dt$, where $\rho > 0$ and $U(\cdot)$ is a nice utility function. This household inelastically supplies a flow $\mathcal{L} > 0$ of labor and produces a flow $\mathcal{E} > 0$ of capital from scratch. Starting from some $K_0 > 0$, the resource constraints are

$$C_t = F(K_t, L_t), \qquad L_t + M_t \leq \mathcal{L}, \qquad DK_t = -\delta K_t + G(K_t, M_t) + \mathcal{E},$$

where $\delta > 0$, and where F and G are both constant returns to scale production functions that are increasing and concave, with smooth marginal products that range throughout $(0, \infty)$. Consider competitive equilibria. Write r_t for the risk-free rate, w_t for the wage, and q_t for the price of capital.

(a) Give the conditions for a competitive equilibrium. Briefly explain.

SOLUTION: Since these are complete markets and all agents have perfect information, we know that the competitive equilibrium allocation is equivalent to solving the planner's problem and we can back out the prices later.

$$\begin{aligned}
\max_{C_t, K_t, L_t} \quad & \int_0^\infty e^{-\rho t}U(C_t)\,dt \\
s.t. \quad & (1) \quad C_t = F(K_t, L_t), \\
& (2) \quad DK_t = -\delta K_t + G(K_t, \mathcal{L} - L_t) + \mathcal{E}, \\
& (3) \quad 0 \leq L_t \leq \mathcal{L},\ C_t \geq 0
\end{aligned}$$

■

(b) Show that in a steady state, the steady state capital stock $K_t = K$ must satisfy

$$K = \frac{\mathcal{E}}{\delta - G(1, m[q])}, \tag{7}$$

$$K = \frac{\mathcal{L}}{l[q] + m[q]}, \tag{8}$$

where $l[\cdot]$ and $m[\cdot]$ are functions of q that are determined by solving (for l and m) the conditions

$$q = \frac{D_1 F(1, l)}{\rho + \delta - D_1 G(1, m)}, \qquad q = \frac{D_2 F(1, l)}{D_2 G(1, m)}. \tag{9}$$

Solution: ■

(c) Eliminate q from (9) and use this to show that l and m must co-move in the region where $G(1, m) < \delta$. You may want to write $g(m) \equiv G(1, m)$ to simplify the algebra. Use this to show that (7) is increasing in q and (8) is decreasing in q.

Solution: ■

Now suppose the flow $\mathcal{E} > 0$ is an average of single units of "startup" capital that arrive at Poisson rates. All the capital that is produced, directly or indirectly, from one particular unit of startup capital is defined to be a firm. Assume that $\delta = \delta_F + \delta_K$, where $\delta_K > 0$ is the rate at which capital inside a firm depreciates continuously, and $\delta_F > 0$ is the rate at which a firm with all its capital is randomly destroyed.

(d) Determine the steady state measure of firms. Determine the stationary distribution of capital across firms when the equilibrium is such that firms grow at a positive rate. How does a reduction in $\mathcal{E} > 0$ affect the mean of this distribution?

Solution: ■

5.11.14 Prelim: Spring 2021 (Growth with a Scale Effect)

There is a unit measure of consumers whose preferences over consumption flows $\{C_t\}_{t \geq 0}$ are determined by

$$\mathcal{U}(C) = \int_0^\infty e^{-\rho t} \ln(C_t)\, dt,$$

where ρ is positive. Consumption is a composite good,

$$C_t = \left(\int_0^{J_t} C_{j,t}^{1-1/\varepsilon}\, dj \right)^{1/(1-1/\varepsilon)},$$

where $\varepsilon > 1$ and J_t is the measure of differentiated commodities at time t. The initial value J_0 is given and positive.

At any point in time, a consumer can choose to be either a worker or an entrepreneur. A consumer with ability vector $(x, y) \in \mathbb{R}^2_{++}$ can supply a flow of x units of labor as a worker, or create new blueprints randomly at an average rate $y \times J_t$ as an entrepreneur. The distribution of these abilities in the population is Ψ. This distribution has full support in $\mathbb{R}^2_{++}$, finite means, and a smooth density ψ. The owner of the blueprint for differentiated commodity j can use labor to produce the commodity. The technology is linear with a unit productivity.

Adopt the following notation. The price at time t of differentiated commodity j is $p_{j,t}$. The price index of all differentiated commodities at time t is P_t. These prices are in units of some arbitrary numéraire. The wage at time t is w_t in units of consumption. The time-t profits that accrue to the owner of a typical blueprint are v_t, also in units of consumption. The price of a blueprint at time t is q_t in units of consumption. The real interest rate at time t is r_t. Markets are complete and consumers face a standard present-value budget constraint.

(a) Determine $[w_t, l_t, v_t]$ in terms of C_t and J_t, where l_t is the amount of labor used per blueprint to produce a differentiated commodity. Also determine the real wage w_t.

Write $\mathcal{L}(s_t)$ for the supply of labor at time t, and $J_t\mathcal{E}(s_t)$ for the aggregate flow of new blueprints created at time t.

> SOLUTION: ■

(b) What is the relative price s_t that governs these factor supplies and makes $\mathcal{L}(\cdot)$ a decreasing function and $\mathcal{E}(\cdot)$ an increasing function? Express these factor supplies in terms of s_t and primitives of this economy.

> SOLUTION: ■

(c) Describe the equilibrium conditions for this economy.

> SOLUTION: ■

(d) What do these equilibrium conditions say about the dynamics of $q_t J_t / C_t$?

> SOLUTION: ■

(e) Determine the equilibrium of this economy. What happens if consumers suddenly (and unforeseen) become more patient?

Solution: ■